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Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Sources and code: <https://github.com/matteodisante/soaring-anomalous-transport>

Abstract

Soaring flight couples persistent horizontal motion to the search for atmospheric lift and the management of altitude. This thesis investigates which stochastic descriptions can account for that motion using paraglider and hang-glider records from the French free-flight federation. It combines empirical observables, probabilistic flight-phase segmentation and a programme of mathematical modelling and Monte Carlo validation.

The data chapter documents recorder channels, cleaning and the limits of the retained trajectories. Global transport is then examined through displacement moments and quantiles, constant-velocity controls, regional anisotropy and velocity memory at several temporal resolutions. A fresh subset analysis over 10–10,000 seconds shows why effective exponents, changes of distribution shape and finite observation windows must be distinguished. Principal-axis analysis describes regional geometry without identifying wind or removing environmental effects. Competing stochastic models are assessed through explicit predictions, with the scope of each exclusion stated.

The segmentation chapter presents the current probabilistic decoder, readable flight examples and quantitative diagnostics. Independent manual annotation remains necessary before accuracy can be claimed. Subsequent work will measure observables conditioned on flight phase, compare solitary and group flight, construct and simulate a physically constrained stochastic model, and investigate thermal landscapes and route optimisation.

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Chapter 1

Introduction

1.1 A flight is a transport process with an energy budget

A soaring pilot travels by repeatedly exchanging height for distance and recovering height in rising air. The atmosphere supplies the energy, but the pilot must locate that supply and decide how to use it. A long displacement is therefore the outcome of several coupled processes: motion through the air, encounters with lift, decisions made with incomplete information, and constraints imposed by wind and terrain. Because the recorded trajectory contains all these contributions, inferring a stochastic description requires more than fitting its geometry: the description must also preserve the physical mechanisms that allow the pilot to remain airborne.

Two elementary relations explain why neither an unconstrained random walk nor a purely geometrical description is sufficient. Writing $\mathbf{r}$ for horizontal position and z for altitude, a local kinematic description is

$$\dot{\mathbf{r}}(t) = \mathbf{v}_{\text{air}}(t) + \mathbf{w}_h(\mathbf{r}(t), z(t), t), \quad (1.1)$$

$$\dot{z}(t) \simeq w_z(\mathbf{r}(t), z(t), t) - s(|\mathbf{v}_{\text{air}}(t)|, \phi(t)). \quad (1.2)$$

Here $\mathbf{w}_h$ and w_z are the horizontal and vertical air velocities, $s > 0$ is the sink rate relative to the air, and ϕ is bank angle. The second relation is a quasi-steady approximation: it makes explicit the competition between lift and aerodynamic sink without attempting to model every manoeuvre. Altitude is therefore a finite resource, and a transport model must account for its use and replenishment if its trajectories are to represent sustained flight.

This physical picture suggests three descriptions of behaviour: directed *transition*, exploratory *search*, and *climb* in lift. They are hypotheses about the organisation of motion, rather than labels directly supplied by a recorder. A pilot can find lift during a transition, climb without completing a circle, or search while losing height. Determining which distinctions the data support is part of the scientific problem.

Soaring control has been studied through the problem of locating and exploiting thermals, including adaptive strategies based on local information [1]. The present work asks a complementary question: what statistical laws emerge over entire flights, and which mechanisms can account for those laws? A direct empirical precedent is the analysis of soaring transport and flight phases by Vilpellet et al. [2]. This thesis revisits that connection using the FFVL archive, explicit data-quality controls, and a separation between measured properties and assumptions used to interpret them.

1.2 The questions and the evidence needed to answer them

1.2.1 Which stochastic models survive the observations?

A first observable is the squared displacement over an interval τ ,

$$M_2(\tau) = \left\langle |\mathbf{r}(t + \tau) - \mathbf{r}(t)|^2 \right\rangle. \quad (1.3)$$

Ordinary diffusion without drift has $M_2 \propto \tau$; directed motion can produce a ballistic law $M_2 \propto \tau^2$. An intermediate slope is informative, but it does not identify a mechanism, since correlated Gaussian motion, persistent motion observed during a crossover, and mixtures of flight behaviours can produce similar second moments. Random-walk frameworks offer competing explanations of anomalous transport [3]. The purpose here is to discriminate between such explanations using more than one curve.

Chapter 3 therefore measures displacement quantiles, the spectrum of moments, the shape of the increment distribution, directional anisotropy and velocity correlations alongside the MSD. Their roles are complementary: quantile ratios test whether the distribution changes shape as the lag increases, while higher moments compare the growth of ordinary and large displacements; velocity correlations resolve persistence in time. Open and closed tasks help expose the effect of the flight's intended geometry. A model is challenged where its predictions disagree with these observations under a comparable measurement protocol.

This logic also sets the limit of an exclusion. A discrepancy with a homogeneous Gaussian model for pooled flights need not exclude Gaussian fluctuations within a flight: different conditions can create a non-Gaussian mixture. Similarly, evidence against a particular constant-speed renewal Lévy walk is not a theorem against every process with finite-speed legs [4]. The chapter specifies the hypotheses being tested and leaves unsupported exclusions open.

1.2.2 How do the flight phases generate global transport?

Although an unsegmented flight supplies constraints on the whole process, it does not identify which behaviour generates a long displacement, a change of heading or a period of persistence. Chapter 4 addresses this missing link with a probabilistic segmentation into transition, search and climb. It states the mathematical model, displays its behaviour on readable trajectory excerpts, and distinguishes internal consistency from accuracy against independent annotations.

The next step will be a chapter devoted to observables conditioned on phase: episode durations, distance and height changes, speed and turning distributions, displacement scaling, and dependencies between successive episodes. A useful model must reproduce both these conditional statistics and the global ones, a stronger requirement than generating a visually plausible sequence of coloured flight segments. Windows that cross phase boundaries must also be retained when reconstructing global observables.

1.2.3 Does information from other pilots change the motion?

A nearby pilot can reveal lift before a follower reaches it. This creates a possible information channel that is absent from a model of independent searchers. A later chapter will compare solitary and group flight using both global and phase-conditioned observables. The operational definition of a group will concern simultaneous proximity and plausible visibility, not membership of a competition. A competition can contain isolated pilots, and pilots outside a competition can travel together.

The comparison will ask where differences occur: in time spent searching, climb selection, transition direction, episode persistence, or the efficiency of converting altitude into distance. Site, weather, wing and pilot characteristics must be controlled because they can produce the

same differences without social interaction. Incomplete recording of nearby pilots will limit what can be called solitary flight. These are conditions for an interpretable comparison, not effects assumed in advance.

1.3 From empirical constraints to a physical model

The thesis combines empirical analysis, mathematical modelling and simulation, with a distinct role for each. Data analysis establishes the constraints that a candidate process must satisfy; mathematics then makes that process precise and derives its predictions, which Monte Carlo simulation tests under finite flight durations, heterogeneous environments and the actual observation protocol.

Model construction will follow the global analysis and the analysis conditioned on phase. Possible ingredients include finite-speed transitions, lift encounters, distributions of episode duration, and dependencies between successive directions or episodes. Renewal independence is one hypothesis to test; correlated random walks provide alternatives when that independence fails [5, 6]. The choice among these ingredients must be justified by the observations and an explicit physical mechanism, rather than fixed by the outline of the thesis. Parameters fitted on one set of flights will be checked on separate flights or days, using several observables jointly.

A reference model without wind and terrain may help isolate mechanisms, but obtaining it requires an inference about the environment: mean ground velocity alone does not determine wind, and rotating a trajectory into its principal axes does not remove orography. Artificially making a covariance matrix isotropic changes the measured object; it does not recreate a flight in still air. The analysis consequently preserves the motion in geographical coordinates and studies environmental strata before asking which properties could survive in a common reference model.

A further part of the programme concerns the spatial organisation of thermals and route choice. It will first estimate the two-dimensional distribution of *visited* lift regions, accounting for where pilots have searched. Recovering the distribution of available thermals requires an observation model: unvisited terrain is not evidence of absent lift. A mathematical description of thermal availability and strength could then support separate optimisation problems, such as minimum-time travel between two locations, a closed route, or maximum distance within a specified flight budget. Wind and, subsequently, possible geometrical constraints from terrain will be introduced in stages. A spatial density alone cannot determine an optimal path without assumptions about thermal strength, spatial dependence, altitude and the pilot's information.

1.4 Structure and current scope

Figure 1.1 shows how the measurements support the later modelling. The present document develops the data, global transport and segmentation chapters. The chapters on conditional observables, social flight, stochastic modelling and route optimisation form the research programme; their results are not claimed here. The immediate priority is to consolidate Chapters 3 and 4, then undertake the solo-group comparison with the necessary observables conditioned on phase.

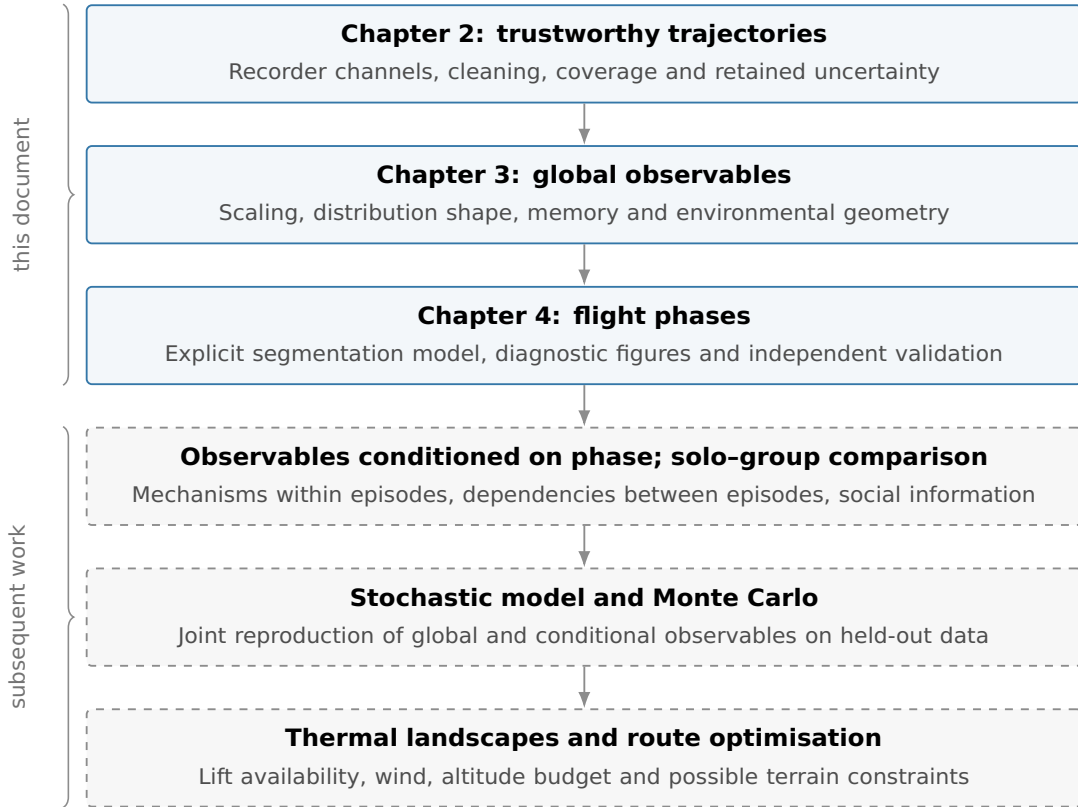


Figure 1.1. Logical structure of the thesis. Solid boxes are developed in this document; dashed boxes describe subsequent work. Model comparison must return to the measured observables, not stop at a fitted trajectory or a single exponent.

Chapter 2 follows the records from acquisition to retained trajectories, including the treatment of missing or inconsistent channels. Chapter 3 then asks what can be learned without assigning phases. Chapter 4 introduces that assignment and identifies what must be checked before its labels can support physical claims. The two appendices state the spectral conventions and the geodetic transformation used by the measurements.

1.5 Chapter summary

The thesis asks which stochastic description can account for soaring trajectories while respecting the physical roles of lift, travel and environmental constraints. The route from observations to a model begins with reliable trajectories and global transport statistics, then uses validated flight phases to identify the mechanisms mixed by those statistics. Comparisons between solitary and interacting flights, followed by simulation and thermal-landscape modelling, extend this programme. Their conclusions remain future work: the present chapters establish the empirical measurements and validation needed to make those comparisons meaningful.

Chapter 2

The dataset

This chapter follows the data end to end: how the raw archive is acquired, verified and kept reproducible (Sec. 2.1); what the records contain and what the collection looks like (Secs. 2.2–2.6); and the pre-processing that turns a raw tracklog into the analysis-ready trajectory the transport analysis consumes (Sec. 2.7). The pipeline summary (Sec. 2.8) brings these steps together; a characterization of the retained dataset closes the chapter (Sec. 2.9).

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata—among them pilot, date, take-off and landing sites, scored distance, duration, and glider, with the full set of fields listed in Sec. 2.3—and, in most cases, the raw GPS tracklog in IGC format (the standard flight-recorder format, Sec. 2.2). A “season” is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

2.1.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package, exposed through two command-line tools, one per discipline, sharing the same subcommands for fetching, downloading, cataloguing, and verifying. The design favours robustness for a long-running, large

job: downloads are resumable and safe to interrupt, a response is accepted only once validated as real IGC content, and download requests are rate-limited so as not to load the server .

2.1.4 Reproducibility

The destination directory is selected through an environment variable or configuration file. The local configuration contains the working disk path; an environment override allows the same acquisition to run elsewhere. Archiving the season XMLs verbatim preserves the acquisition provenance.

2.1.5 Integrity

Because the data sit on an external drive, integrity is re-checked independently after download, rather than trusted at download time . Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass the same content check applied at download — a readable file whose first record is an A header and which carries at least one B fix record (Sec. 2.2). These are the tracklogs actually fetched; they cover a subset of the flights that carry a track at all (Sec. 2.5).

2.1.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data are the main analysis target; the hang-glider data are collected in parallel and enable the cross-type comparison of Sec. 2.4 (the counts for both are given in Sec. 2.5). Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Sec. 2.4). A large sailplane source with a machine-readable interface is *WeGlide*, which exposes flight metadata and IGC tracks through an API. The default access is rate-limited (of order sixty requests per day), which rules out a bulk export of the archive; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. A comparable research-access request has been sent to XContest; as of 20 July 2026 neither WeGlide nor XContest has replied. All of these can be acquired with the same approach, each producing a catalog in the same format.

2.2 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [7], decoded from fixed character positions; every downloaded track is checked at acquisition to be IGC content — an A record followed by at least one B record (Sec. 2.1).

Each line is a *record* whose type is set by its first character: A the flight-recorder identifier, H header fields (date, pilot, glider, datum, ...), I the declaration of optional extra fields, and B the GPS fixes that form the body of the file. A B record decodes to a UTC time, latitude and longitude on the WGS84 datum, a fix-validity flag, and *two* altitudes in metres — barometric (pressure) then GNSS. The flag describes the GNSS solution: A denotes a three-dimensional fix, whereas V denotes either a two-dimensional fix or an absence of GNSS data [7]. Thus V invalidates the adopted GNSS altitude but does *not* certify that the horizontal coordinates remain correct. The cleaning retains those coordinates provisionally and subjects them to the horizontal and frozen-lock checks below. A pressure reading in the same B record is a separate measurement; its numerical presence alone does not establish that the sensor is working.

Decoding can succeed numerically on a corrupted byte that is not a legal encoding, so the parser also checks format validity at decode time (a valid time, valid coordinate fields, a valid hemisphere letter), dropping the record otherwise, on the same footing as a record of the wrong length. This is distinct from the *plausibility* checks of fix-level cleaning (Sec. 2.7.2), which target fixes that decode validly but are physically implausible given the rest of the trajectory.

The sequence of B records is a time-stamped three-dimensional trajectory at a sampling period typically of a few seconds, not guaranteed uniform and varying across recording devices — a point the analysis must handle explicitly.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

Table 2.1. Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

Field(s)	Content / caveat
date	flight date
pilot, club	pilot and club
take-off, landing, department	the two sites and the French department
aile	wing model name (raw XML field; French for “wing”)
aile_class	wing class, one system per discipline (Sec. 2.4)
flight type	the <i>scoring</i> category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, ... (<i>caveat</i> : a scoring label, not a glider or manoeuvre type)
distance, points	scored distance and points
duration, speed	duration and average speed, where present

From these, the pipeline builds a single derived table, the **catalog** (`catalog.csv`), one row per flight across all seasons: the metadata above plus a `local_path` linking each flight to its track on disk and a `downloaded` flag, so a flight traces to its file and back to its metadata and public page without an external lookup. A smaller **season index** (`seasons_index.csv`) summarises each season by flights declared, carrying a track, and downloaded, and is the source of the statistics below. Both are regenerated from the archived XMLs on demand (`build-catalog`).

2.4 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. The reference study distinguishes three aircraft families: paragliders with a flexible canopy, hang gliders with a framed flexible or rigid wing, and sailplanes with a rigid airframe. Their speed ranges, glide performance and turning geometry differ, which motivates separate analyses rather than a single pooled transport law [2].

The category is set by the *data source*: the FFVL competition is split by discipline (Sec. 2.1), so the present dataset spans both paragliders and hang gliders, already enough for the cross-type comparison of [2]. Sailplanes are collected by different portals and not yet in hand.

Within each discipline, the catalogue field `aile_class` provides equipment groups (Table 2.2). For paragliders it combines EN and older LTF labels, for example “C ou 2”; the

Table 2.2. Class labels recorded in `aile_class`, one system per discipline. EN labels summarize test-based recovery characteristics; FAI classes distinguish control and design. They are used as equipment groups, without assigning pilot ability. Sources: the FAI Sporting Code, Common Section 7 [9] (§1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the consulted 2012 final draft [8].

Class	Notes
Paragliders — harmonized EN/LTF catalogue groups	
EN A	maximum passive safety, extremely forgiving; any level of training
EN B	good passive safety, forgiving; ceiling of the FAI <i>Standard</i> sub-class
EN C	moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI <i>Sport</i> sub-class
EN D	demanding, potentially violent reactions; for pilots practised in recovery
CCC	CIVL Competition Class racing wings
tandem / non-certified	two-seater and uncertified wings, pooled in this field
Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)	
Delta Class 1	weight-shift as the sole method of control
Delta Class Sport	sub-class of Class 1: type-certified production wing with a king post
Rigide Class 2	movable aerodynamic surfaces as the primary control
Rigide Class 5	as Class 2, but in the roll axis only; no fairings

analysis harmonizes these catalogue codes under the short labels EN A, B, C and D, with CCC and tandem/non-certified entries kept distinct. This is a grouping convention, not an independent reassessment of a wing’s certificate.

The EN categories describe responses in prescribed flight-safety tests and the demands of recovery, as set out in the consulted 2012 final draft [8]. They do not define a measured ladder of cruise speed or glide ratio. Equipment class can be associated with performance and pilot selection, but those associations must be checked in the observations. The published 2013 standard and subsequent amendments were not compared here, so the table is not presented as a current certification specification.

The FAI hang-glider classes principally distinguish design and control systems [9]. Our stratified comparisons require reproducible group membership; they do not assume that classes are equally spaced on a performance scale. Neither scheme measures an individual pilot’s experience. Blank and placeholder entries remain unclassified, and a tandem designation alone does not establish whether that wing has an EN certificate. Chapter 3 uses the paraglider classes to define two operational experience proxies, beginners (EN A/B) and experts (EN C/D/CCC), while retaining this distinction between equipment requirements and individual skill.

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights,

of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.3. FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.4. FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the “With GPS” column of Tables 2.3 and 2.4.
- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights. GNSS altitude is used for every admitted flight, with the channel recorded per flight (Sec. 2.7.1).

- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries carry the placeholder date 0000-00-00, written as a sentinel at acquisition. Such entries cannot support a calendar-conditioned comparison until a date is recovered independently; the relative-time preprocessing does not itself enforce this metadata exclusion. The count is small: 8 of 203,343 paraglider entries carry the sentinel (0.0039%), 0 of 9,259 hang-glider ones, and the largest within-season fraction is only 0.17% of its entries (2001-2002).

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude. The pipeline then proceeds in a fixed order:

- (i) choose the altitude channel (Sec. 2.7.1);
- (ii) clean individual fixes (Sec. 2.7.2);
- (iii) trim the ground phases (Sec. 2.7.3);
- (iv) filter out unfit whole flights (Sec. 2.7.4);
- (v) convert to a local Cartesian frame (Sec. 2.7.5);
- (vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);
- (vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output consists of fix, segment and attempted-flight tables, accompanied by a table of suspect intervals retained for later sensitivity analysis. What each file holds and how the disk is organised is set out in Sec. 2.8.

Steps (i)–(iv) act on the *raw geographic* coordinates, since none of them needs a direction. A horizontal distance comes straight from (φ, λ) through the great-circle (haversine) formula of Eq. (2.2); a speed is one such distance over an elapsed time; an altitude is read as logged. The conversion (v) to the metric East–North–Up frame (Sec. 2.7.5) then runs on the cleaned, trimmed track, with the origin at the first fix that survives trimming. Its reliability remains subject to the cleaning and archive checks; passing them does not establish the true position. Steps (vi)–(vii) come last because they produce *vector* quantities: 3D position, velocity and acceleration components, which need the Cartesian axes step (v) provides.

Every step below is governed by a handful of numerical thresholds. Each is stated where it is argued, and the whole set is gathered, with the symbols used here, in Table 2.8 in the pipeline summary (Sec. 2.8). All of them are *working values*, specified independently of the transport exponents and audited through their removal fractions (Table 2.7).

Status of the numerical snapshot. The census, transport measurements and phase analysis use the archive regenerated with pipeline version 2.2.0. It applies the local paired-barometer requirement, the 10 m s^{-1} vertical-speed threshold and the propagation of reconstructed altitude through derivative-estimation windows. The checks establish agreement with these numerical rules, not the absence of every recorder error. The witness audit below concerns the isolated version-2.0.1 correction. The targeted comparison reran the legacy and corrected rules on 2,388 candidates: 189 flights changed. Frozen-run counts changed in 188 cases; interior-ground counts changed in 2 cases (the categories can overlap). The corrected witness retained 22,468 additional horizontal fixes; 0 flight admission decisions changed. These counts describe the

targeted set, not detector error rates. They do not replace regeneration of the archive and its observables.

2.7.1 Choice of altitude channel

The analysis uses the logged *GNSS* altitude for every admitted flight. This avoids switching between pressure altitude and GNSS within the population, but it does not establish a common vertical datum. IGC-approved recorders specify ellipsoidal height; the published CIVL specification document for hang-gliding and paragliding recorders prescribes geoid-referenced GNSS altitude [7, 10]. The published CIVL document contains draft annotations, and recorder compliance across this historical archive has not been established. The parser currently preserves the numerical field without resolving these conventions. Throughout the analysis, z therefore means *logged GNSS altitude*.

Pressure altitude also depends on atmospheric pressure and temperature. The channel choice is a consistent data-processing convention, not evidence that GNSS is always more accurate. A constant height offset cancels from increments; a spatially varying geoid correction or a changing sensor bias does not. Absolute-height comparisons need a recorder-datum audit before they can be interpreted geodetically.

Comparing the barometric and GNSS channels. The spectral comparison starts from 3,000 raw flights sampled per discipline. It retains flights with at least 95% pressure-altitude presence and a median cadence within 0.25 s of the pooled mode of median intervals rounded to seconds. Within each flight, both spectra use the same longest block of paired finite, nonzero readings with a valid GNSS flag. Duplicate or backward timestamps and gaps larger than 1.5 modal intervals break that block; the earliest block wins a tie in fix count. Each channel is then linearly resampled at the modal interval, with at least 256 raw fixes and 256 grid values required. Thus the comparison neither interpolates across missing-channel runs nor represents every sampled recorder. Appendix 2.A defines the spectral estimator. Figure 2.1 shows the resulting comparison. Panel (a) shows one representative flight over a 30 min window: the two raw channels track each other with a varying offset. Panel (b) is their difference after removing its mean. The remaining variation shows that a constant offset alone does not align these channels, even within this window. Such a change can include pressure and temperature effects, spatially varying reference-surface differences, and GNSS error; these traces alone do not separate them. Panel (c) is the per-frequency *median* Welch spectrum of each channel (Appendix 2.A), pooled over both disciplines with the 10 to 90 percentile band across flights: at high frequency the two medians nearly coincide, both compatible with contributions from metre-resolution recording and sensor noise. Coincident median spectra do not establish equal accuracy on individual flights.

Variation between flights. Similar medians can conceal different upper tails. We therefore measure each flight’s median PSD in the 0.35 Hz to 0.50 Hz band and compare it with the median of these values across barometric records. Of the 2,008 paired flights contributing to the spectrum, 6.5 % carry a GNSS floor above three times the barometric channel’s typical one, and 2.1 % above ten times it. Multipath is one possible GNSS error source [11]; this comparison does not attribute the measured excess to a unique physical mechanism.

Availability. A flight is admitted only when its GNSS channel is both present – covering at least 95 % of its fixes – and alive – spanning at least 30 m (Sec. 2.7). These operational criteria reject missing or nearly constant channels without identifying the underlying sensor failure; a genuine small altitude excursion can also fail the range criterion. Figure 2.1d shows where the population actually sits against the presence cutoff: nearly every flight is at or near

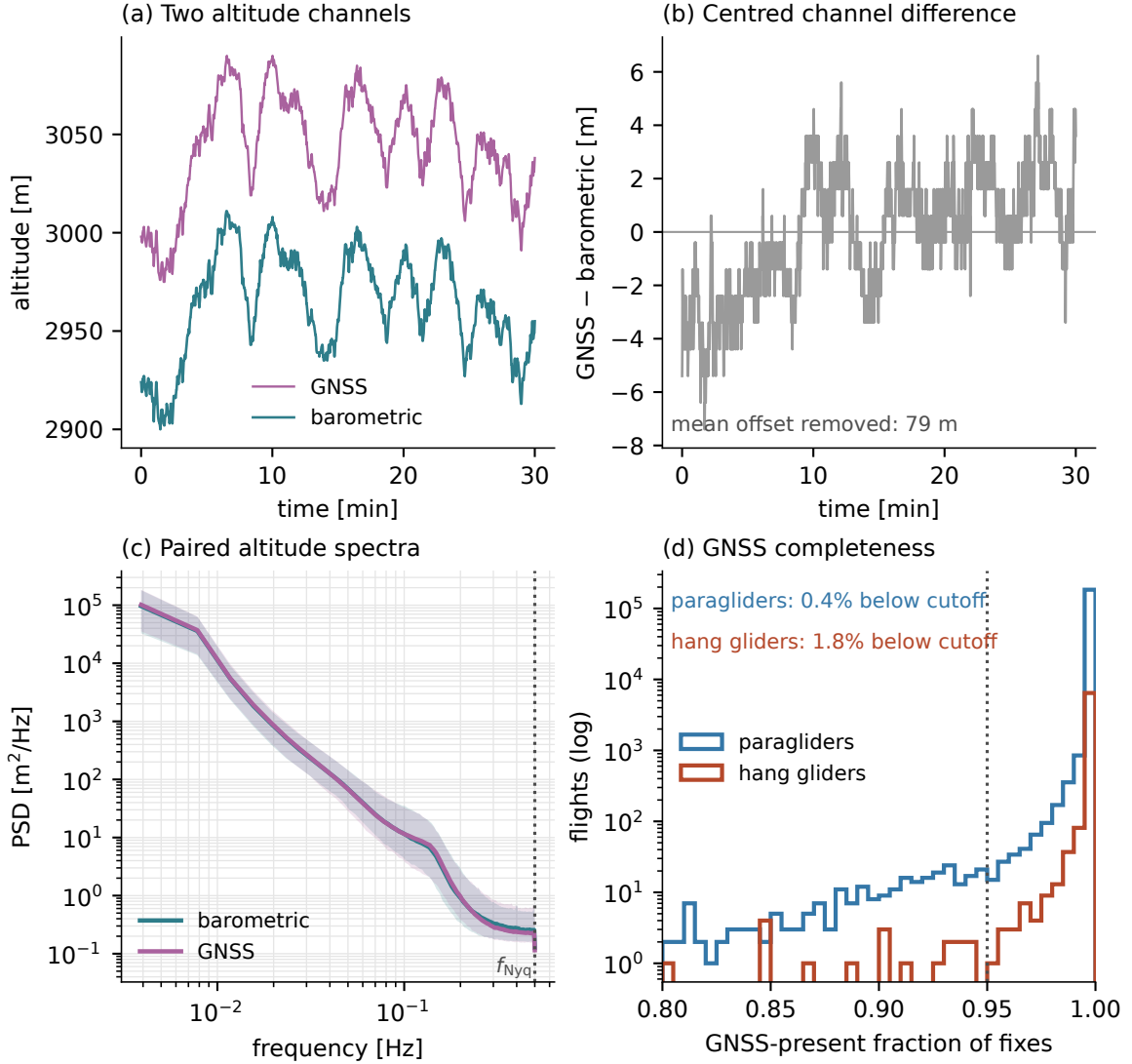


Figure 2.1. Barometric vs GNSS altitude noise. (a) One representative flight over a 30 min window, the two raw channels on a shared axis. Their separation combines reference-datum differences, atmospheric effects and sensor errors. (b) Their difference after removing its mean; the remaining variation is visible within the 30 min window. (c) Per-frequency *median* Welch spectrum of each channel (Appendix 2.A), pooled over the two disciplines, with the 10 to 90 percentile band across flights. The medians are close at high frequency; the bands show variation between recorded flights, not uncertainty on a mean spectrum. The dotted line marks the Nyquist frequency $f_{Nyq} = 1/(2\Delta t)$. (d) Each flight's GNSS-presence fraction, per discipline, zoomed to 0.80 to 1.0 on a log flight count: nearly every flight sits at or near full presence, with a modest population thinning out on approach to the 95 % admission cutoff (dotted). Below it a flight fails the adopted completeness criterion and is excluded, without substitution from the barometer (Sec. 2.7.1).

full presence, with only a thin population approaching it from below. A flight below either bound fails the adopted vertical-data quality requirement and is excluded from this analysis; this does not imply that every one of its coordinates is unusable. The presence criterion alone excludes 0.4 % of paraglider and 1.8 % of hang-glider flights. Both channels fall below their presence threshold in 0.14 % and 1.13 %, respectively; this is not a count of channels with every reading absent. Within an admitted flight, a fix with missing GNSS altitude keeps its horizontal coordinates. Only its altitude is marked missing; resampling reconstructs that channel and marks it `z_reconstructed`. There is no substitution from the barometer. The affected-flight fractions are 6.1 % for paragliders and 10 % for hang gliders; the median number of missing altitudes among affected paraglider flights is 5.

A witness must be available where it is used. The pressure channel has a separate role: it can corroborate a suspected frozen GNSS position and an interior ground interval (Secs. 2.7.2 and 2.7.3). Its flight-level eligibility requires at least 95 % nonzero finite readings and a range of at least 30 m. The corresponding archive fractions, 72.1 % and 84.8 %, describe eligibility, not coverage of every candidate interval.

At fix i , define $b_i = 1$ when a finite, nonzero pressure altitude accompanies the horizontal position in the same decoded record, and $b_i = 0$ otherwise. For a candidate interval I , the barometric test is available only if

$$B_I = B_{\text{flight}} \prod_{i \in I} b_i = 1. \quad (2.1)$$

Before duplicate timestamps are merged, zero pressure values are converted to missing: a duplicate pair $\{0, 1000\}$ supplies its one available 1,000 m reading, not an invented 500 m average. If both are missing, the merged witness remains missing. This uses simultaneous readings, not interpolation across time.

A globally usable barometer with even one missing reading inside I cannot certify that interval: the barometric criterion abstains. The missing values are neither interpolated for this decision nor treated as zeros. The pairing needed here is *position and pressure at the same timestamp*; requiring a valid GNSS altitude as well would remove the corroborating pressure record precisely when the GNSS altitude fails.

Requiring both sensors on *every flight before cleaning* would select a different population of recording devices and discard usable horizontal trajectories. Instead, GNSS-only flights remain eligible and the weaker recorder-declaration rule is stated explicitly below. A sensitivity analysis restricted to $B_I = 1$ would test how much a result depends on this weaker evidence.

2.7.2 Fix-level cleaning

Cleaning separates evidence of corruption from the action taken on the trajectory. A horizontal defect changes which positions can be used; an altitude defect need not. The three cases in Fig. 2.2 organize this distinction:

1. **Off-the-trend horizontal defect.** A position spike or an out-and-back jump may receive a Hampel flag. Deletion depends independently on the impossibility gate: the position is unreachable at the speed bound and removal of its block permits a reachable rejoin. A flag alone never deletes a fix.
2. **On-the-trend defect.** Repeated or backward timestamps and frozen positions can agree geometrically with their neighbours. They require time and recorder checks. Duplicate timestamps are merged; frozen runs produce a mandatory segment boundary.

3. **Vertical defect.** A missing or implausible altitude is invalidated while its horizontal fix remains. Reconstruction is recorded separately from horizontal interpolation, so later vertical analyses can identify it.

A retained fix is not guaranteed correct. Slowly varying errors and plausible-looking corruption can escape every local test; smoothing does not prove that these errors are small. The operational aim is to remove positively identified defects while making removals, reconstruction and remaining uncertainty inspectable.

Geometry on the raw coordinates. The cleaning runs on the raw geodetic coordinates (φ_i, λ_i) , before the Cartesian conversion (Sec. 2.7.5), so distances between fixes are great-circle (haversine) distances [12] on a sphere of mean radius $R_\oplus = 6,371$ km:

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos \varphi_i \cos \varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \quad (2.2)$$

giving the horizontal speed $v_{xy,i} = \Delta s_i / (t_{i+1} - t_i)$ and, from the adopted altitude channel, the vertical velocity $v_{z,i} = (h_{i+1} - h_i) / (t_{i+1} - t_i)$; the same distance also yields the whole-flight *path length* $L = \sum_i \Delta s_i$ and *extent* $D = \max_i \Delta s(\text{fix}_0, \text{fix}_i)$. The spherical approximation is ample: replacing the WGS84 ellipsoid by a sphere perturbs an inter-fix distance by about 0.3% at the mid-latitudes of the main study regions¹, about 3 cm on a 10 m step, far below the metre-scale noise of consumer GNSS receivers.

The Hampel identifier. Let the window of fix k be the fixes within $\pm w$ of t_k , and $\tilde{\varphi}_k, \tilde{\lambda}_k$ the component-wise window medians. The residual is the great-circle distance from that median point,

$$r_k = \Delta s((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)), \quad (2.3)$$

with local scale $\text{MAD}_k = \text{med}\{r_j : |t_j - t_k| \leq w\}$, and the fix is flagged when

$$r_k > \max(k \sigma_k, \varepsilon_{\min}), \quad \sigma_k = 1.4826 \text{MAD}_k, \quad (2.4)$$

σ_k a robust local scale [13]. The factor 1.4826 is the usual one-dimensional Gaussian MAD normalization; for these two-dimensional radial residuals it is a scale convention, not an exact standard deviation or a calibrated false-alarm probability. The flag requires both a relative departure and an absolute floor $\varepsilon_{\min}$ of a few times the GPS noise. The working values are $k = 5$, a half-window $w = 20$ s and $\varepsilon_{\min} = 15$ m, with the test evaluated only where the window holds at least 5 fixes. A flag does not delete: what deletes is the absolute v_{xy} bound, which removes a fix when

- it is unreachable from the last accepted fix at that bound, *and*
- the removal of its block lets the track rejoin within it.

The identifier flags and attributes, and supplies the per-flight anomaly count, but is not required for deletion: a genuine tight thermalling circle produces the same local-median departure as a spike, at a fraction of the speed a spike would imply, so the protection lies in the speed bound that does the deleting rather than in the flag.

¹The ellipsoid's local radius of curvature runs from the meridian value $M(\varphi)$ to the prime-vertical value $N(\varphi)$ (Eq. (2.B.5)), which the haversine formula replaces by the single radius $R_\oplus$. At $\varphi \approx 45^\circ$, the latitude of the Alps sites that launch most of this archive (Sec. 2.9), $N(45^\circ) = 6,388.83$ km and $M(45^\circ) = 6,367.38$ km against $R_\oplus = 6,371$ km: a step is shortened by approximately 0.28% running east-west and stretched by approximately 0.06% running north-south, together the $\lesssim 0.3\%$ above. More generally the deviation is of order the flattening $f = 1/298.257 \approx 0.34\%$ at any latitude, up to 0.56% for a meridional step at the equator and -0.45% near the poles. Derived in Appendix 2.B.

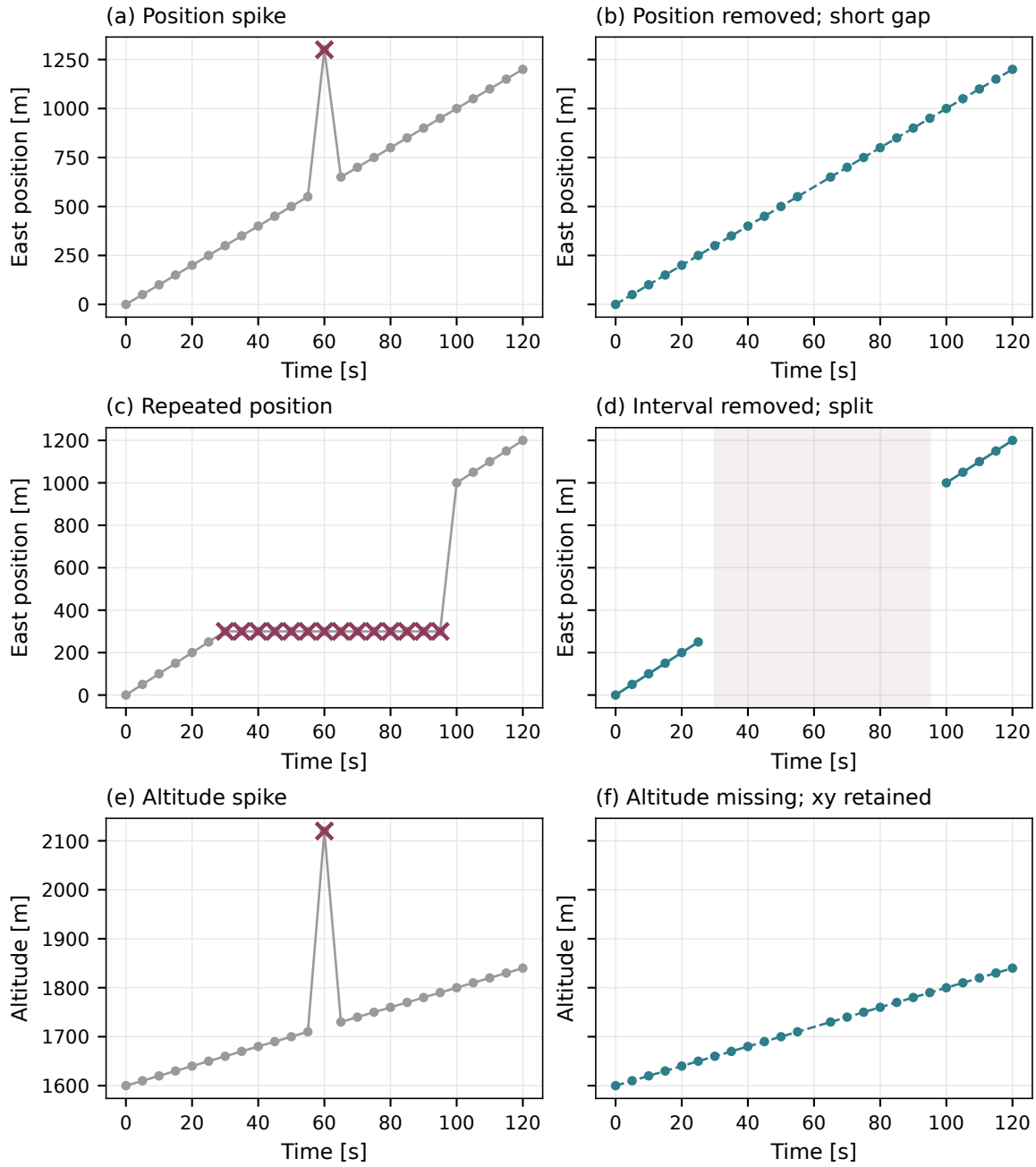


Figure 2.2. Three synthetic defects and their distinct consequences. Crosses mark the suspect readings; dots show the retained information. A horizontal spike is deleted only after the speed/rejoin test; a repeated-position interval is cut and split; an altitude spike leaves the horizontal fixes in place. Dashed lines show possible reconstruction, performed later at resampling. These examples illustrate the rules and do not estimate their accuracy on the archive. Repeated positions also require the duration and witness conditions described in the text.

What the median vertical-speed test measures. Write the observed altitude as $h_i = z_i + \epsilon_i$. Even a smooth true trajectory produces noisy finite differences,

$$v_{z,i}^{\text{obs}} = \frac{h_{i+1} - h_i}{t_{i+1} - t_i} = \bar{v}_{z,i}^{\text{true}} + \frac{\epsilon_{i+1} - \epsilon_i}{t_{i+1} - t_i}. \quad (2.5)$$

For independent errors of variance σ_z^2 at uniform cadence Δt , the noise term has variance $2\sigma_z^2/\Delta t^2$. This illustrative calculation explains why one large derivative at a short cadence is weak evidence of a defect; independence is not assumed for real GNSS noise.

The sustained-excess statistic is

$$\tilde{v}_j = \text{med}\{|v_{z,i}^{\text{obs}}| : |m_i - m_j| \leq w_v, v_{z,i}^{\text{obs}} \text{ finite}\}, \quad m_i = \frac{1}{2}(t_i + t_{i+1}), \quad (2.6)$$

with $w_v = 5$ s. It measures the typical *magnitude* of the local observed vertical increments, not a signed climb rate or a noise-free velocity. Each value belongs to a *step*, located at its midpoint m_i ; a record of N fixes therefore supplies $N - 1$ values. Figure 2.3 shows the calculation: select the midpoints within the window, take the absolute speeds, sort them, and read the middle value. For $2r + 1$ finite values, the median exceeds a bound only if at least $r + 1$ values do. For an even number, the median is the average of the two central values.

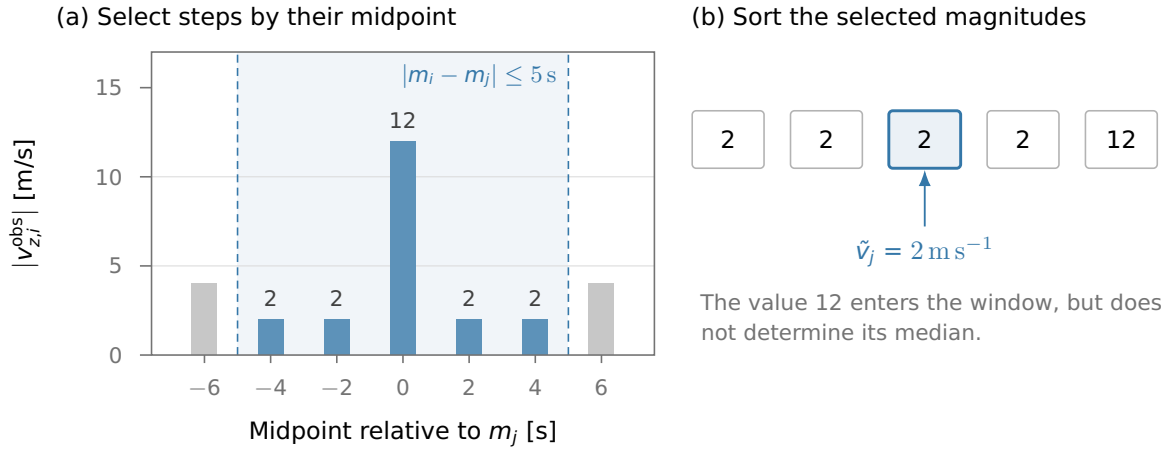


Figure 2.3. How Eq. (2.6) is evaluated. This schematic uses fixes every 2 s and a half-window of 5 s: five step midpoints lie inside it. Grey bars are outside the window and are excluded, regardless of their height. Missing or non-finite speeds are also excluded. Taking absolute values makes an ascent and a descent of equal magnitude contribute equally. These are illustrative values, not flight data.

With fewer than 5 finite steps, the implementation does not use the median. Writing n_j for that count, the quantity used by the cleaning is

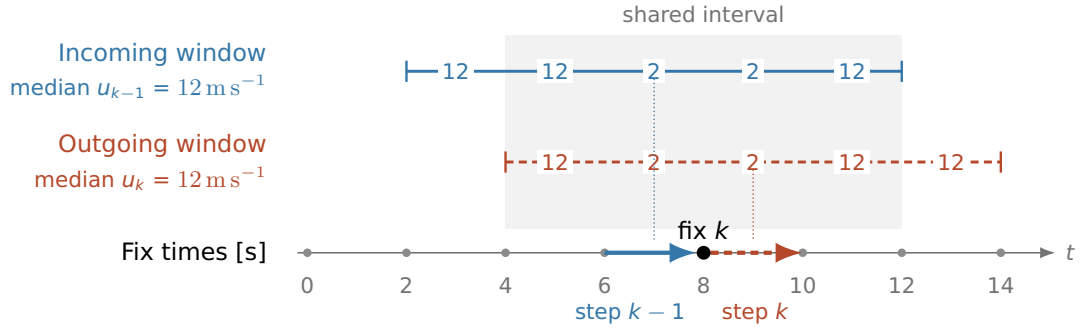
$$u_j = \begin{cases} \tilde{v}_j, & n_j \geq 5, \\ |v_{z,j}^{\text{obs}}|, & n_j < 5. \end{cases}$$

The second branch is the sparse-window fallback. It can occur at a slow recording cadence, near missing altitudes, or at the ends of a record. The sustained-excess rule marks the altitude of an interior fix k as missing when

$$u_{k-1} > v_z^{\text{max}} \quad \text{and} \quad u_k > v_z^{\text{max}}, \quad v_z^{\text{max}} = 10 \text{ m s}^{-1}. \quad (2.7)$$

Step $k - 1$ arrives at fix k and step k leaves it. Their statistics must *both* exceed the bound; equality does not trigger the rule (Fig. 2.4). When both windows are populated, this tests the surrounding speeds. It does not require either of the fix's own increments to exceed the bound.

(a) Two windows meet at one decision



(b) Apply the AND condition

u_{k-1} [m/s]	u_k [m/s]	Both $> 10 \text{ m s}^{-1}$?	Result of this rule
2	2	No	Keep altitude
2	12	No	Keep altitude
12	2	No	Keep altitude
12	12	Yes	Mark altitude missing

Figure 2.4. How Eq. (2.7) assigns a window-based decision to a fix. In this schematic, fixes are 2 s apart and both half-windows are 5 s. The incoming window spans [2, 12] seconds and the outgoing one [4, 14] seconds; the shaded interval is shared. Each window contains five step midpoints; numbers on the bars are their speed magnitudes in m/s. Both medians are 12 m s^{-1} , even though the two steps touching fix k have magnitude 2 m s^{-1} . The altitude is therefore marked missing at a 10 m s^{-1} bound: this is a test of the neighbourhood, not proof that fix k is corrupt. The table gives four example decisions. Other tests can still invalidate a retained altitude; this rule retains the timestamp and horizontal position in every case.

The windows share observations, so agreement is not independent confirmation, nor does it identify which altitude caused the excess.

Figure 2.5 contrasts a change of climb rate, an isolated sensor spike on a low-speed background, and a sustained excessive interval. With sufficient window support, the isolated spike leaves the median low and needs a separate out-and-back test: both bounding steps exceed $v_z^{\max}$ and have opposite signs. Genuine sustained motion is retained only while it remains below the chosen threshold. A sustained observed excess above it is censored; whether it arose from noise or exceptionally strong physical motion cannot be decided from the median alone. Correlated noise, clustered errors and a long missing interval therefore remain relevant to phase segmentation.

An unreturned vertical step. A level jump can escape both tests: on a sufficiently low-speed background, one large step need not raise the window median, and no opposite-sign return triggers the spike rule. The counter `n_alt_level_shift` marks an isolated excessive step when none of the finite altitude readings among the next five fix positions returns within 30% of its jump amplitude. This short horizon does not establish a permanent offset, and the count precedes later trimming and segment selection; some candidates also satisfy another altitude censoring rule.

In the retained-flight metadata, 47,341 paraglider and 1,349 hang-glider records have at least one such candidate before those later steps. These counts cannot be interpreted as the number of flights with a surviving sensor defect. A targeted check of two high-count and two seeded positive-count records per discipline reproduced the counters and found candidate times within usable phase-feature windows in all eight examples. The quality masks therefore do not guarantee that this ambiguity has disappeared. Candidates are not automatically repaired: a

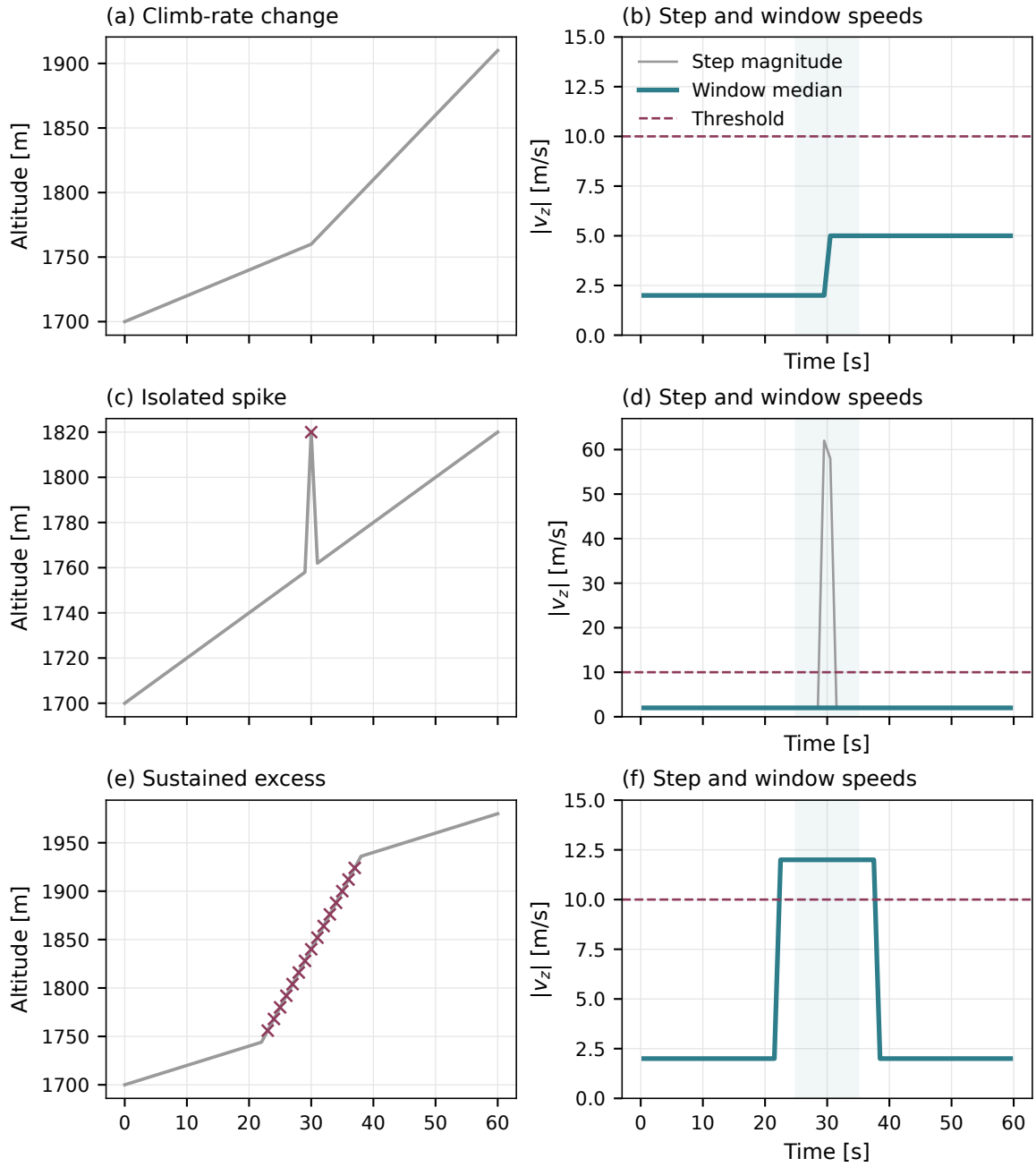


Figure 2.5. The median test at a one-second cadence, using the configured window and speed threshold. Left: synthetic observed altitude. Right: absolute per-step speed (grey), window median (solid), and the bound (dashed). A change between plausible climb rates remains below the bound; an isolated spike barely affects the median and is handled by the separate spike rule; a sustained excessive interval raises the median. Crosses mark altitudes invalidated by the actual implementation.

persistent offset would affect absolute altitude and produce a transient after differentiation and smoothing, whereas an isolated large increment can also reflect real descent or ordinary measurement noise. A phase-level sensitivity comparison is still needed before claiming that these candidates have negligible influence.

The vertical-speed threshold is placed at 10 m s^{-1} , near the change from a steep decrease to a more slowly decaying tail in Fig. 2.6(b). Around 9 m s^{-1} to 10 m s^{-1} , the paraglider density (blue) also begins to exceed the hang-glider density (terracotta). These visual features motivate an empirical cleaning threshold: the transition is gradual, and neither the change of slope nor the crossing establishes a physical maximum or proves that every larger value is an error. Values in this tail are treated as candidate outliers by the two-window and spike rules; only the altitude is invalidated, never the horizontal fix. The full diagnostic sample has 0.069 % of window medians above this threshold, compared with 0.014 % above the previous 30 m s^{-1} threshold. These are exceedance fractions, not removal rates. Inspection of real examples and sensitivity to the cut remain necessary, especially for extreme descents. The altitude and horizontal-speed bounds are also checked against the observed histograms. The Hampel and frozen-lock parameters are likewise working values (Table 2.8); removal fractions quantify their impact, not their classification accuracy.

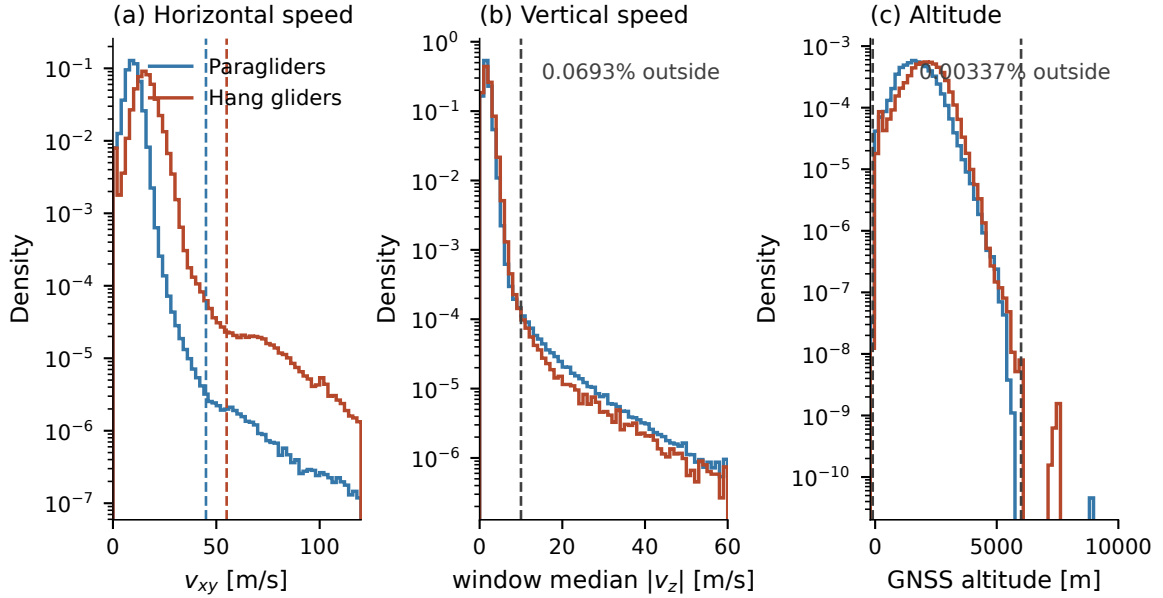


Figure 2.6. Fix-level cleaning: auditing the absolute bounds. Per-fix distributions of the quantities the absolute bounds test—bounds that serve both the out-of-range rule on the vertical channel and, through the v_{xy} bound, the impossibility gate that deletes an off-the-trend fix on the horizontal one (Sec. 2.7.2) – (a) horizontal speed v_{xy} , (b) the windowed GNSS $\text{med } |v_z|$ of Eq. (2.6), the quantity the bound of Table 2.5 actually acts on, (c) GNSS altitude – pooled over a large paraglider subsample and the complete hang-glider population, with the adopted bounds dashed (Table 2.5); the y -axis is logarithmic so the sparse tail the bounds act on is visible. All x -axes are linear, so the distance from ordinary speeds to the threshold is directly readable. Panel (b) emphasizes the interval through 60 m s^{-1} ; its exceedance annotation includes the full sample. The x -axis of panel (c) is extended to 10,000 m regardless of the sample or the bound. Panel (a) spans 0 m s^{-1} to 120 m s^{-1} . Histogram densities use the full finite-sample denominator; the displayed tail is not renormalized. Binning choices and sample sizes: paragraph *Diagnostic sample* of Sec. 2.7.2 (p. 14). This figure audits the *absolute* bounds only; the Hampel and frozen-lock detectors have no per-fix distribution to read a threshold off, and require the example inspection and sensitivity assessment described below. The annotated fractions count bound exceedances, not fixes actually removed: deletion also depends on the rejoin rule and vertical invalidation on neighbouring windows.

Removal policy. A speed belongs to the step *between* two fixes, so a bound exceedance alone does not identify the erroneous endpoint. The horizontal rule removes an unreachable block when its removal permits a rejoin within the bound. This convention uses the local geometry to assign the removal; it does not prove which recorded positions are wrong. What happens next depends on the channel that fails the rule, with reconstruction deferred to the single resampling step (Sec. 2.7.6). An altitude failure marks only that channel missing, retaining the horizontal position subject to its own tests. A rejected horizontal position or timestamp removes the whole fix, so the gap rule can decide whether to bridge the resulting hole or split the trajectory (Sec. 2.7.6). The same channel-specific convention is used for a V flag: GNSS altitude is invalidated, while horizontal coordinates remain subject to the geometric tests. This is an adopted treatment of the declaration, not evidence that those coordinates are accurate.

An impossible step is a boundary, never flown path. After fix-level cleaning, any surviving pair whose inferred horizontal speed exceeds the bound is separated by a segment boundary. Where a block is kept because nothing could rejoin it, the step by which the track returns from it is itself impossible and, left alone, enters a segment as flown motion: a jump of tens of kilometres inside one second, in the archive. Such a step is a transition whose course is unknown, so it is made a segment boundary on the same footing as a long gap (Sec. 2.7.6), counted per flight as `n_boundaried`.

The forward scan does not re-evaluate its initial anchor, so an erroneous first fix can survive this local procedure and displace the origin of the later Cartesian frame. A separate flight-level test therefore rejects a track containing a fix beyond the distance reachable from its first retained position (Sec. 2.7.4). The reach test acts on spherical distances before projection and smoothing, with a 50 m tolerance. The written Cartesian trajectories undergo separate diagnostic checks (Sec. 2.9); those operations do not preserve the reach inequality as an exact invariant.

Frozen-position runs and the evidence for GNSS lock loss. A logger may repeat its last position after losing GNSS lock while timestamps continue. An unremoved run then resembles a period of spatial confinement. The detector uses three joint conditions,

$$D_I < \delta_{xy}, \quad t_{\max I} - t_{\min I} \geq \tau_{\text{freeze}}, \quad W_I = 1, \quad (2.8)$$

where D_I is the great-circle diagonal of the latitude-longitude bounding box of candidate run I , evaluated after longitude unwrapping. This is the implemented extent proxy, not an exact all-pairs diameter. The working values are $\delta_{xy} = 15$ m and $\tau_{\text{freeze}} = 60$ s.

Let E_I denote exact equality of both coordinate fields throughout the run, d_I the fraction of its fixes with a V flag or missing GNSS altitude, and Δb_I the median pressure altitude of the final quarter minus that of the first quarter. With B_I defined in Eq. (2.1), the witness is

$$W_I = E_I \vee \begin{cases} |\Delta b_I| < \delta_z, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1, B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0, \end{cases} \quad \delta_z = 5 \text{ m}. \quad (2.9)$$

The middle branch is essential: incomplete local pressure coverage is not evidence of a flat trace. An exact coordinate repeat remains a separate signature even if the pressure channel is missing. Once the run is removed, the trajectory is always split there; no straight-line interpolation represents its unobserved path.

These criteria support a practical detector, not a proof of lock loss. Equal quantized coordinates can also occur for a stationary receiver; with sparse raw sampling, a long span between a few fixes is not proof of continuous repetition throughout the interval. A small net

pressure change can hide an up-and-down excursion, and a working barometer can continue climbing during a real horizontal GNSS freeze. Moreover, pressure and GNSS are physically distinct sensors but their recorded failures need not be statistically independent if the logger freezes both channels. The word *witness* therefore means corroborating evidence.

For scale, a coordinated turn in still air has

$$R = \frac{v_{\text{air}}^2}{g \tan \phi}, \quad T_{\text{turn}} = \frac{2\pi R}{v_{\text{air}}}. \quad (2.10)$$

At $v_{\text{air}} = 10 \text{ m s}^{-1}$ and bank angle $\phi = 35^\circ$, $2R \simeq 29 \text{ m}$ exceeds δ_{xy} , while a sustained 1 m s^{-1} climb changes pressure altitude far more than δ_z . These are reference scales, not universal minimum diameters or guarantees for every wind, bank angle and recorder cadence.

The remaining errors have clear directions. A small genuine manoeuvre can be cut if all signatures happen to agree; a jittering freeze can be kept without an adequate witness; a short freeze is deliberately kept below τ_{freeze} . A single missed short run has limited duration, but multiple missed runs can accumulate. Their rates cannot be inferred from the number the detector removes: independent inspection or labelled injections are required.

Completeness, the integrity gate, and validation. The segmentation (Ch. 4) needs a defined position and altitude at every time within a *segment*, a maximal stretch between two splits, not everywhere; cleaning does not guarantee this fix by fix, and the guarantee is established one step later, at resampling (Sec. 2.7.6). Cleaning also counts how much it had to do over the airborne window (defects in a ground phase that trimming removes anyway must not condemn the flight): a flight whose cleaning removed or reconstructed more than 10 % of its fixes is dropped whole.

Validation must distinguish three questions. Removal fractions measure how much the pipeline changes the data. Threshold sweeps measure sensitivity to working values. Neither measures false positives or false negatives. Those require inspected real intervals or known injected defects, with the limitation that injection tests establish performance only for the injected shapes and backgrounds. The synthetic regressions and the real examples below check explicit failure modes; a representative, independently labelled audit remains needed for an archive-wide error rate.

2.7.3 Trimming of ground phases

Outer ground phases (take-off and landing). The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis, using the horizontal speed v_{xy} alone. With $v_0 = 5 \text{ m s}^{-1}$ and $T_0 = 30 \text{ s}$ (Table 2.8), the retained airborne segment is $[t_{\text{on}}, t_{\text{off}}]$ with

$$t_{\text{on}} = \min\{t : v_{xy} > v_0 \text{ on } [t, t + T_0]\}, \quad t_{\text{off}} = \max\{t : v_{xy} > v_0 \text{ on } [t - T_0, t]\}, \quad (2.11)$$

the same sustained-speed rule read forward and time-reversed. Here v_{xy} is the speed inferred on each observed step, not a continuously measured velocity. Interior slow periods do not define these outer boundaries, although slow flight at either end can be clipped. A flight with no sustained stretch at all is discarded.

The persistence time T_0 reduces sensitivity to brief speed excursions. It does not bound take-off or landing error by T_0 : a sustained fast ground movement can be retained, while a long airborne interval below v_0 at either end can be removed. The trimmed fraction is recorded per flight, and such cases require direct inspection.

Defect	Detected by	Action
Off the trend — <i>the horizontal position itself is corrupt</i>		
Position spike, out-and-back jump	two-stage: the Hampel identifier [14] only <i>flags</i> a candidate and attributes the anomaly count; what actually <i>deletes</i> is the impossibility gate alone — unreachable from the last accepted fix at the absolute v_{xy} bound ($45/55 \text{ m s}^{-1}$, para/hang), <i>and</i> its block's removal lets the track rejoin	delete the fix; the short gap is bridged at resampling
Position discontinuity (re-acquisition offset)	impossible step that no bounded removal rejoins	split at the step, like a long gap
Impossible step still standing	any surviving step past the v_{xy} bound, whatever the scan resolved (postcondition)	make it a segment boundary
Corrupt block closing the record	a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most w	delete the block, instead of splitting
On the trend — <i>record structurally wrong, horizontal position agrees with its neighbours</i>		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	$D_I < \delta_{xy}$ for $\geq \tau_{\text{freeze}}$, witnessed (Eqs. 2.8–2.9): paired baro. with small net change or identical coordinates; V/zero-alt where no barometer witnesses	mark as gap, <i>split</i> there
Out of range — <i>horizontal fix intact; altitude unavailable or suspect</i>		
Missing / out-of-band altitude, sustained $ v_z $ excess, or $ v_z $ spike	separate rules for each condition: <i>missing</i> — altitude channel absent; <i>out-of-band</i> — outside $[-100, 6000] \text{ m}$, a fixed working band needing no neighbour; <i>sustained excess</i> — both adjacent window statistics exceed 10 m s^{-1} , with the sparse-window fallback (Eq. 2.7); <i>spike</i> — an out-and-back pair past that same bound	drop the altitude only; the fix stays
Policy: altitude-only invalidation retains the timestamp and horizontal coordinates. Position and time rules may remove complete fixes. These are operational decisions, not known sensor truth. Altitude reconstruction occurs at resampling (Sec. 2.7.6).		

Table 2.5. Fix-level decisions by defect class. Horizontal flags, horizontal deletions, and altitude invalidations are distinct operations. The bounds are working values, audited in Fig. 2.6 and collected in Table 2.8; operational details are in Sec. 2.7.2. No detector establishes that every retained fix is correct.

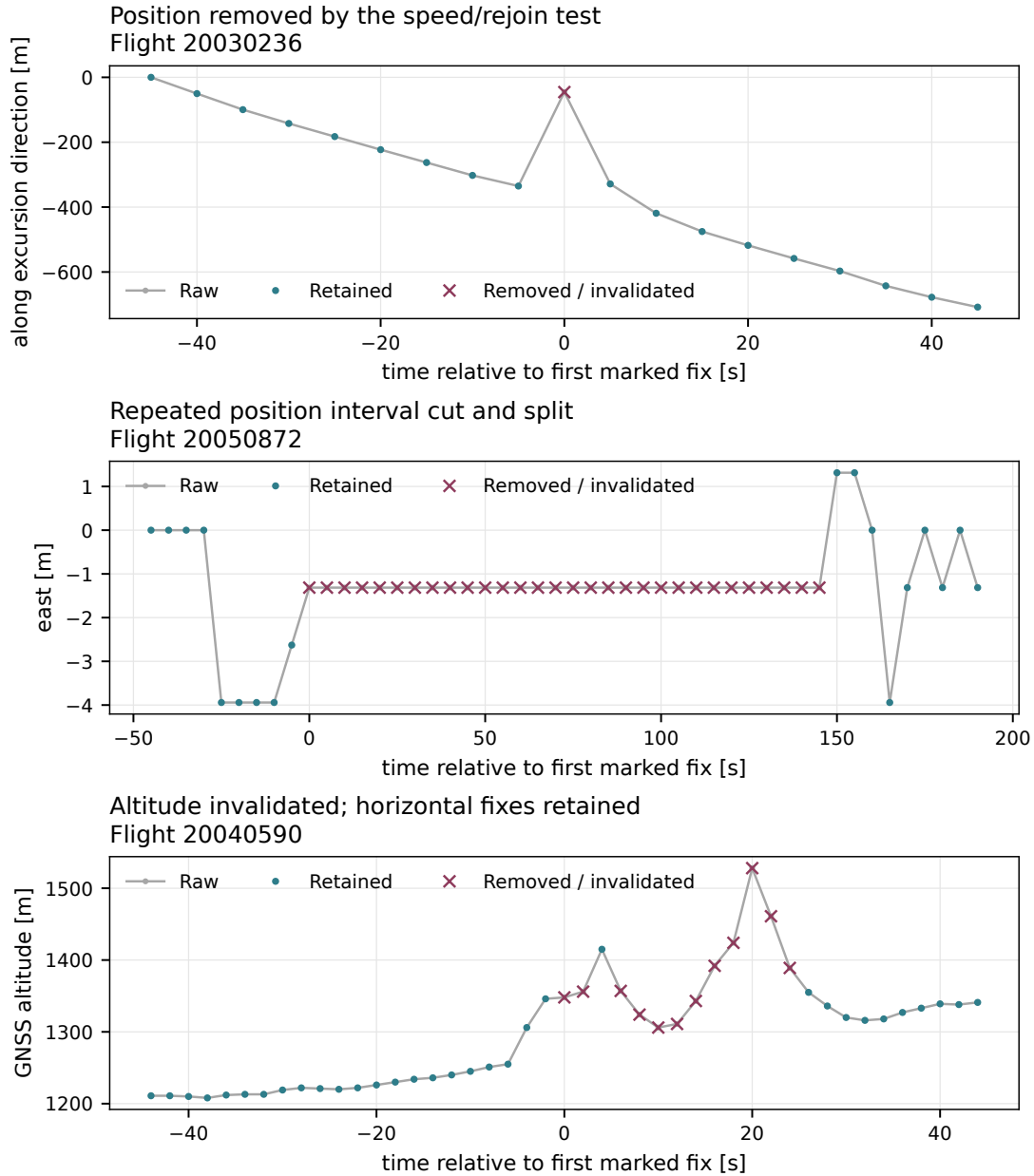


Figure 2.7. Short examples extracted from the interior of raw flight records and reprocessed with the current cleaning implementation. Top: the horizontal coordinate projected along the detected excursion direction, making the removed spike visible. Middle: a collapsed-position interval; crossed marks are removed and the surviving sides split. Bottom: GNSS altitude invalidation, with all horizontal fixes retained. Flight identifiers and plotted raw values are saved with the figure. The examples document detector decisions; no independent sensor truth is available for these intervals.

Interior ground stints (mid-flight landings). A mid-flight landing and relaunch recorded in one log would otherwise survive as an interior near-zero-speed stint and read downstream as a false long wait in the tail of $\psi(\tau)$. An interior-ground detector, built like the frozen-lock rule, cuts a stint only when the horizontal and barometric conditions agree:

- *Sustained low speed.* Successive observed steps remain below v_0 for at least $T_{\text{ground}} = 10$ min. This working persistence threshold reduces cuts of short slow intervals; it does not prove that a longer interval occurred on the ground.
- *A flat barometer.* With complete paired pressure coverage throughout the candidate stint (Eq. 2.1), its detrended central span stays within $\delta_z^{\text{gr}} = 5$ m over the whole stint. Specifically, an ordinary least-squares trend is removed and the 95th-minus-5th percentile span of the residuals is compared with this bound. Pressure supplies a second measured channel; independence of recorder errors is not assumed. The central span reduces sensitivity to isolated extremes, although short excursions and finite-sample variability remain possible.

The fitted pressure-altitude slope must also satisfy $|\hat{\beta}| \leq \delta_z/\tau_{\text{freeze}}$: removing a linear trend alone would erase a steady real climb. Where the barometric witness is absent or locally incomplete, the guard *abstains* outright rather than fall back to sustained stillness alone – which is exactly the soaring-into-wind failure mode the second signature exists to prevent – at the cost of never flagging a mid-flight landing on those flights .

A cut stint is excised and the flight split there, exactly as at a long gap (Sec. 2.7.6). The flight-level cuts of Sec. 2.7.4 are not re-applied to the individual segments, but to the *parent* flight, its airborne duration and path length summed over its segments, since a segment is short because a gap happened to fall there, which says nothing about whether the flight is genuinely cross-country. Trimming therefore runs before the flight-level cuts in the pipeline order (step (iii) before step (iv)), so a duration or path length is only meaningful once the ground phases are gone.

Shorter intervals satisfying the same speed and barometric conditions are *flagged* when their duration reaches 60 s, the configured minimum passed to this stage . They remain in the trajectories. The stored flags support a future sensitivity analysis with and without these observations; that comparison has not yet been carried out.

2.7.4 Flight-level filtering

The following thresholds define the population analysed in Chapters 3 and 4. The retained-fraction curves in Fig. 2.8 show how sensitive sample size is to each choice. A plateau does not establish that excluded records are defective or that selection is unbiased. Duration selection, for example, changes which flights can contribute at long lags and can therefore change the estimated scaling.

The criteria act on the trimmed parent flight. Recorded duration T_{rec} and path length L sum only within contiguous blocks, excluding forced boundaries and gaps longer than g_{max} (Sec. 2.7.6). Elapsed span $T_{\text{span}} = t_{\text{last}} - t_{\text{first}}$ includes those gaps .

- (1) *Duration:* $T_{\text{rec}} \geq 40$ min and $T_{\text{span}} \leq 16$ h. The upper bound screens long recording sessions, including loggers left running after landing. The raw census contains 9 paraglider records spanning 16 h to 166 h, and none among hang gliders. These raw counts are not post-trimming removal counts.
- (2) *Path length:* $L \geq 20$ km restricts short-distance records. In the raw diagnostic population, among flights above the duration floor, 236 of 184,541 paraglider records and 7 of 6,638 hang-glider records fall below this path threshold. Neither duration nor path length alone identifies a flight as soaring.

- (3) *Altitude activity*. Where finite altitude is available, the adopted channel's whole-flight range must reach 600 m. This selects appreciable vertical excursion; soaring at nearly constant altitude can fail the cut. The threshold is therefore a sample restriction, not a classifier of soaring versus non-soaring. The cut, and its diagnostic in Fig. 2.8c,f, act on the adopted GNSS channel over the whole population (Sec. 2.7.1) .
- (4) *Mean speed*: $L/T_{\text{rec}} \leq v_{xy}^{\text{max}}$, with consistent length and time units. If every included step already satisfies the same bound, its duration-weighted mean does too. This is a consistency check on the accumulated quantities, not independent evidence of data quality .
- (5) *Reach*. No fix may lie further from the first than the bound could have carried it in the elapsed time, $\Delta s(\text{fix}_0, \text{fix}_i) \leq v_{xy}^{\text{max}} t_i + \varepsilon_{\text{reach}}$, with $\varepsilon_{\text{reach}} = 50$ m and t_i measured from the first trimmed fix. The test acts before projection and smoothing, for every i , not as the whole-flight extent against the whole duration, since that weaker form would license a fix 100 km out ten seconds into the record on the grounds that the flight went on for three hours. This is what catches a corrupt first fix once its step is no longer summed into L (Sec. 2.7.2): it removes 217 paraglider and 5 hang-glider flights, order 10^{-3} of either ensemble in the stored snapshot. The written Cartesian trajectories have separate checks with operational tolerances (Sec. 2.9).
- (6) *Glider class* (metadata). Flights whose `aile_class` cannot be normalised (Sec. 2.4) are dropped from class-resolved sub-analyses but kept in the pooled one; stratification by discipline is already possible.

A further criterion, sampling regularity, is applied later, where a flight's time base is examined (Sec. 2.7.6).

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates: resampling and smoothing operate on the coordinates *componentwise* (Sec. 2.7), and latitudes and longitudes cannot play that role, being angles on a curved surface whose componentwise differences are not displacements. The mapping proceeds in two steps.

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6,378,137$ m, flattening $f = 1/298.257223563$, $e^2 = f(2 - f)$) [15,16],

$$\begin{aligned}
 N(\varphi) &= \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, & X &= (N + h) \cos \varphi \cos \lambda, \\
 & & Y &= (N + h) \cos \varphi \sin \lambda, \\
 & & Z &= (N(1 - e^2) + h) \sin \varphi,
 \end{aligned} \tag{2.12}$$

$N(\varphi)$ the prime-vertical radius of curvature and h the ellipsoidal height along the normal, derived step by step in Appendix 2.B.

Geodetic versus geocentric latitude. The latitude φ is *geodetic*: the angle between the equatorial plane and the ellipsoid *normal* (Figure 2.9), not the geocentric latitude ψ measured to the Earth's centre O , since the normal does not pass through it. The gap $\varphi - \psi \approx (e^2/2) \sin 2\varphi$ (Appendix 2.B) peaks at $\varphi = 45^\circ$ and reaches $\approx 0.19^\circ$, which if confused would misplace a fix by up to ~ 20 km. No conversion is needed here: WGS84 — hence the GPS receiver and the IGC B records (Sec. 2.2) — already reports geodetic latitude, and Eq. (2.12) with $N(\varphi)$ is precisely the transform that consumes it; these are also the (φ_0, λ_0) that orient the local frame of Step 2 (Figure 2.10).

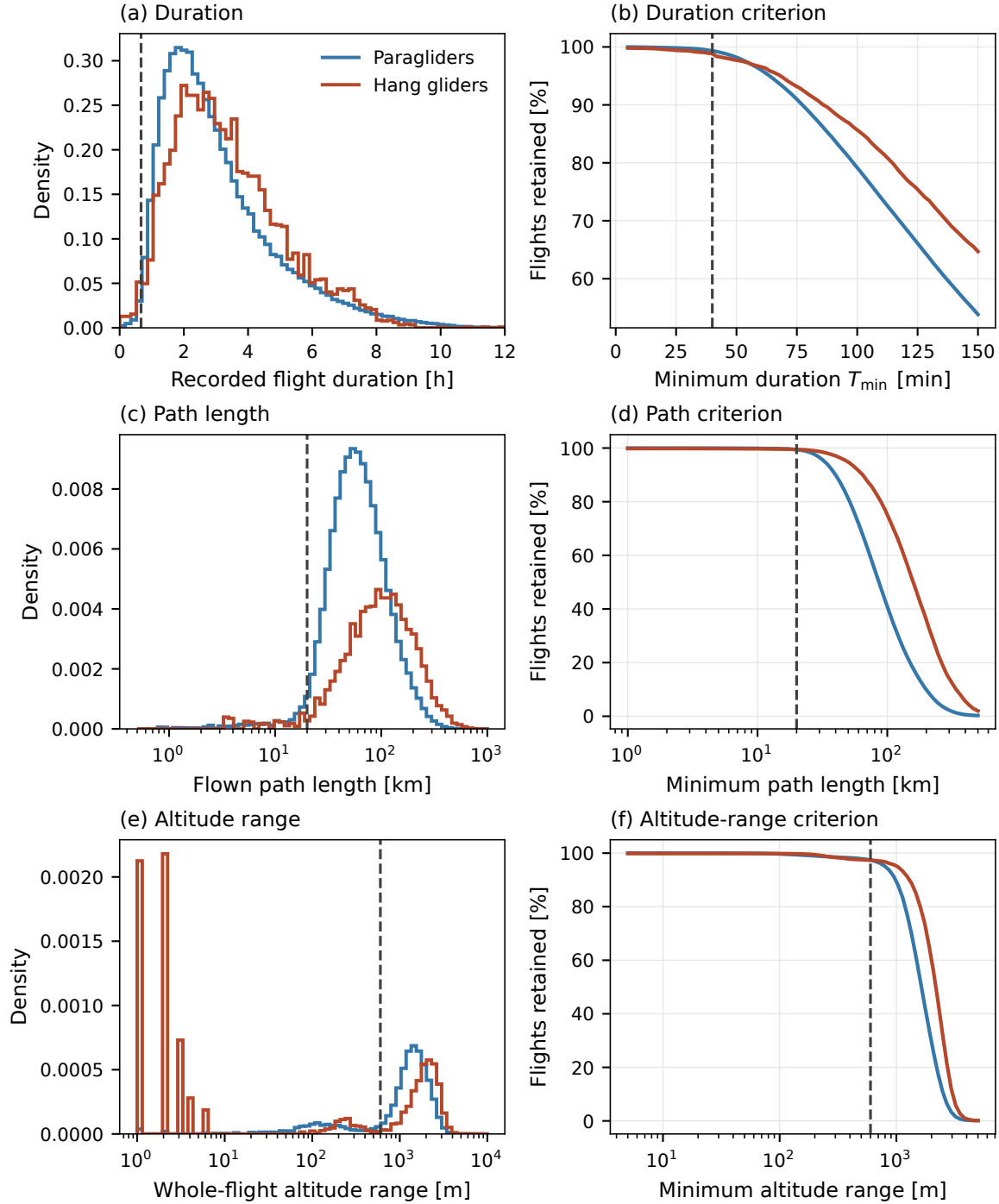


Figure 2.8. Raw flight-level diagnostics by discipline. Left column: (a) recorded duration, (c) flown path length and (e) GNSS altitude range. Right column: retention under the corresponding single criterion, evaluated separately on these raw summaries. Dashed lines mark the working thresholds ($T \geq 40$ min, $L \geq 20$ km and altitude range ≥ 600 m). The altitude panels require the configured minimum GNSS coverage. Histogram densities retain their full finite-sample denominators when tails extend outside the displayed range. These marginal raw diagnostics are not the retention fractions of the sequential cleaned-flight cuts.

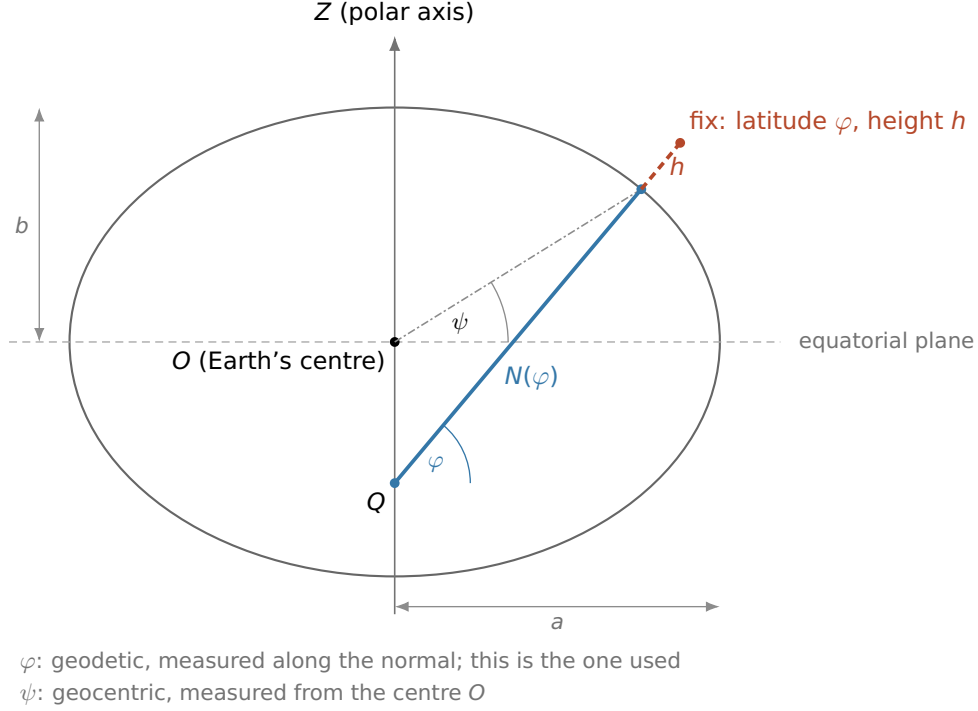


Figure 2.9. Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84 $f \approx 1/298$ would be indistinguishable from a circle): the semi-axes a and b ; the geodetic latitude φ , the angle between the ellipsoid normal and the equatorial plane; the height h along that normal; and $N(\varphi)$, the distance along the normal from the surface to the point Q on the polar axis. The normal does not pass through the centre O , so φ differs from the geocentric latitude ψ (dash-dotted). Longitude λ is out of the page here; it is shown in Figure 2.10.

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the *trimmed* track as the origin — close to but not the take-off point, since the ground phase has been removed (Sec. 2.7.3) — the ECEF displacement $\Delta \mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

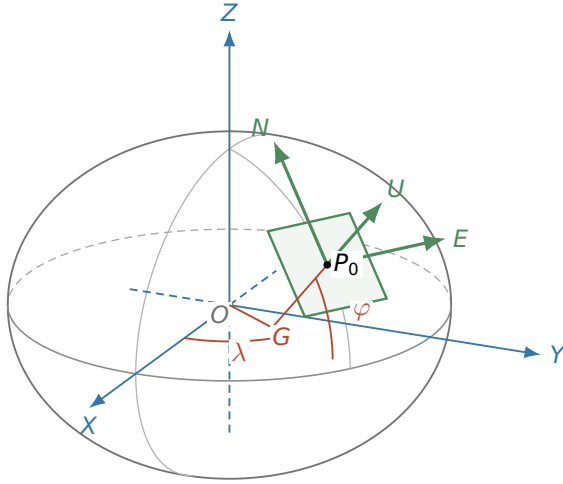
$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix} \Delta \mathbf{X}, \quad (2.13)$$

derived in Appendix 2.B: an orthogonal matrix of unit determinant, so lengths and angles are preserved and only the axes change. This ENU frame is the local Cartesian frame the whole analysis is carried out in (E east, N north, U along the local vertical); the ECEF coordinates of Step 1 are only an intermediate representation, no observable is ever expressed in them. The rotation preserves the three-dimensional ECEF chord. Keeping only (E, N) projects that chord onto the tangent plane and introduces a different approximation. For a point at surface distance d from the origin on a sphere,

$$r_{EN} = R_{\oplus} \sin(d/R_{\oplus}), \quad d - r_{EN} = \frac{d^3}{6R_{\oplus}^2} + O(d^5/R_{\oplus}^4). \quad (2.14)$$

The horizontal shortening is about 0.03 m at 20 km and 0.51 m at 50 km; it reaches about 33 m at 200 km. The often-used $d^3/(24R_{\oplus}^2)$ expression instead compares an arc with its full three-dimensional chord. Neither estimate is a uniform error bound for every pair of fixes in a fixed frame.

(a) The global frame and the two angles



(b) The tangent frame, close up

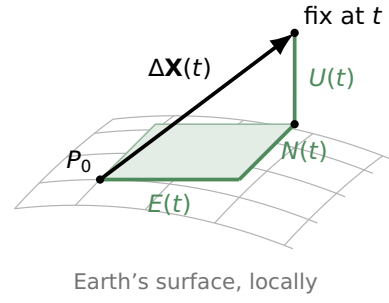


Figure 2.10. The ENU rotation: it turns a vector's global ECEF components into local East, North, Up ones. (a) The global ECEF frame (X, Y, Z) in blue, centred at the Earth's centre O ; the two red angles that locate the origin fix P_0 (the first fix of the trimmed track, Sec. 2.7.5): longitude λ , measured in the equatorial plane from the prime meridian, and geodetic latitude φ , the tilt of the ellipsoid normal, measured at the point G where the normal meets the equatorial plane; and the local ENU frame in green, tangent at P_0 . Flattening exaggerated as in Figure 2.9: U follows the normal $G-P_0$, not the radius from O . The green arrows are the rows of the rotation matrix, as unit vectors. (b) Close-up of the tangent plane: a later fix's ECEF displacement $\Delta\mathbf{X}(t)$ is the space diagonal of the box whose edges are its ENU components $E(t)$, $N(t)$, $U(t)$.

The transform uses the logged GNSS altitude as its height input. Since the recorder's vertical datum is not yet harmonized, this is an approximation to ellipsoidal height. A height error δh contributes a horizontal error of order $|\delta h| \sin(d/R_\oplus) \simeq |\delta h| d/R_\oplus$: for example, 100 m at 50 km gives about 0.8 m. This is an illustrative scale, not an archive-wide upper bound. The vertical coordinate remains $z(t) = h_{\text{GNSS}}(t)$ rather than the rotation's U component, which also contains the curvature of the reference surface. The origin altitude is not subtracted; vertical velocities depend on increments, whereas absolute height retains the datum limitation stated in Sec. 2.7.1.

2.7.6 Uniform sampling within a flight

The smoothing and derivative estimates in the next section require equally spaced observations. Each flight keeps its native interval, $\Delta t = \text{median}_i(t_{i+1} - t_i)$: a flight recorded every second stays at one second, and a flight recorded every five seconds stays at five seconds. Figure 2.11 shows these intervals in the archive.

Fill a short gap; split at a long one. For a gap $g = t_{i+1} - t_i$ between surviving fixes, interpolation is allowed when $g \leq g_{\text{max}}$, with

$$g_{\text{max}} = \min(10 \Delta t, \max(20 \text{ s}, 2\Delta t)). \quad (2.15)$$

The relative limit scales with the recording interval; the 20 s cap limits the time reconstructed at common cadences. The $2\Delta t$ floor allows one missed fix even for slower loggers. Thus $g_{\text{max}} = 10 \text{ s}$ at $\Delta t = 1 \text{ s}$, and $g_{\text{max}} = 20 \text{ s}$ at $\Delta t = 5 \text{ s}$. A longer gap starts a new segment. Boundaries imposed by cleaning also force a split. The cap is a working choice: it limits interpolation without guaranteeing that all manoeuvres inside a short gap are resolved.

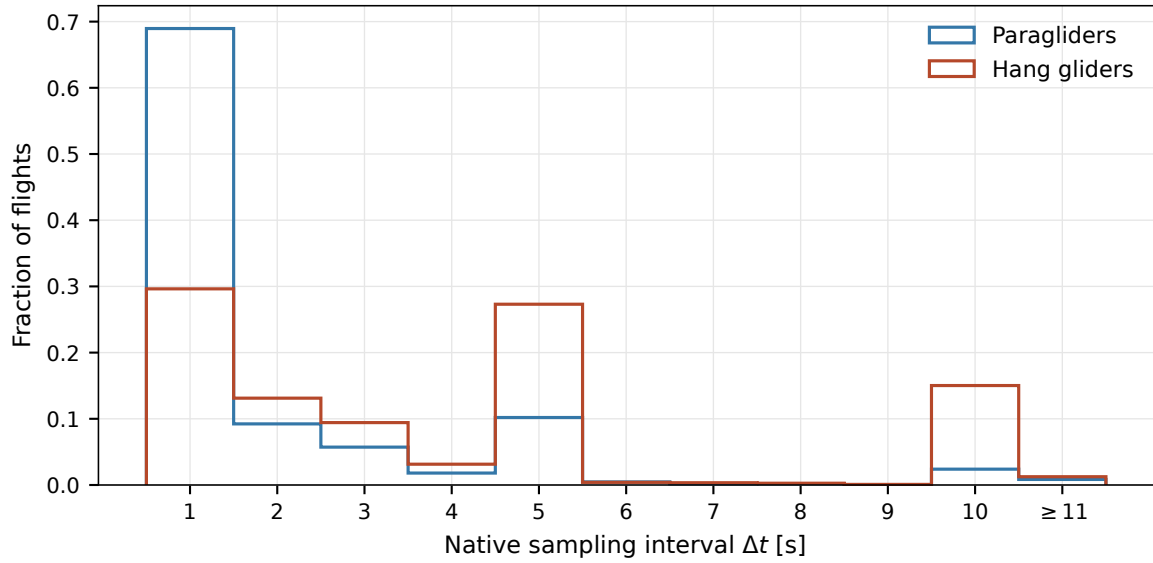


Figure 2.11. Native sampling interval per flight, measured as the median inter-fix time in the raw archive. Bars give the fraction of flights in one-second bins; the last bin also includes intervals beyond 11 s. Paragliders concentrate at 1 s, while hang gliders are spread across several cadences. Resampling preserves each flight’s native interval.

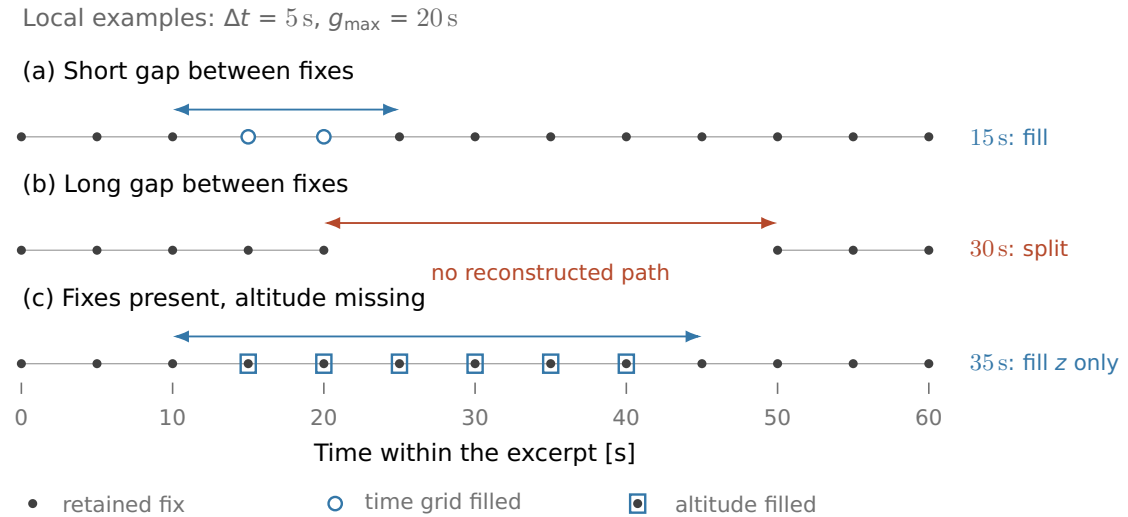


Figure 2.12. Three local examples of resampling. Circles mark new grid times in a short gap; a long gap leaves separate segments. Squares mark reconstructed altitude at existing fixes. The third case has complete temporal coverage: the current rule fills the altitude hole even when it exceeds $g_{\max}$.

Reconstructing the coordinates. Within segment s , the grid is $t_k^{(s)} = t_{\text{start}}^{(s)} + k\Delta t$, ending at or before its last fix. An already uniform, complete series is unchanged. Otherwise, E and N are evaluated with a shape-preserving cubic interpolant (PCHIP) [17], and z with linear interpolation. Linear filling avoids overshoot in altitude but can smooth away a climb–glide transition. Outside the first and last finite altitude readings, the implementation carries the nearest endpoint value.

A missing altitude at an existing fix is treated separately (Fig. 2.12c). It opens no time gap, so the horizontal trajectory stays connected and the altitude is filled without a duration limit. The flag `interpolated` records grid times with no fix within $\Delta t/2$; `z_reconstructed` additionally marks those whose nearest fix had no altitude. Smoothing propagates altitude reconstruction through the derivative windows into `z_derivative_reconstructed`. These flags allow the phase analysis to exclude unsupported vertical dynamics (Chapter 4); numerical completeness alone does not establish accuracy.

Which segments are retained. A segment must span at least 90 s, have at most 10 % of its grid times flagged as `interpolated`, and yield finite values in all three coordinates. The temporal-coverage gate does not limit altitude-only reconstruction. A rejected segment leaves the other segments available; the parent’s flight-level cuts are not reapplied.

Segments retain the flight’s spatial origin and elapsed time. Time-lagged displacement pairs and analysis windows are formed within segments. Splitting therefore changes the available coverage and can truncate flight phases; it must be accounted for when comparing observables. Figure 2.13 describes gaps in the raw archive; observables that depend on the recording interval, such as fix-to-fix turning angles, require a common temporal scale or a comparison within cadence groups.

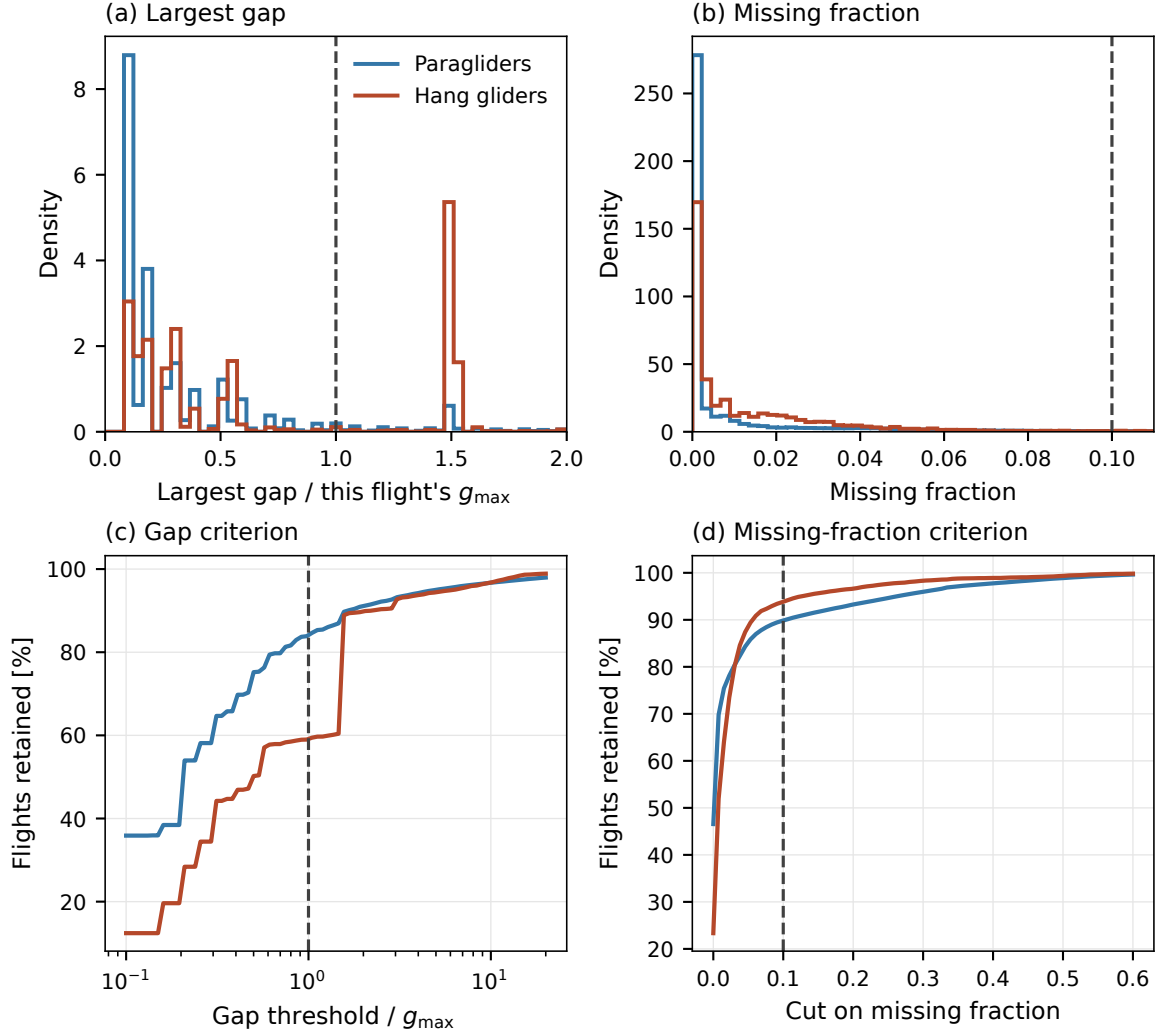


Figure 2.13. Sampling gaps in the raw archive, before cleaning and splitting. (a)/(b): distributions of the largest gap relative to $g_{\max}$ and of the missing fraction on a native-rate grid. (c)/(d): fractions of flights satisfying each criterion separately as its threshold varies. These are diagnostics of whole raw flights; the operational missing-fraction gate acts on segments after splitting, so these curves are not final flight-retention rates.

2.7.7 Smoothing and differentiation

Position errors are amplified by differentiation. We estimate position, velocity and acceleration from the same local polynomial, fitted independently to E , N and z within each retained segment. For $w = 2m + 1$ equally spaced samples around fix i , the Savitzky–Golay fit [18] is

$$P_i(u) = \sum_{k=0}^p a_k(i) u^k, \quad u = -m, \dots, m, \quad (2.16)$$

$$\mathbf{a}(i) = \arg \min_{\mathbf{a}} \sum_{u=-m}^m [x_{i+u} - P_i(u)]^2. \quad (2.17)$$

Here u is a sample offset. The physical quantities at the window centre are

$$\hat{x}_i = a_0(i), \quad \hat{v}_i = \frac{a_1(i)}{\Delta t}, \quad \hat{a}_i = \frac{2a_2(i)}{\Delta t^2}. \quad (2.18)$$

The weights. With $V_{u,k} = u^k$ and $\mathbf{x}_i = (x_{i-m}, \dots, x_{i+m})^\top$, the mathematical least-squares solution is

$$\mathbf{a}(i) = (V^\top V)^{-1} V^\top \mathbf{x}_i. \quad (2.19)$$

The inverse is a formula for the weights, not a recommendation to form a matrix inverse numerically. The implementation uses SciPy’s Savitzky–Golay coefficients. For fixed (w, p) , row k defines weights $c_k(u)$ with

$$a_k(i) = \sum_{u=-m}^m c_k(u) x_{i+u}. \quad (2.20)$$

This forward-offset convention is a correlation; convolution uses the reversed weights. For an odd derivative the reversal changes their signs. Table 2.6 gives the forward-offset weights, before conversion to seconds.

Table 2.6. Interior Savitzky–Golay weights for $w = 5$, $p = 3$. Velocity and acceleration weights must be divided by Δt and Δt^2 , respectively.

	sample offset u				
	−2	−1	0	1	2
c_0	−3/35	12/35	17/35	12/35	−3/35
c_1	1/12	−8/12	0	8/12	−1/12
$2c_2$	2/7	−1/7	−2/7	−1/7	2/7

The order $p = 3$ allows nonzero acceleration and locally varying acceleration. A quadratic already estimates velocity and acceleration; a cubic is a modelling choice, not a mathematical necessity. On a symmetric window, adding a cubic term changes the first-derivative weights but leaves the value and second-derivative weights of the quadratic fit unchanged.

Window length and cadence. A raw spectral diagnostic suggests a change in slope around 0.2 Hz (Fig. 2.15). It motivates the working timescale

$$\tau_c = 1/f_c = 5 \text{ s}, \quad w = \max\left(\left\lceil \frac{\tau_c}{\Delta t} \right\rceil_{\text{odd}}, 5\right). \quad (2.21)$$

Horizontal and vertical timescales are separate configuration entries, currently equal. The code converts this timescale to a *number of samples*: the distance in time between the first and last sample of a window is $(w - 1)\Delta t$. Thus $w = 5$ spans 4 s at a 1 s cadence, and 20 s at a 5 s cadence. Slower recording therefore changes the physical smoothing scale and can remove motion that faster loggers resolve. Comparisons of short-time observables must control for cadence.

The spectral knee is neither a fitted receiver-noise model nor the cutoff frequency of this filter. For the value weights, the interior frequency response is

$$G(f) = \sum_{u=-m}^m c_0(u) e^{2\pi i f u \Delta t}, \quad S_{\hat{x}}(f) = |G(f)|^2 S_x(f). \quad (2.22)$$

Its shape follows from the kernel (Fig. 2.16). Rounding a field to steps of size q has variance $q^2/12$ only under a uniform-error approximation; a flat quantization spectrum additionally requires temporal independence. Similar high-frequency spectra do not establish those assumptions or identify the source of the observed knee.

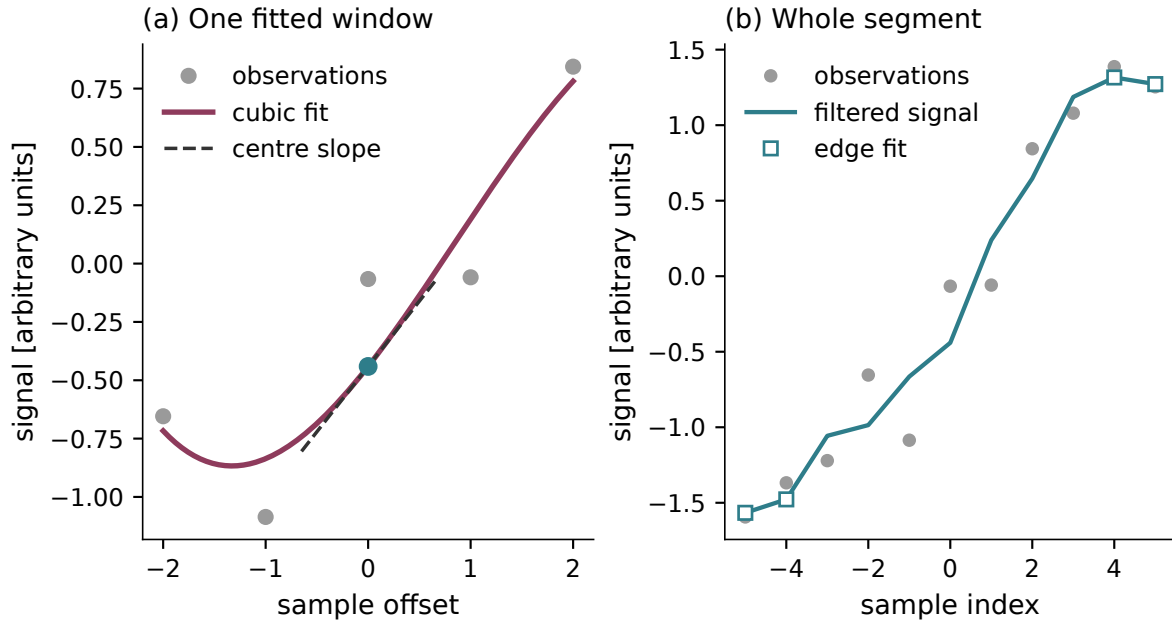


Figure 2.14. A cubic fit to a synthetic signal. Left: one five-sample window, its fit and the fitted value and slope at the centre. Right: the output of the implemented filter over the same series; squares mark the two edge samples at each end. Synthetic values are $\sin(0.28j) + \varepsilon_j$, with independent $\varepsilon_j \sim \mathcal{N}(0, 0.35^2)$.

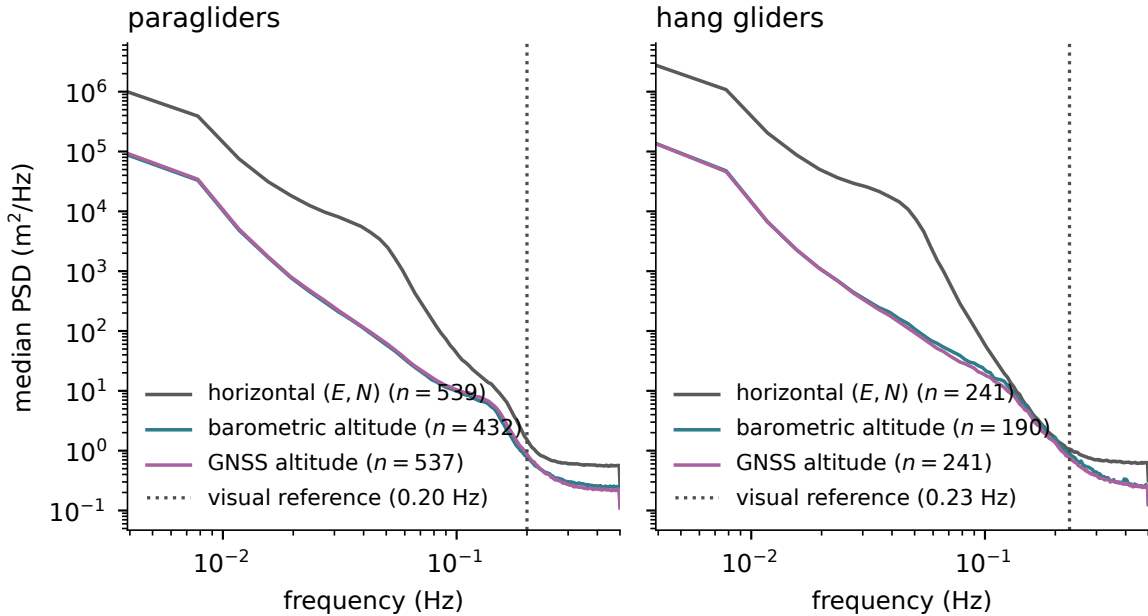


Figure 2.15. Median raw Welch PSDs, by discipline and coordinate channel, on a seeded sample of up to 900 flights per discipline. Spectra use recording intervals near 1 s; legend counts give the usable flights. Horizontal coordinates use a local equirectangular approximation before cleaning, and the curve pools both component spectra from each admitted flight. Channel eligibility is assessed separately, so the three medians need not represent identical flights or intervals. The vertical dotted lines are visual reference frequencies, 0.20 Hz and 0.23 Hz, not estimates with an uncertainty interval. Recording precision, sensor error, interpolation and real motion can all affect these spectra.

What smoothing changes in an MSD. For an interior sample with independent, zero-mean measurement error of variance σ^2 per coordinate, the filtered error is $\eta_i = \sum_u c_0(u) \varepsilon_{i+u}$. Extending the weights by zero outside the window gives its covariance and its contribution to a one-coordinate MSD:

$$C_\eta(\ell) = \sigma^2 \sum_u c_0(u) c_0(u - \ell), \quad (2.23)$$

$$B(\ell) = \mathbb{E}[(\eta_{i+\ell} - \eta_i)^2] = 2\sigma^2 \left[\sum_u c_0(u)^2 - \sum_u c_0(u) c_0(u - \ell) \right]. \quad (2.24)$$

For $w = 5$, $p = 3$, $\sum c_0^2 = 17/35$. At lags $\ell \geq 5$ the noise contribution is therefore $34\sigma^2/35$ per coordinate, compared with $2\sigma^2$ before smoothing. A horizontal MSD adds the contributions of both coordinates. At shorter lags, overlapping windows correlate the errors, so the bias is lag dependent. These expressions require independent errors; they are not an empirical GNSS calibration. The filter also changes the true motion unless it is polynomial of degree at most p on every relevant window. A Hurst exponent alone supplies no bound on that change.

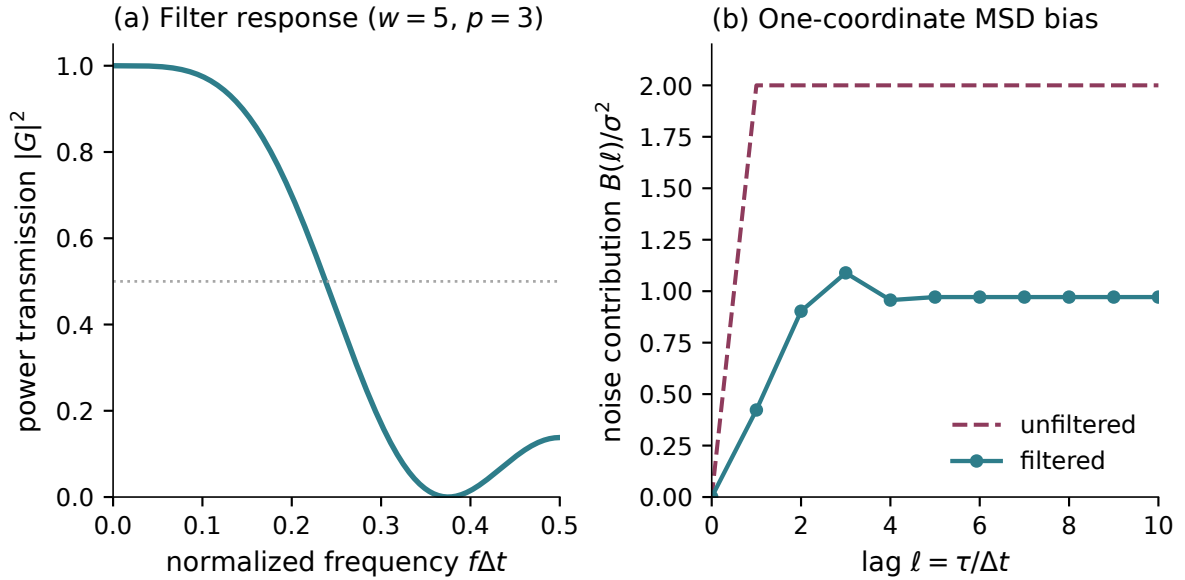


Figure 2.16. Exact effects of the five-sample cubic filter away from segment edges. Left: power transmission $|G(f)|^2$ against normalized frequency $f\Delta t$. Right: additive one-coordinate MSD from independent unit-variance measurement noise. The remaining bias depends on lag and does not equal the unfiltered $2\sigma^2$.

Edges and validation. At either end of a segment, the filter fits the nearest full window and evaluates that polynomial inside its fitted interval. These are off-centre evaluations, not extrapolation beyond the observations. The first and last m samples carry an **edge** flag because their weights and error variance differ from the interior. The filter can overshoot despite earlier cleaning, so speed bounds are not guaranteed by smoothing.

The kernel identities and derivative units are checked in automated tests. Archive-level validation remains separate: after rebuilding the cleaned archive, compare the MSD and phase features at neighbouring odd window lengths, within each cadence class, and with edge and reconstructed samples excluded. A flat residual spectrum is not a sufficient criterion: even white input noise has residual PSD $|1 - G(f)|^2 S_\varepsilon(f)$. Until these comparisons are reported, the adopted window is a working choice rather than a certified optimum.

2.7.8 Notation

Throughout:

- **Horizontal position.** $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$, in this local frame. Before smoothing, the first trimmed fix defines $\mathbf{r}(0) = 0$; the filtered output can differ slightly at this edge.
- **Vertical position.** $z(t)$, the adopted altitude channel (Sec. 2.7.1). Unlike $\mathbf{r}$, it is *not* re-zeroed: it is kept at its measured value, so the height at the start of free flight is retained (Sec. 2.7.5).
- **Time.** t , the elapsed flight time, with the clock reset to $t = 0$ at the start of free flight (the first fix of the trimmed track, Sec. 2.7.3). A segment created by a split at a long gap keeps its parent flight's origin and clock (Sec. 2.7.6).
- **Velocity.** $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, the horizontal velocity, with magnitude the horizontal speed $v_{xy} = |\mathbf{v}|$; $v_z = \dot{z}$, the vertical velocity.
- **Displacement.** $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$, the horizontal displacement over a lag t .

A terminological rule holds throughout the document: a bare “position” means $\mathbf{r}$, the horizontal one; a statement about the full three-dimensional point says *3D position* explicitly.

2.8 Pipeline summary

Figure 2.17 gathers the processing steps of Sec. 2.7 in the order they are applied, from raw IGC tracks to the retained trajectories. Each stage identifies the records it acts on, its effect and the section that explains it. Section 2.9 then describes the resulting dataset.

Two decisions organise the chain.

- **The order.** Every check on the scalar record as logged (times, geodetic coordinates, altitude) comes first (stages 1–4), and the vector geometry is built only from what survives (stages 5–7). Converting coordinates does not clean a record: it recomputes each fix in a new frame, corrupt or not. Running it before the cleaning stages would therefore buy nothing, and would cost rework whenever trimming moved the fix the frame is anchored to, or filtering dropped the record outright.
- **The level.** Each stage acts per fix, per flight or per segment, and every parameter it adapts is read off the record it is acting on rather than off the archive around it. No flight's cleaning therefore depends on which other flights happen to be in the dataset. What is shared across flights is the thresholds of Table 2.8, held fixed for this run.

What the chain retains. Run over the whole archive, the cascade retains 156,305 of 186,052 paraglider flights (84.0 %) and 6,093 of 6,716 hang-glider flights (90.7 %). Table 2.7 says where the rest went. The numbers are a *cascade*, not a set of marginal effects: a flight counted against one criterion was still standing when that criterion tested it, so the table reads in pipeline order and the counts do not add up to what each cut would remove on its own (those are the `\StatScan*` numbers quoted in Secs. 2.7.4 and 2.7.6, measured on the raw population).

Within the retained flights, cleaning deletes 1.83 % of paraglider fixes and 0.72 % of hang-glider ones, and invalidates an altitude on a further 0.055 % and 0.035 %. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68 % and 0.96 % of fixes are flagged and *kept*. The flag records local disagreement, while the impossibility gate controls horizontal deletion (Sec. 2.7.2). Resampling reconstructs 0.58 % and 0.31 % of grid

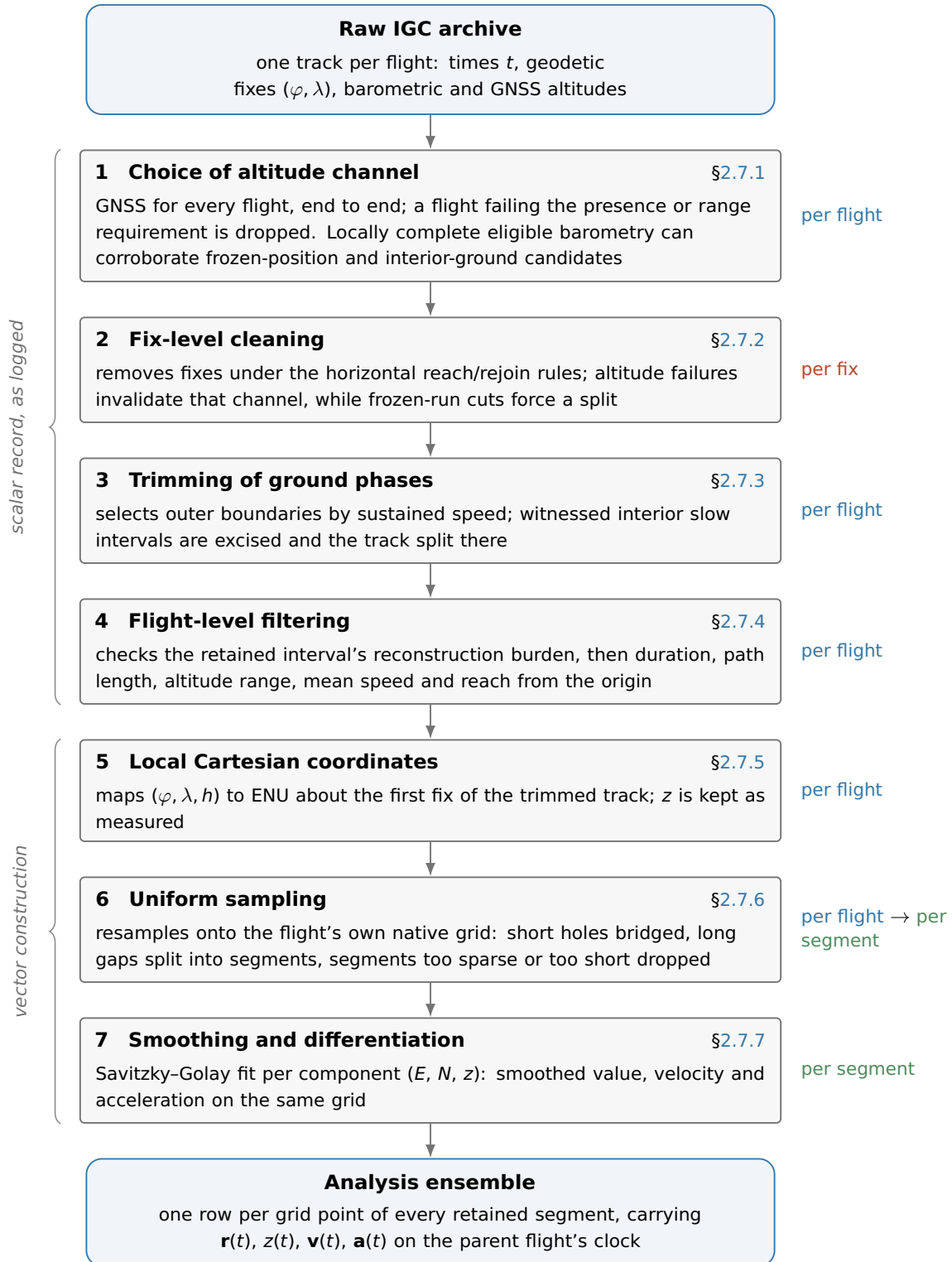


Figure 2.17. The processing chain, from the raw archive to the analysis ensemble. Each stage carries its effect and the section that argues the choice; the badge to the right is the level it acts at (*per fix*, *per flight*, *per segment*). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record about to be repaired or dropped. The thresholds each stage applies are listed, with their symbols, in Table 2.8.

points. A split is common but not the rule: 6.8% of retained paraglider flights and 16.0% of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.11 as Sec. 2.7.6 anticipated.

Table 2.7. The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* counts flights that survived earlier criteria; *share* is the fraction removed from that remaining population. Resampling leaves no retained segment in 20,899 paraglider flights (11.8% of those reaching that step) and 194 hang-glider flights (3.1%). These conditional removal fractions differ from the separate raw-population diagnostics in Secs. 2.7.4 and 2.7.6.

Why the flight was dropped	paragliders			
	standing	removed	share (%)	remaining
unreadable track	186,052	27	0.01	186,025
no usable GNSS altitude	186,025	697	0.37	185,328
never airborne	185,328	197	0.11	185,131
integrity gate	185,131	550	0.30	184,581
shorter than the duration cut	184,581	2,084	1.13	182,497
longer than the duration cut	182,497	3	0.00	182,494
path below the floor	182,494	173	0.09	182,321
altitude activity below the floor	182,321	4,899	2.69	177,422
out of reach of its own first fix	177,422	217	0.12	177,205
no segment survived resampling	177,205	20,899	11.79	156,306
no segment carries the smoothing window	156,306	1	0.00	156,305
retained				156,305

Why the flight was dropped	hang gliders			
	standing	removed	share (%)	remaining
no usable GNSS altitude	6,716	135	2.01	6,581
never airborne	6,581	1	0.02	6,580
integrity gate	6,580	53	0.81	6,527
shorter than the duration cut	6,527	79	1.21	6,448
path below the floor	6,448	1	0.02	6,447
altitude activity below the floor	6,447	155	2.40	6,292
out of reach of its own first fix	6,292	5	0.08	6,287
no segment survived resampling	6,287	194	3.09	6,093
retained				6,093

Table 2.8 lists the working thresholds and their symbols by stage.

Table 2.8. Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; configured values are read from `configs/preprocessing.yaml`. The minimum smoothing window is fixed by the polynomial-order rule in Sec. 2.7.7. The listed sections explain each criterion.

Symbol	Value	What it controls
Altitude admission (Sec. 2.7.1)		
—	95 %	minimum fraction of finite, nonzero GNSS readings; a flight below this completeness threshold is excluded

continued on the next page

Table 2.8 (continued)

Symbol	Value	What it controls
—	30 m	minimum GNSS range among finite, nonzero readings; a flight below it is excluded, without establishing a sensor failure
—	95 %	minimum barometric presence for witness eligibility; each candidate also requires complete local support
—	30 m	minimum barometric range for the same witness eligibility
Fix-level cleaning (Sec. 2.7.2)		
k	5	robust- σ multiplier of the Hampel test, Eq. (2.4)
w	20 s	half-width of the Hampel window
$\varepsilon_{\min}$	15 m	absolute floor of the Hampel test: no flag below this residual, whatever the local scale
—	5 fixes	smallest window population for which the test is evaluated
$v_{xy}^{\max}$	45 m s ⁻¹ (para), 55 m s ⁻¹ (hang)	absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)
$ v_z ^{\max}$	10 m s ⁻¹	cutoff for the two adjacent window statistics in Eq. (2.7); also used by the out-and-back spike rule
w_v	5 s	half-width of that window
—	5 finite steps	smallest window population for which the median is estimable; below it the per-step value stands in
$[h_{\min}, h_{\max}]$	$[-100, 6000]$ m	sensor-plausibility window on the altitude
δ_{xy}	15 m	diameter within which a frozen-lock run must stay
δ_z	5 m	altitude flatness tolerance of the same rule
τ_{freeze}	60 s	minimum duration before a run can be cut by the frozen-position rule
—	10 %	share of fixes above which the cleaning is declared to have failed and the flight is dropped
Ground trimming (Sec. 2.7.3)		
v_0	5 m s ⁻¹	ground-speed threshold used to identify take-off and landing candidates
T_0	30 s	how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)
T_{ground}	10 min	minimum duration of an interior ground stint before it is excised
—	5 m	altitude flatness tolerance over such a stint
Flight-level filtering (Sec. 2.7.4) — discards whole flights that cannot support the statistics		
—	40 min	minimum recorded airborne duration, accumulated within blocks
—	20 km	minimum flown path length
—	16 h	maximum elapsed span, including gaps; records above it are excluded
—	600 m	minimum whole-flight altitude range
—	$v_{xy}^{\max}$ (above)	plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)

continued on the next page

Table 2.8 (continued)

Symbol	Value	What it controls
—	$v_{xy}^{\max}$ (above)	reach bound applied at every fix: $d_i \leq v_{xy}^{\max}(t_i - t_0) + 50$ m, where d_i is its distance from the first trimmed fix
Resampling (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split		
$g_{\max}$	$\min(10 \Delta t, \max(20 \text{ s}, 2\Delta t))$	longest gap that is interpolated instead of splitting the flight
—	10 %	largest share of grid points a retained <i>segment</i> may have had to interpolate
—	90 s	minimum retained segment duration; smoothing additionally requires enough grid samples for its window
Smoothing (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives		
p	3	polynomial order of the filter
τ_c	5 s (horizontal), 5 s (vertical)	working timescale setting the window through Eq. (2.21); equal values are adopted for the two channels, without identifying a physical correlation time
—	5 samples	minimum odd window for the adopted cubic fit. The temporal span is $(w - 1)\Delta t$, with w given by Eq. (2.21)

The analysis ensemble, defined. The *analysis ensemble* is the set of trajectories written to `fixes.parquet`: one row per grid point of every *retained* segment of every *retained* flight, carrying the elapsed time t in the parent clock and the local Cartesian coordinates (E, N, z) with their first two derivatives. A flight is retained when it passes stages 1–4 of Figure 2.17 and at least one of its segments survives resampling and smoothing. Each retained segment must support channel reconstruction, pass its duration and density criteria, and contain the required smoothing window. That leaves 156,305 paraglider flights of 186,052 (84.0 %) and 6,093 hang-glider flights of 6,716 (90.7 %). Subsequent estimators draw their stated subsets from this archive, with additional requirements on cadence, segment length, lag support or reconstruction flags. Their contributing populations can therefore differ. Every segment keeps its parent’s origin and clock, so a position is always measured from that flight’s take-off and a time is always elapsed since it, whether or not the record is continuous up to that point. Section 2.9 characterises this ensemble, Chapter 3 analyses it, and `docs/guide/data-on-disk.md` states which columns each estimator reads and what a segment boundary means to it.

2.9 Preliminary characterization

This section describes the retained flights: their duration, path length, recording interval and geographical coverage. These measurements require no phase segmentation. Directional differences and their relation to geography are examined with the transport observables in Sec. 3.3.

Trajectory statistics. For each retained flight: airborne duration T , total horizontal path length, number of valid fixes and native sampling interval Δt . Their distributions (Fig. 2.18) describe the dataset the results rest on and allow unusually large values to be inspected. The native interval Δt determines the per-flight smoothing window (Sec. 2.7.7). Over the 156,305 retained paraglider flights the median airborne duration is 2.63 h (interquartile range 1.77 h to 3.94 h), the median flown path 87 km (57 km to 134 km) and the median record 7,138 fixes;

69.9% of them log at 1 Hz. The 6,093 hang-glider flights have larger median duration and path length—3.07 h and 151 km in the median—and slower-logging, with a median Δt of 3 s against 1 s and only 28.0% at 1 Hz.

The dataset audit counts 0 non-finite coordinates and 0 non-monotone flight clocks in the paraglider audit. Its largest filtered interior step speed is 49 m s^{-1} , with 32 interior steps above the working bound but below the verifier’s gross-anomaly threshold. These checks identify structural failures and large kinematic anomalies. Finite coordinates and bounded speeds do not exclude plausible-looking recorder errors.

Panels (a) and (b) describe the segments retained after resampling. Their duration and path length can be smaller than those of the trimmed parent flight tested in Sec. 2.7.4, because short, sparse or unreconstructable segments can be discarded. The resulting distributions include 530 paraglider flights (0.34%) and 5 hang-glider flights below the original 40 min or 20 km threshold. The parent-flight quantities passed those cuts; the cuts are not reapplied to the shorter retained trajectory.

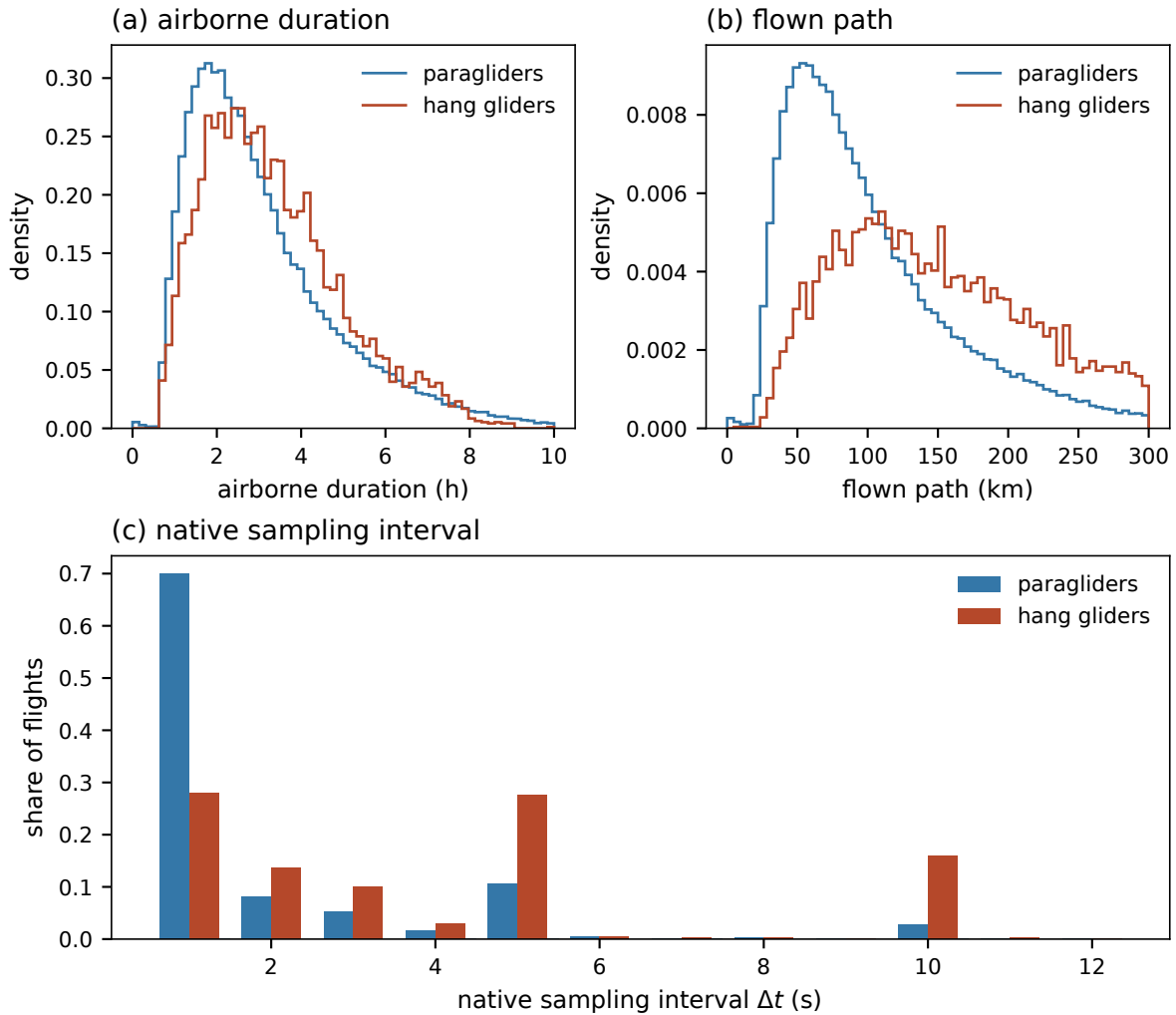


Figure 2.18. The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval Δt , per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.13 and 2.11. Duration and path are measured over the segments a flight keeps after resampling (Sec. 2.7.6), not over the trimmed flight the cuts of Sec. 2.7.4 tested, which is why a thin tail falls below those cuts.

Retention across recording intervals and seasons. Table 2.7 reads the discards as a cascade: end to end it leaves 84.0 % of attempted paraglider flights standing against 90.7 % of hang-glider ones.

The largest paraglider loss occurs when no segment survives resampling: 20,899 flights, or 11.8 % of those reaching that stage. This conditional fraction is 4.4 times the fraction removed by the altitude-range criterion (2.7 %). The recorded reason combines segments that are too short, too sparse or cannot be reconstructed; it does not attribute every rejected flight to the missing-fraction threshold alone.

Among flights with a recorded native-cadence estimate, the share with no surviving segment ranges across the six most common cadences from 0.0 % to 22.4 %. The missing-fraction and minimum-segment-duration bounds remain working choices that require sensitivity checks. Figure 2.19b gives retention across all criteria, between 69.8 % and 91.9 % over the 20 seasons with at least a thousand flights. Similar aggregate rates can conceal changes in the cadence, region and pilot composition of the retained population.

For hang gliders, loss of all segments is also the largest conditional fraction, 3.1 %, but is only 1.3 times that of the altitude-range cut (2.4 %). Its separation from the other criteria is therefore smaller than in the paraglider archive.

Where the flights are. The distribution of take-off sites is presented across three separate panels to account for the geographic coverage of the CFD: Figure 2.20 displays these sites against coastline and national borders.

- **Metropolitan France.** 94.7 % of retained paraglider flights launch here, against 96.7 % of hang-glider ones — the hang gliders are slightly more domestic.
- **La Réunion.** 3.2 % (4,946 flights), the one overseas department with a substantial share.
- **Elsewhere.** 2.2 % (3,379 flights): the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,338 distinct take-off sites, where distinct sites are launches rounded to a hundredth of a degree (~ 1 km); by share of paraglider launches:

- **Alps.** 58.8 %.
- **Pyrenees.** 7.9 %.
- **Massif Central.** 8.2 %.
- **Channel Coast.** 3.2 %: a box over low-relief terrain (Normandy, Picardy, the coastal Nord), not one carved out by excluding the other three.
- **Outside the massifs.** 16.6 %: everywhere else in metropolitan France, a residual defined by exclusion and not a terrain claim.

These groups are longitude and latitude boxes, drawn on Fig. 2.20a; membership is assigned from the first recorded position.

Altitude at the beginning of the record. Figure 2.21 groups flights by the altitude of their first raw fix, using the bands < 300 m, 300 m to 800 m, 800 m to 1,500 m and $\geq 1,500$ m adopted in the regional classification of Hernández-Aguayo, Cristelli and Benzaquen [19]. These altitude bands are a simplification, not the full mountain classification of Kapos et al. [20].

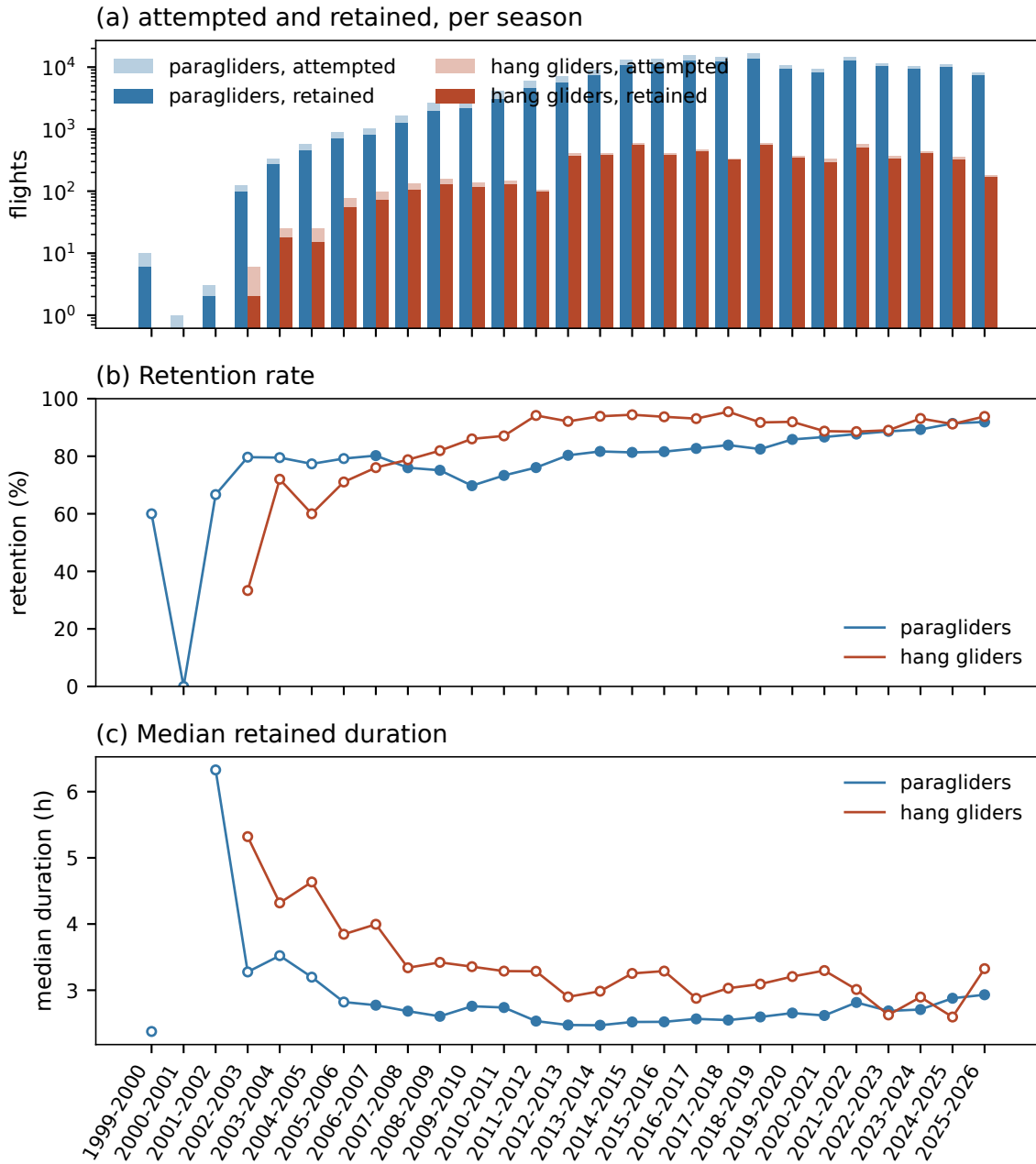


Figure 2.19. The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights. The aggregate retention rate is approximately flat across the larger seasons; this does not establish equal selection within cadence or region groups.

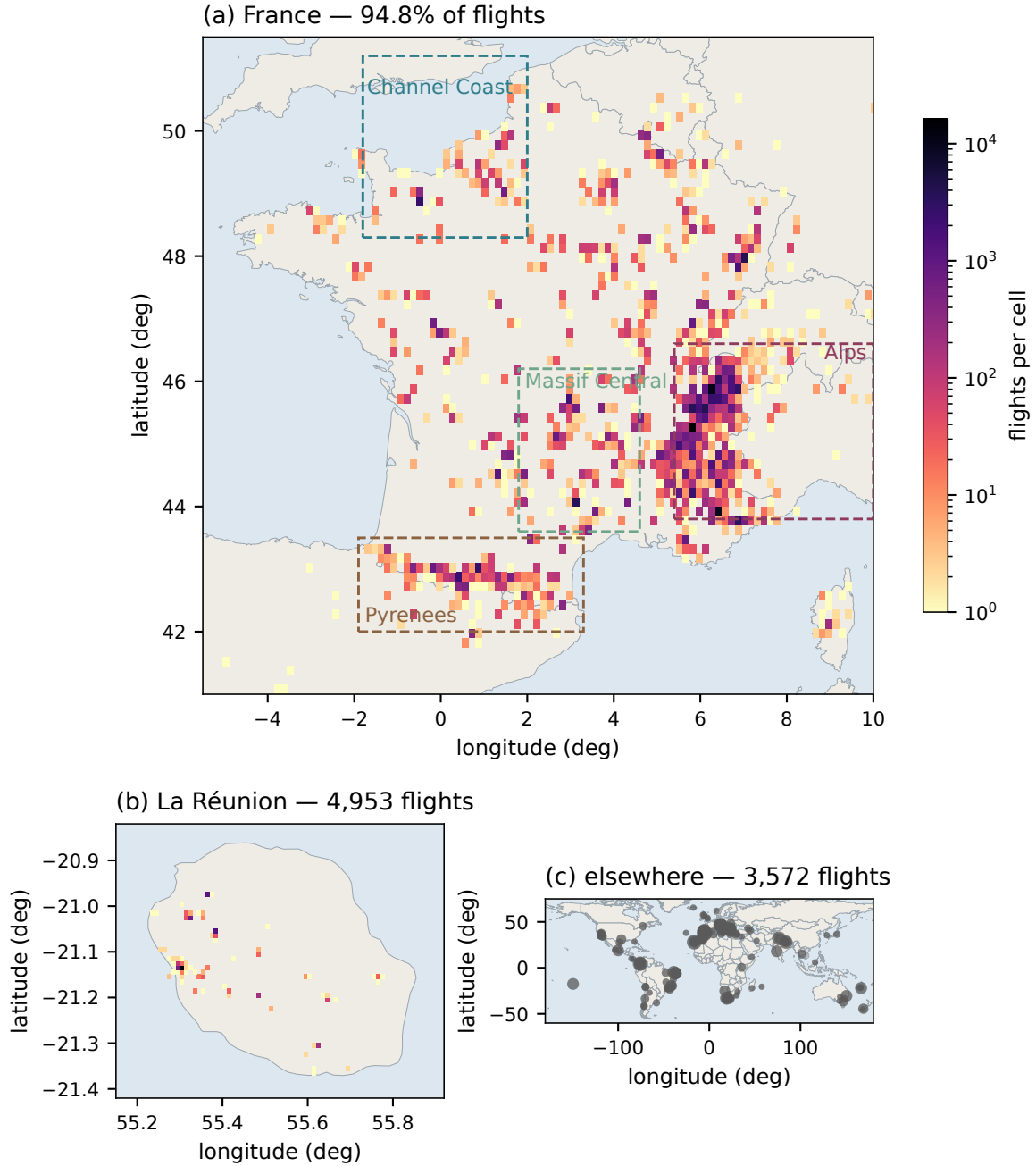


Figure 2.20. Take-off sites of the retained ensemble, on three frames, pooling both disciplines throughout. (a) Metropolitan France, density on a 0.15° cell; the dashed boxes are the orographic groups used as strata in Fig. 3.9 (Sec. 3.3). (b) La Réunion, on a 0.01° cell, the one overseas department with a substantial share. (c) Everything else, one marker per 0.5° cell with the area set by its count.

The present altitude is a recorder measurement before trimming, not a terrain-model elevation: the logger can start away from launch, already airborne, or with an unsettled sensor. GNSS is preferred, with a pressure-altitude fallback in this exploratory map only. The historical terrain names in its legend identify the bands; they do not establish local relief, slope or mountain geometry. Datum and atmospheric offsets can also move a fix across a band boundary. 99.9 % of the retained paraglider ensemble classifies this way. By share of the classified flights: 7.6 % Plains, 11.2 % Hills, 50.7 % Low mountains and 30.5 % High mountains.

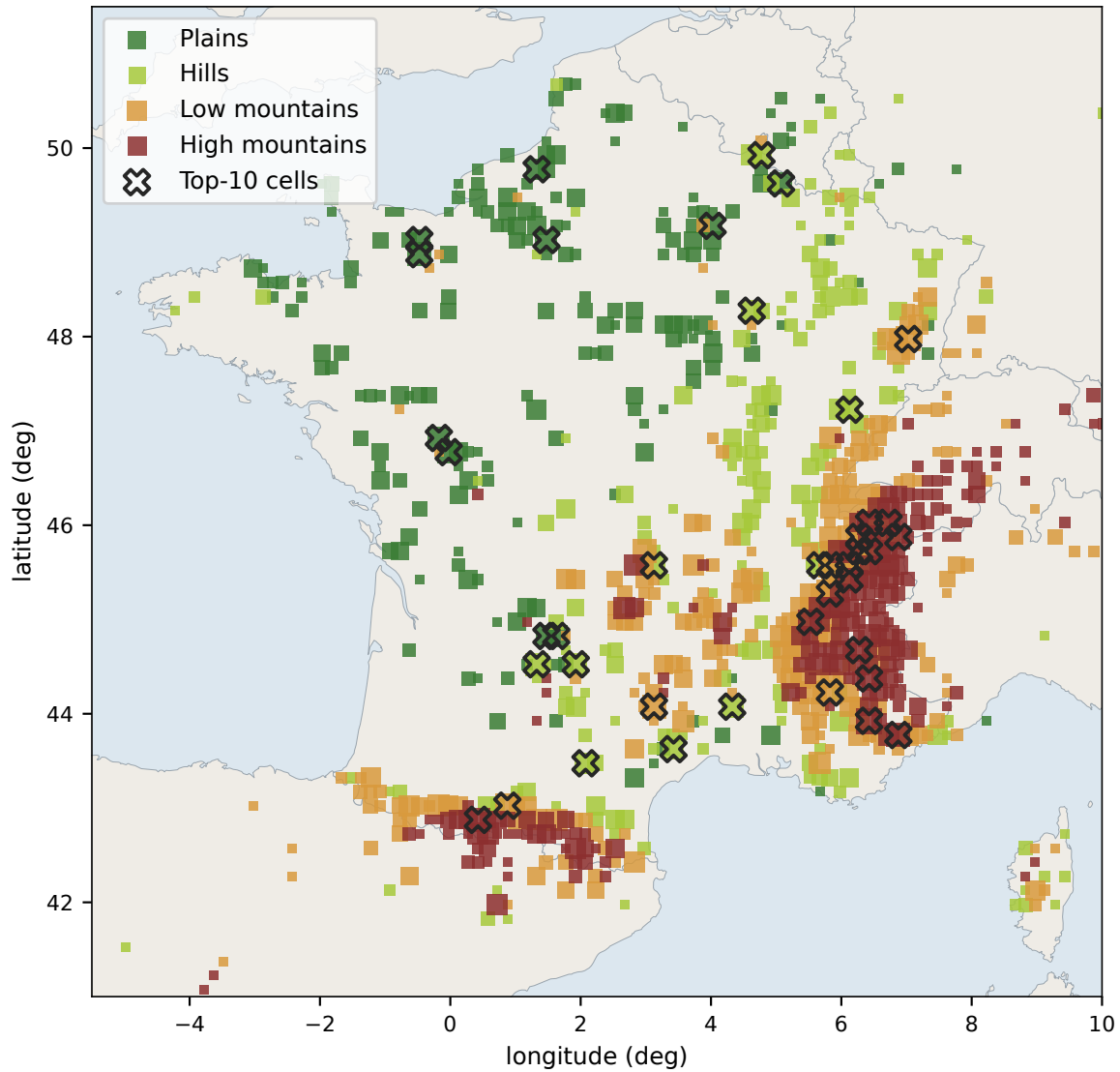


Figure 2.21. First recorded positions of the retained ensemble (both disciplines pooled), grouped by logged altitude using the elevation bands of [19]. These are approximate launch-altitude groups, not measured terrain classes. One marker per occupied 0.15° cell, coloured by its majority class and sized by the logarithm of its flight count; crosses mark the 10 busiest cells of each class.

2.10 Chapter summary

The recorder archive has been converted into trajectories with explicit temporal support, local horizontal coordinates and traceable quality decisions. Cleaning separates isolated position defects, repeated-position runs and sustained recording failures; it uses a barometric witness

only where that channel is available over the interval being tested. The 10-m/s windowed vertical-speed bound is an operational selection criterion, not a measured physical ceiling for every soaring flight.

Gaps, reconstructed altitude and derivative support remain visible in the retained data. Uniform sampling makes within-segment lag statistics possible. The masks exclude flagged reconstruction and derivative support from phase assignment, although an unflagged recording defect can still affect the features. The geographical and temporal composition of the archive also remains heterogeneous. Chapter 3 therefore treats weighting, lag support and environmental conditioning as part of the definition of each observable, rather than as incidental details of its calculation.

Chapter 3

Global transport

Whole flights combine circling, gliding and the search for lift, whose accumulated effect appears in the displacement statistics. These statistics constrain any model of the motion: how far a glider moves over a given time, how variable that distance is, which directions it favours and how long its velocity remains correlated. We measure these properties before assigning phase labels, then compare them with predictions of candidate stochastic processes. Chapter 4 examines how the motion decomposes into individual behaviours.

The archive contains 156,305 paraglider and 6,093 hang-glider flights in the retained transport snapshot. A reproducible diagnostic subset is used for measurements spanning 10 s to 10,000 s, including regional principal axes and velocity correlations at several resolutions. Subset results and archive-wide descriptions are identified separately.

3.1 The measured displacement and its time scales

Let $\mathbf{r}_f(t) = (E_f(t), N_f(t))$ denote the horizontal position of flight f in its local east–north frame. A displacement is

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f(t, \tau) = |\mathbf{X}_f(t, \tau)|. \quad (3.1)$$

East and north are two coordinates; east and west are opposite signs of the same coordinate. The radial displacement R combines both coordinates and is not a third independent component.

For a continuous segment of duration T_f , the time-averaged MSD is

$$\overline{\delta_f^2}(\tau) = \frac{1}{T_f - \tau} \int_0^{T_f - \tau} R_f(t, \tau)^2 dt. \quad (3.2)$$

Only windows lying entirely within a retained segment are admitted, so an increment never crosses a data gap. For flights with several segments, combining the segment sums and their numbers of admissible windows gives a flight-level mean; averaging further over the population requires a choice of weights:

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \overline{\delta_f^2}(\tau), \quad M_2^{\text{window}}(\tau) = \frac{\sum_f n_f(\tau) \overline{\delta_f^2}(\tau)}{\sum_f n_f(\tau)}. \quad (3.3)$$

The first gives each contributing flight equal weight, whereas the second gives each admissible window equal weight and consequently weights longer flights more strongly. Both averages still combine flights made under different meteorological and geographical conditions.

The archive time average in Figure 3.1 uses a third convention: it gives each contributing *segment* equal weight. A flight split into several segments can therefore contribute several times.

Segments with fewer than eight fixes are omitted from this measurement. Within an admitted segment, all admissible starting fixes enter the mean. The diagnostic subset used later keeps one segment per flight and uses starting times $0, \tau, 2\tau, \dots$; its first-difference windows do not overlap. Its V_1 and V_2 curves give each flight equal weight, whereas the baseline quantiles and higher moments pool windows. Section 3.2.2 adds quantile controls with fixed flights, weights and origins. In discrete time, a segment of n fixes at interval Δt contributes

$$\widehat{\delta^2}(k\Delta t) = \frac{1}{n-k} \sum_{j=0}^{n-k-1} |\mathbf{r}_{j+k} - \mathbf{r}_j|^2, \quad 1 \leq k < n. \quad (3.4)$$

The subset is evaluated on a 10 s grid; flights whose native interval exceeds that value are excluded. Each admitted flight contributes its longest retained segment. Its sampling design is therefore distinct from a uniform draw of all flights.

For completeness, Figure 3.1a retains the launch-synchronised average $\text{MSD}_{\text{launch}}(t) = \langle |\mathbf{r}_f(t) - \mathbf{r}_f(0)|^2 \rangle_f$. Its difference from the time average is unsurprising: these flights occur in different locations, seasons and years, and are not established to be independent realisations of one common law. It is a descriptive archive average, without a process exponent inferred from it.

3.1.1 Growth over ten seconds to ten thousand seconds

The comparison interval is 10 s to 10,000 s. For a positive observable $Y(\tau)$ we distinguish its fitted slope and its local slope,

$$\log Y(\tau) \simeq a + b \log(\tau/\tau_0), \quad b_{\text{loc}}(\tau) = \frac{d \log Y}{d \log \tau}, \quad (3.5)$$

Logarithms of an observable refer to its numerical value in the stated units, equivalently $\log(Y/Y_0)$ for a fixed reference Y_0 with the same dimensions. Here $\tau_0 = 1$ s; changing reference units alters the intercept, not b . The subset and duration reports fit ordinary least squares in $\log_{10} Y$, giving each admitted lag equal weight regardless of its number of contributing windows. A fit over three decades provides a compact description of the curve, but if the local slope changes substantially, that single value is an *effective exponent* rather than evidence of an exact power law.

At the lower end, the native cadence and smoothing window can affect the displacement; 10 s is not automatically outside that influence for every recorder. At the upper end, a flight must be long enough to contribute, and the number of windows per flight decreases. Coverage, local slopes and the open–closed split accompany the fit; any comparison restricted to longer flights describes that selected population.

Sampling uncertainty and sensitivity to the analysis interval require separate treatment. Repeated windows from one flight, and flights from the same site and day, can share information; a final interval estimate should therefore resample whole site–day groups with their complete curves. This applies bootstrap resampling [21] at the cluster level, conditional on treating those clusters as exchangeable. Changing the fitted interval instead tests sensitivity to curvature, smoothing and selection. That sensitivity is reported separately from sampling uncertainty; adding it to a standard error does not create a confidence interval. The present subset plots are descriptive measurements, not calibrated hypothesis tests.

3.1.2 What a second difference removes

A constant velocity contributes a quadratic term to an uncentred MSD. If $\mathbf{r}(t) = \mathbf{v}_d t + \mathbf{Y}(t)$ and the residual increment has zero mean, then

$$\mathbb{E}|\mathbf{X}(t, \tau)|^2 = |\mathbf{v}_d|^2 \tau^2 + \mathbb{E}|\mathbf{Y}(t + \tau) - \mathbf{Y}(t)|^2. \quad (3.6)$$

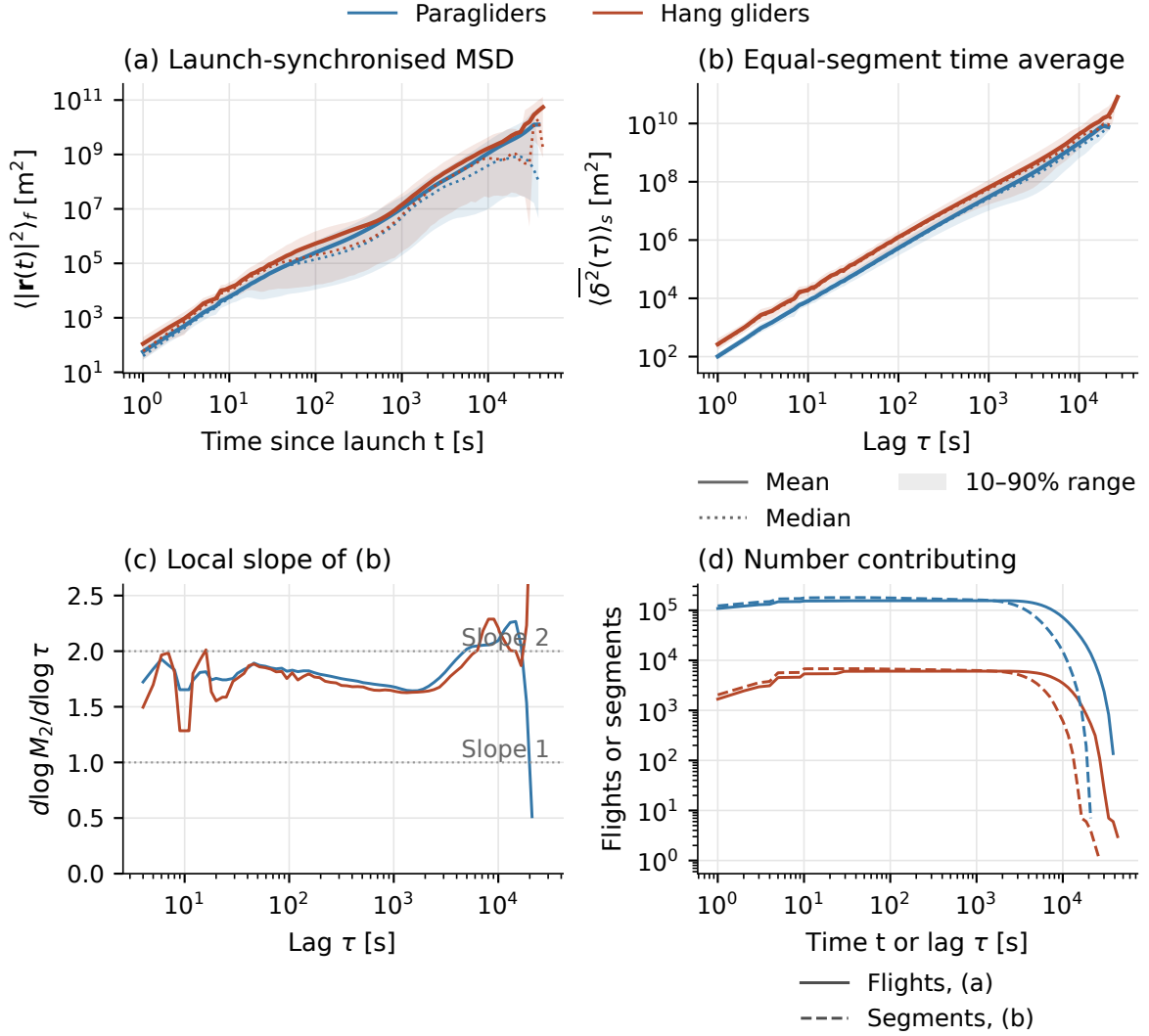


Figure 3.1. Archive displacement summaries. (a) Launch-synchronised average, retained as a description of the heterogeneous collection. (b) Time-averaged MSD, averaged equally over retained segments. Solid lines are means, dotted lines medians, and shading the 10–90% range across contributing flights in (a), or segments in (b); shading is not an uncertainty band on the mean. (c) Local slope of (b). (d) Contributing flights for (a) and segments for (b). Panel (a) uses time since launch; panels (b,c) use within-segment lag.

Without the zero-mean assumption there is also a cross term. The two contributions add as functions of lag; their fitted exponents do not add.

A useful control is the second difference,

$$A_2 \mathbf{r}(t; \tau) = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad (3.7)$$

$$V_2(\tau) = \langle |A_2 \mathbf{r}(t; \tau)|^2 \rangle. \quad (3.8)$$

Since $A_2(\mathbf{r}_0 + \mathbf{v}_d t) = 0$, this observable is exactly insensitive to a constant velocity. It avoids estimating drift from the endpoints, which would impose a return constraint on the residual trajectory. Its cost is a window of length 2τ . We retain it for this specific purpose, alongside $V_1(\tau) = \langle R^2 \rangle$.

To relate the second-difference slope to H , suppose the residual is a centred finite-variance process with stationary increments and $\mathbb{E}|\mathbf{Y}(t + \tau) - \mathbf{Y}(t)|^2 = K\tau^{2H}$. Writing adjacent

increments as $\mathbf{I}_1$ and $\mathbf{I}_2$ gives $2\mathbb{E}[\mathbf{I}_1 \cdot \mathbf{I}_2] = K[(2\tau)^{2H} - 2\tau^{2H}]$, hence

$$V_2(\tau) = K(4 - 2^{2H})\tau^{2H}. \quad (3.9)$$

For $K > 0$ and $0 < H < 1$, the prefactor is positive, so half the slope of V_2 estimates H under these assumptions. At $H = 1$ the second difference vanishes in mean square and its logarithmic slope is undefined. Fractional Brownian motion with $0 < H < 1$ is one example [22]. The prefactor follows by expanding the square of the difference of adjacent increments; it is not an additional fit parameter that changes the exponent.

On a smooth curved trajectory, $A_2\mathbf{r} = \ddot{\mathbf{r}}\tau^2 + O(\tau^3)$, so $V_2 \propto \tau^4$ at sufficiently small lag. Half that slope is not the Hurst exponent of a stationary-increment position process. Agreement between differences of order two and three is a consistency check; it does not prove that all environmental trends have disappeared. In particular, $b_1 - b_2$ is not a measured fraction of transport due to wind or course geometry.

Table 3.1 reports the effective slopes. Both disciplines have $b_1 > 1$ on this interval. However, selecting the same long segments at every lag shifts the estimate. This comparison reveals sensitivity to the selected population, but does not separate changing lag membership from the different behaviour of longer flights. At 10,000s, plain increments retain 461 paraglider flights and 264 hang-glider flights, supplying only 582 and 325 non-overlapping windows; second differences retain 107 and 56 flights. A three-decade interval therefore does not provide equally strong information at both ends.

Table 3.1. Effective equal-flight slopes on 10s to 10,000s. The fixed cohort contains selected segments of at least 20,000s. These are descriptive slopes without calibrated confidence intervals.

Discipline	Population	Flights	b_1	b_2
paraglider	all selected	955	1.702	2.028
	fixed long cohort	107	1.776	2.088
hang glider	all selected	479	1.662	1.985
	fixed long cohort	56	1.734	2.020

Values of b_2 close to or above two cannot be read directly as a stationary-increment Hurst exponent near or above one. Equation (3.9) becomes degenerate at $H = 1$. The observed curvature, short-scale smoothness and changing support must be resolved before the constant-drift control can yield an intrinsic H .

3.1.3 Open and closed flights: geometry survives detrending

A declared closed task rewards return towards the starting region. An open task allows continued progress away from it. This difference can change long-lag displacements even if speed and short-time behaviour are similar. Task classes are compared separately in Figure 3.2; a declaration is not a measurement that a particular retained segment closes exactly.

A circular trajectory makes the distinction precise. For $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$ with period $T_c = 2\pi/\omega$,

$$V_1(\tau) = 4a^2 \sin^2\left(\frac{\pi\tau}{T_c}\right), \quad V_2(\tau) = 16a^2 \sin^4\left(\frac{\pi\tau}{T_c}\right). \quad (3.10)$$

The displacement initially grows, then decreases as the lag approaches a complete return. The second difference also retains this geometry. Removing constant drift cannot remove closure, a turn, or a spatially varying forcing.

The selected paraglider tasks include 391 open and 559 closed flights; hang gliders contribute 171 and 308. Their effective b_1 values are respectively 1.752 versus 1.652, and 1.702 versus

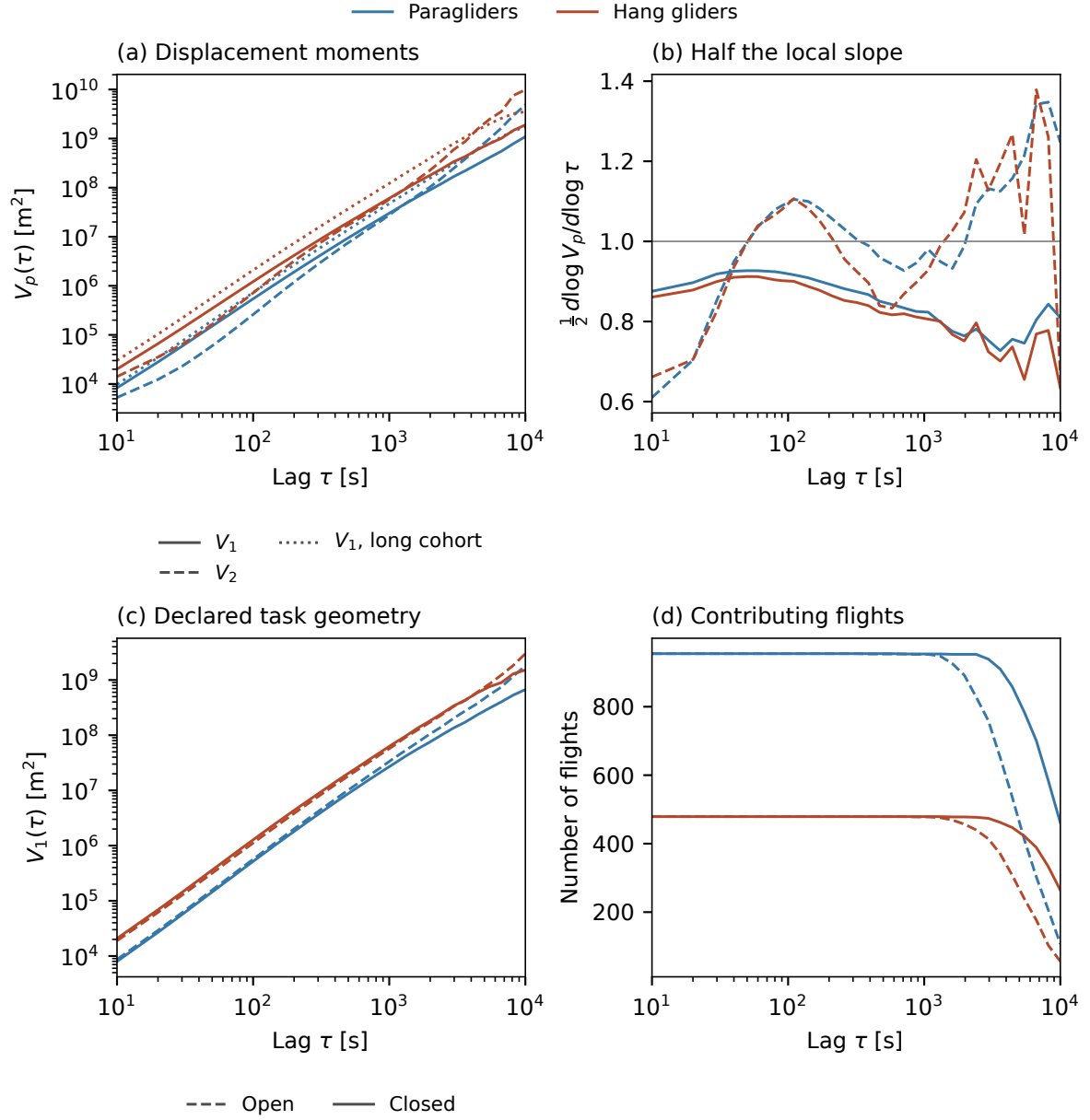


Figure 3.2. Displacement and constant-velocity control over 10 s to 10,000 s in the diagnostic subset. The figure separates the second moments from their local slopes and the declared open and closed tasks. Coverage is shown explicitly; the support of a second difference is smaller because it spans 2τ . Slopes refer to this interval and this subset.

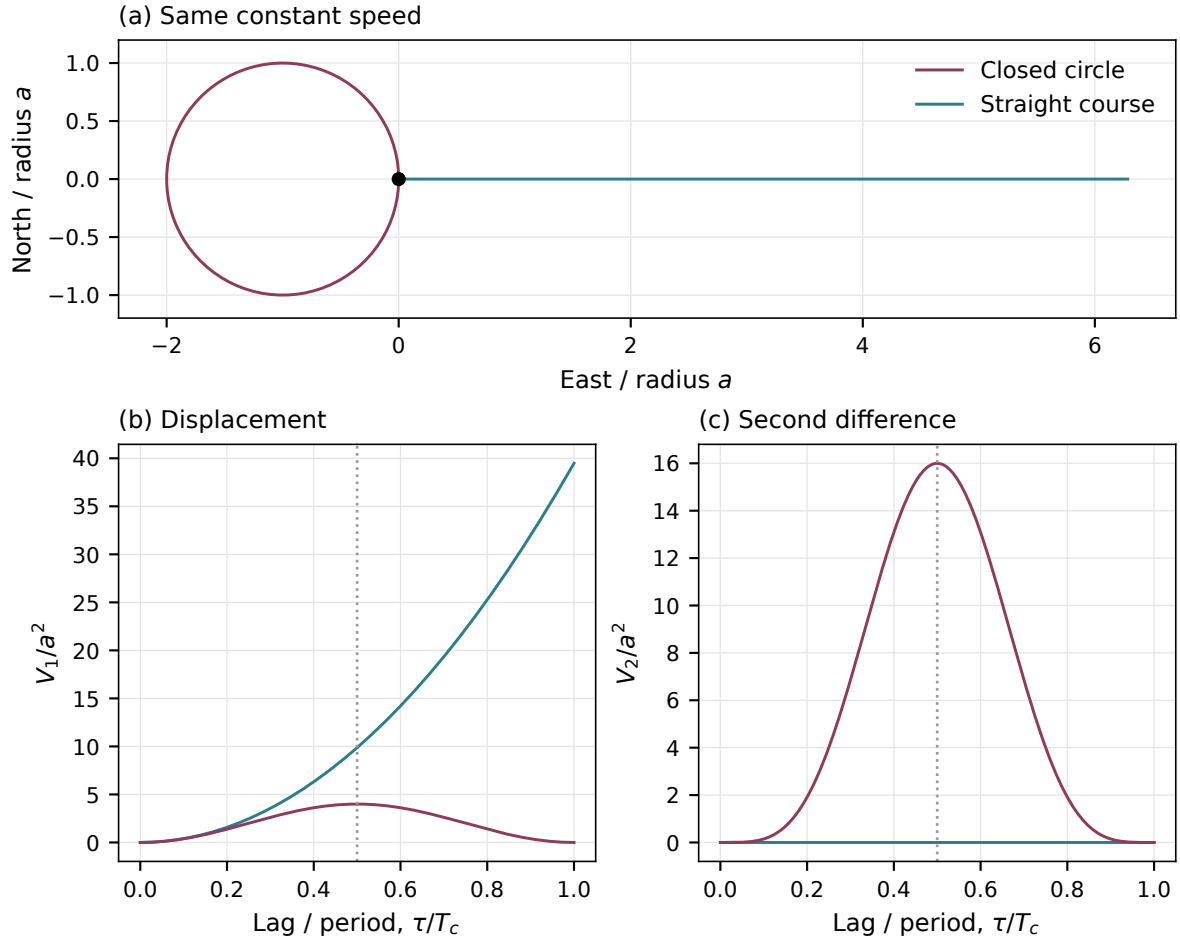


Figure 3.3. Exact geometrical illustration, without flight data. A straight course and a circle have the same speed. The circular path is continued periodically when evaluating second differences. Its displacement vanishes after one period; its second difference is also sensitive to the return. Therefore a change of slope need not indicate a change of stochastic process, and second differencing does not neutralise a closed task.

1.641. At order two the ordering reverses: 1.968 versus 2.054 and 1.893 versus 2.004. This is consistent with different route geometry affecting the two observables differently; it is not a quantitative decomposition into drift and fluctuations.

A complementary geometrical descriptor is the closure ratio

$$\chi_f = \frac{|\mathbf{r}_f(T_f) - \mathbf{r}_f(0)|}{\int_0^{T_f} |\dot{\mathbf{r}}_f(t)| dt} \in [0, 1]. \quad (3.11)$$

In the subset, median χ_f is 0.318 for open paraglider tasks and 0.027 for closed ones; hang-glider medians are 0.192 and 0.026. The geometric contrast therefore survives checking the actual paths, while the nonzero open-task medians remain far below the value one of a straight flight.

The ratio is defined for a continuous observed trajectory of positive path length: small χ_f indicates little net progress relative to distance travelled, not necessarily a declared closed task. Comparing both descriptors, matching duration and region, and studying τ/T_f as well as physical lag would separate task geometry from changing composition more sharply. The current task split is evidence of an association; it does not isolate a causal effect of the scoring rule.

3.2 What the displacement quantiles reveal

3.2.1 From self-similarity to a measurable prediction

For an H -self-similar process starting at the origin, $(\mathbf{r}(ct_1), \dots, \mathbf{r}(ct_n)) \stackrel{d}{=} c^H(\mathbf{r}(t_1), \dots, \mathbf{r}(t_n))$ for every $c > 0$ and every finite choice of times. Self-similarity therefore constrains joint distributions, beyond the MSD alone. With stationary increments, it implies the one-lag prediction

$$\mathbf{X}(t, \tau) \stackrel{d}{=} (\tau/\tau_0)^H \mathbf{X}(t, \tau_0). \quad (3.12)$$

Writing $a_\tau = (\tau/\tau_0)^H$, and assuming the corresponding densities exist, the two-dimensional and radial laws obey different Jacobians:

$$p_{\mathbf{X}}(\mathbf{x}; \tau) = a_\tau^{-2} F(\mathbf{x}/a_\tau), \quad p_R(r; \tau) = a_\tau^{-1} f(r/a_\tau). \quad (3.13)$$

The radial law follows by integrating the joint density over angle, including the polar Jacobian r . It does not require isotropy. In particular, a two-dimensional density at the origin would scale as a_τ^{-2} , not as a_τ^{-1} .

Define the radial quantile at probability p by

$$Q_p(\tau) = \inf\{r : \Pr[R(t, \tau) \leq r] \geq p\}, \quad 0 < p < 1. \quad (3.14)$$

Equation (3.12) implies

$$Q_p(\tau) = a_\tau Q_p(\tau_0), \quad H_p = \frac{d \log Q_p}{d \log \tau} = H \quad \text{when } 0 < Q_p(\tau_0) < \infty. \quad (3.15)$$

The median probes typical displacement; the lower quartile probes relatively small displacements; the upper quartile and 90th percentile probe progressively larger ones. They use ranks rather than powers of distance, complementing the moments without requiring a fourth moment. They remain sensitive to sampling, flight composition and finite support. Equality of several quantile slopes is necessary for the scaling form, but does not prove self-similarity of the full process.

3.2.2 Keeping the population fixed across lags

At a given lag, all probability levels refer to the same empirical distribution. The 25th percentile is not an average over a class of slow flights, nor is the 90th percentile an average over fast flights. Different flights can supply observations near those ranks, and their rank order can change with lag. This is compatible with studying a distribution; it does not track individual objects moving from one quantile to another.

A separate problem arises when the distribution itself changes because its sampling population changes. If flight f supplies $n_f(\tau)$ admissible increments, pooling them defines

$$\hat{F}_\tau^{\text{pool}}(x) = \sum_{f \in \mathcal{F}_\tau} \frac{n_f(\tau)}{\sum_g n_g(\tau)} \hat{F}_{f,\tau}(x), \quad \hat{F}_{f,\tau}(x) = \frac{1}{n_f(\tau)} \sum_k 1\{R_{fk}(\tau) \leq x\}. \quad (3.16)$$

Both membership $\mathcal{F}_\tau$ and weights can depend on lag: a short flight eventually contributes nothing, while the number of non-overlapping windows decreases differently between flights. Quantiles are nonlinear functionals of a distribution, so changing these weights can change H_p even when every flight follows its own unchanged scaling law. Their effects cannot be removed by averaging the individual quantiles.

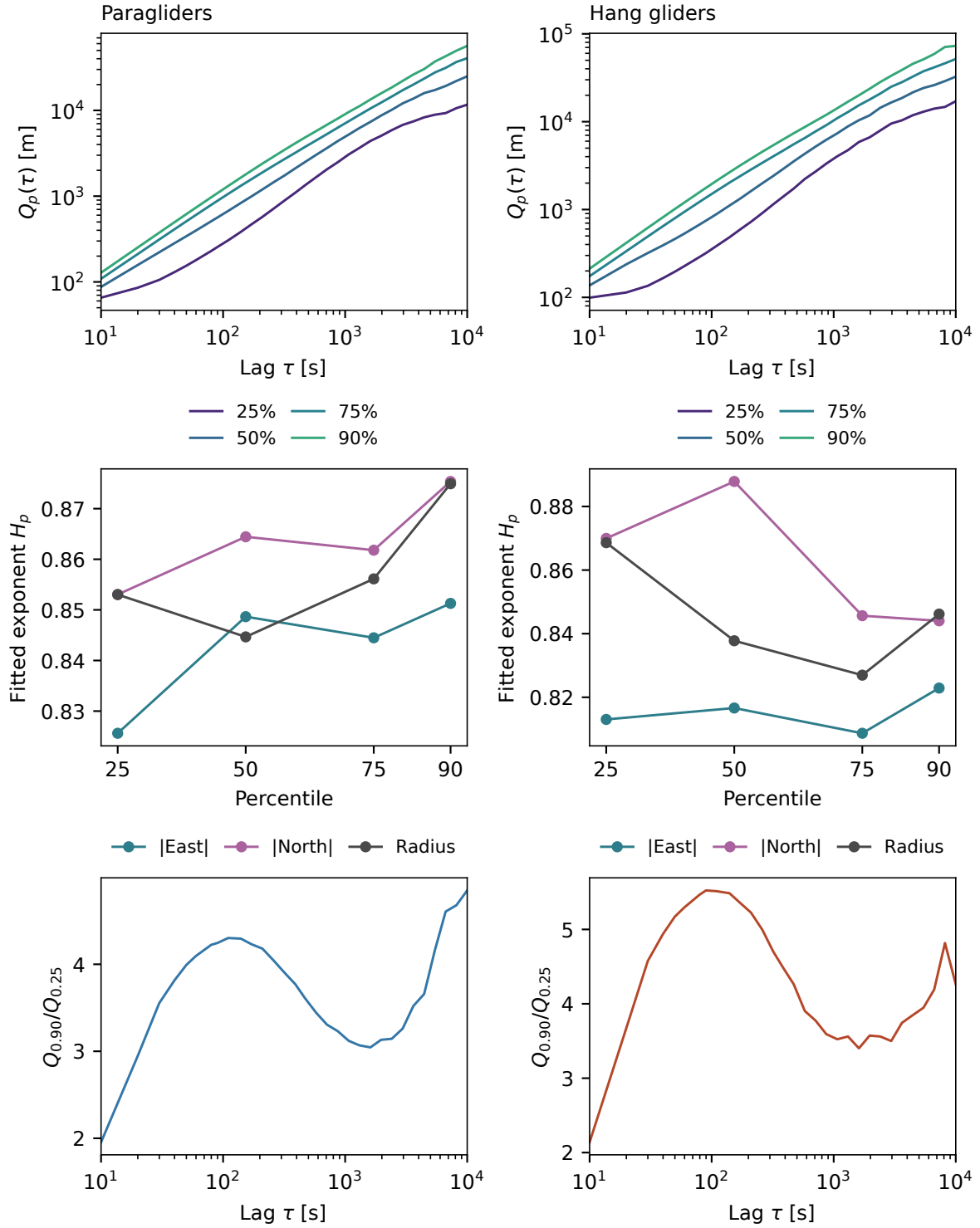


Figure 3.4. Displacement quantiles in the diagnostic subset over 10 s to 10,000 s, with east and north treated separately from radial displacement. Exponents at individual probability levels and ratios of radial quantiles test whether one scale factor suffices. A common exponent would preserve the quantile ratios; a systematic change indicates shape evolution or a change in the sampled population.

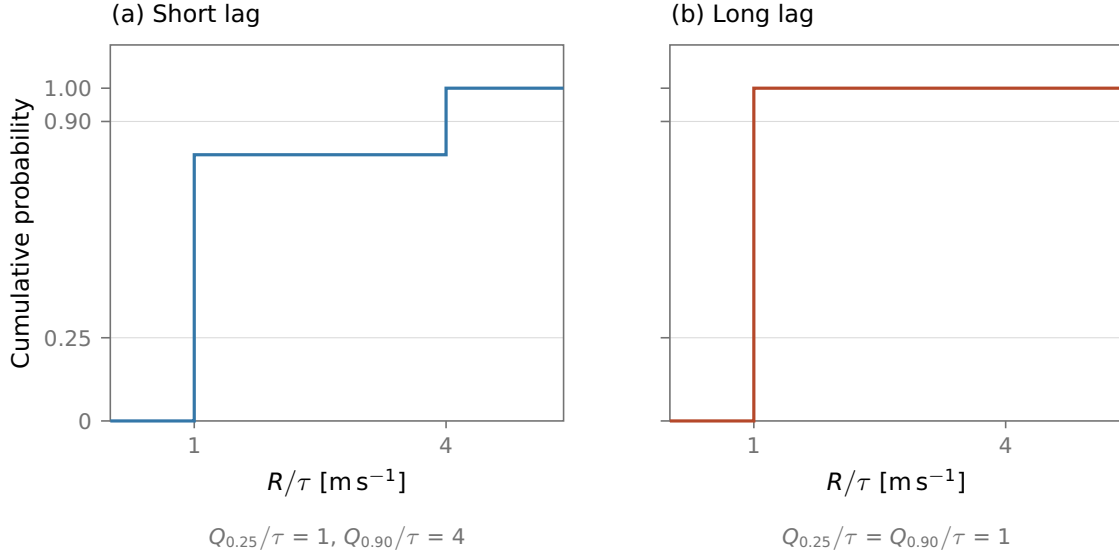


Figure 3.5. A sampling effect can imitate different quantile growth rates. Consider two straight flights at 1 m s^{-1} lasting 20,000 s each and two at 4 m s^{-1} lasting 5,000 s each. At a 10-s lag, the fast flights supply 1/5 of the non-overlapping windows, so the 90th percentile of R/τ is 4. At 10,000 s they supply none, and that percentile is 1. Every flight is exactly ballistic, yet the pooled upper quantile grows more slowly than τ between these lags. A fixed cohort of the two long flights has $Q_p(\tau) = (1 \text{ m s}^{-1})\tau$ at every displayed probability level. The example is constructed, not measured from the archive.

We therefore compare four sampling conventions in Fig. 3.6:

1. All available flights, with the original pooled-window weights.
2. A fixed cohort of segments at least 20,000 s long, retaining pooled weights.
3. The same fixed flights, giving each flight equal weight at every lag.
4. The same flights and equal weights, now using the same starting times at every lag.

For the last convention, let $\tau_{\max} = 10,000 \text{ s}$ and keep every starting time t_{fk} on the 10-s grid satisfying $t_{fk} + \tau_{\max} \leq T_f$. Then

$$\hat{F}_{\tau}^{\text{fixed}}(x) = \frac{1}{|\mathcal{F}_*|} \sum_{f \in \mathcal{F}_*} \frac{1}{n_f^*} \sum_{k=1}^{n_f^*} 1\{R_f(t_{fk}, \tau) \leq x\}, \quad \hat{Q}_p(\tau) = \inf\{x : \hat{F}_{\tau}(x) \geq p\}. \quad (3.17)$$

The set of flights, origins and their weights are now independent of τ , and the same weighted observations define every percentile. All four conventions use this inverse empirical-CDF definition without interpolation between order statistics. The common origins overlap at most lags, which preserves the population but does not make increments independent. The third convention isolates flight weights; the fourth also controls which part of each flight supplies the starting observations.

The controls change the result substantially. In the changing paraglider pool the 90th-percentile slope exceeds the lower-quartile slope, whereas all three fixed-cohort conventions give the reverse ordering. Equalising flight weights has a smaller effect than selecting that cohort, and fixing origins changes the estimates again. For the fully controlled population, $H_{0.25} = 0.979$ and $H_{0.90} = 0.892$ in paragliders; the corresponding hang-glider values are 0.995 and 0.864. The ordering is not strictly decreasing across all four ranks: the 90th-percentile estimate lies above the 75th in both disciplines.

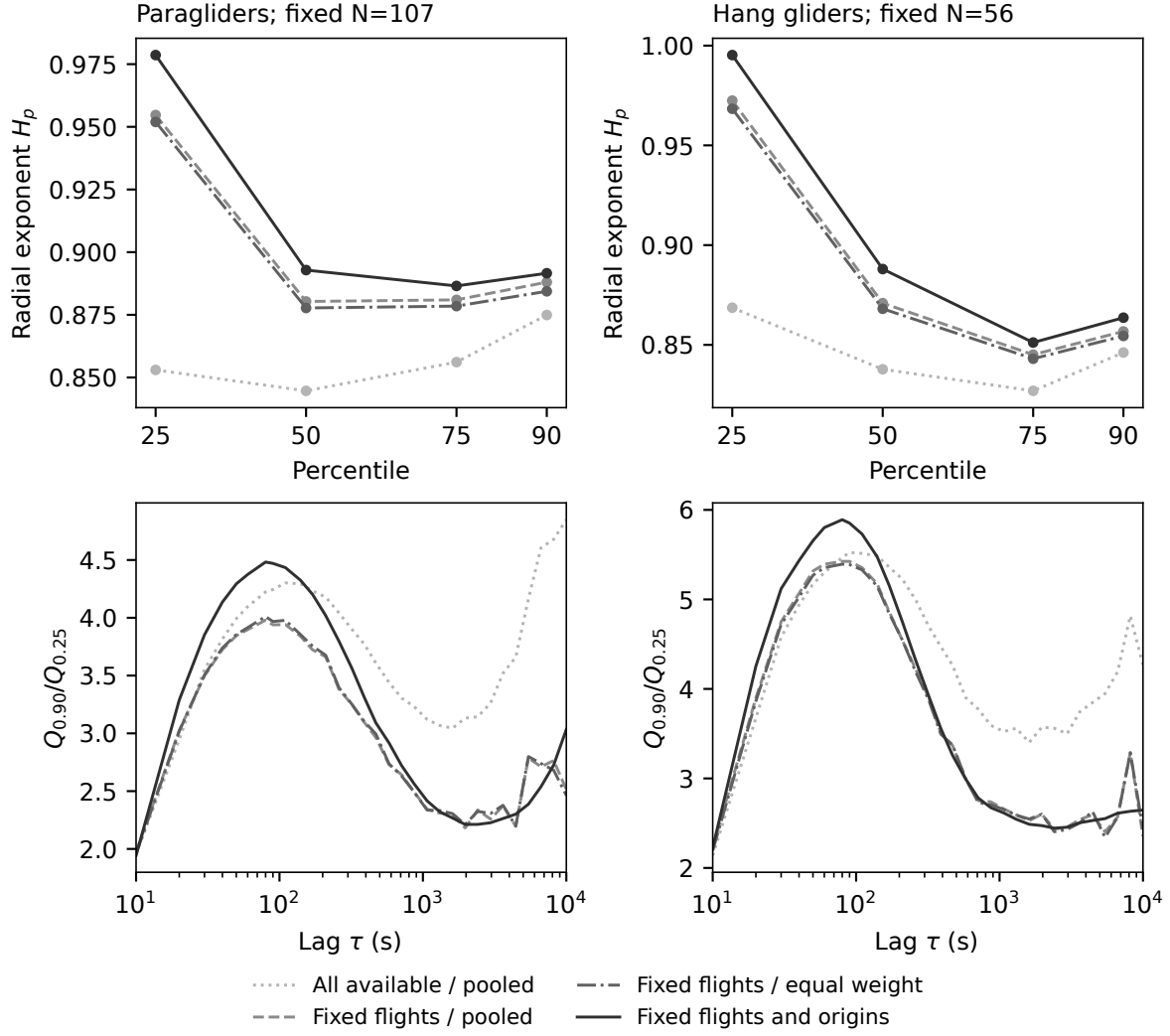


Figure 3.6. Controls for population changes in the quantile analysis. Top: radial quantile slopes under four sampling conventions, fitted on their common positive supported lags. Bottom: the corresponding $Q_{0.90}/Q_{0.25}$ curves. The fixed cohorts contain 107 paraglider and 56 hang-glider flights; the fully controlled convention retains respectively 157,493 and 82,774 origins at every lag. These overlapping origins are not independent replicates. Each displayed estimate requires at least two flights and 30 windows, which are support checks rather than guarantees of statistical precision.

This contrast cannot arise from flights entering or leaving the fully controlled sample, or from their weights changing with lag. It remains a result for long selected flights, however, and can include changes of phase along those flights, route geometry, environmental heterogeneity and sampling uncertainty. The cohort requires a continuous segment of at least about five and a half hours; it cannot represent short flights. Neither the many overlapping origins nor a common fit interval removes that selection.

3.2.3 How quantile growth changes relative spread

The interpretation of an ordering such as $H_{0.25} > H_{0.90}$ follows directly from

$$\frac{d}{d \log \tau} \log \left(\frac{Q_{0.90}(\tau)}{Q_{0.25}(\tau)} \right) = H_{0.90}(\tau) - H_{0.25}(\tau). \quad (3.18)$$

A negative right-hand side means that the ratio of large to small displacements decreases. Both quantiles may increase in absolute distance. The lower one increases faster in *proportion*, so the

Table 3.2. Radial quantile slopes before and after fixing flights, their weights and time origins. Within each discipline both columns use the same lag set, common also to the intermediate controls and the component quantiles: 10 s to 10,000 s for paragliders and 10 s to 10,000 s for hang gliders. These are descriptive slopes, without calibrated confidence intervals; coverage of the target interval does not establish that a single power describes it.

Percentile	Paragliders		Hang gliders	
	Changing pool	Fixed population	Changing pool	Fixed population
25	0.853	0.979	0.869	0.995
50	0.845	0.893	0.838	0.888
75	0.856	0.887	0.827	0.851
90	0.875	0.892	0.846	0.864

distribution becomes less spread out by this relative measure. For example, a slope difference of -0.10 maintained over a tenfold increase of lag multiplies the ratio by $10^{-0.10} \simeq 0.79$.

A fitted slope requires a different reading from this local derivative. On a common lag set, ordinary least squares is linear in the logarithmic observations, so the fitted slope of $\log(Q_{0.90}/Q_{0.25})$ equals the difference between the two fitted quantile slopes. Its sign does not require the ratio to be monotone. Indeed, in the fully controlled curves the ratio rises from about 1.94 to 4.48 in paragliders and from 2.23 to 5.89 in hang gliders between 10 s and 80 s, then declines towards a minimum around 2,000 s. At 10,000 s it is about 3.03 and 2.65, respectively, above its initial value. The negative fitted slope therefore summarizes the long declining part after the early maximum; it is not evidence of monotonic narrowing over all three decades. The ratio curves convey this evolution more faithfully than a single number H_p .

This is not a statement that the left tail becomes flatter. Quantiles describe cumulative probability, whereas local flattening concerns a probability density and requires more information. Nor does the 90th percentile represent the whole distribution: ten per cent of increments exceed it. Finally, even with fixed flights and origins, the observations occupying a given rank can change; a small displacement at one lag is not being followed as an individual object.

If H_p changes with p in the controlled population, fitting a common H to collapse the curves hides some change of distribution shape over that interval. A nearly straight moment spectrum can coexist with this failure: moments integrate across the distribution and need not resolve changes at specific probability levels. Neither observation licenses calling the process exactly monofractal. The useful model target is the measured family $Q_p(\tau)$, including its departures from common scaling.

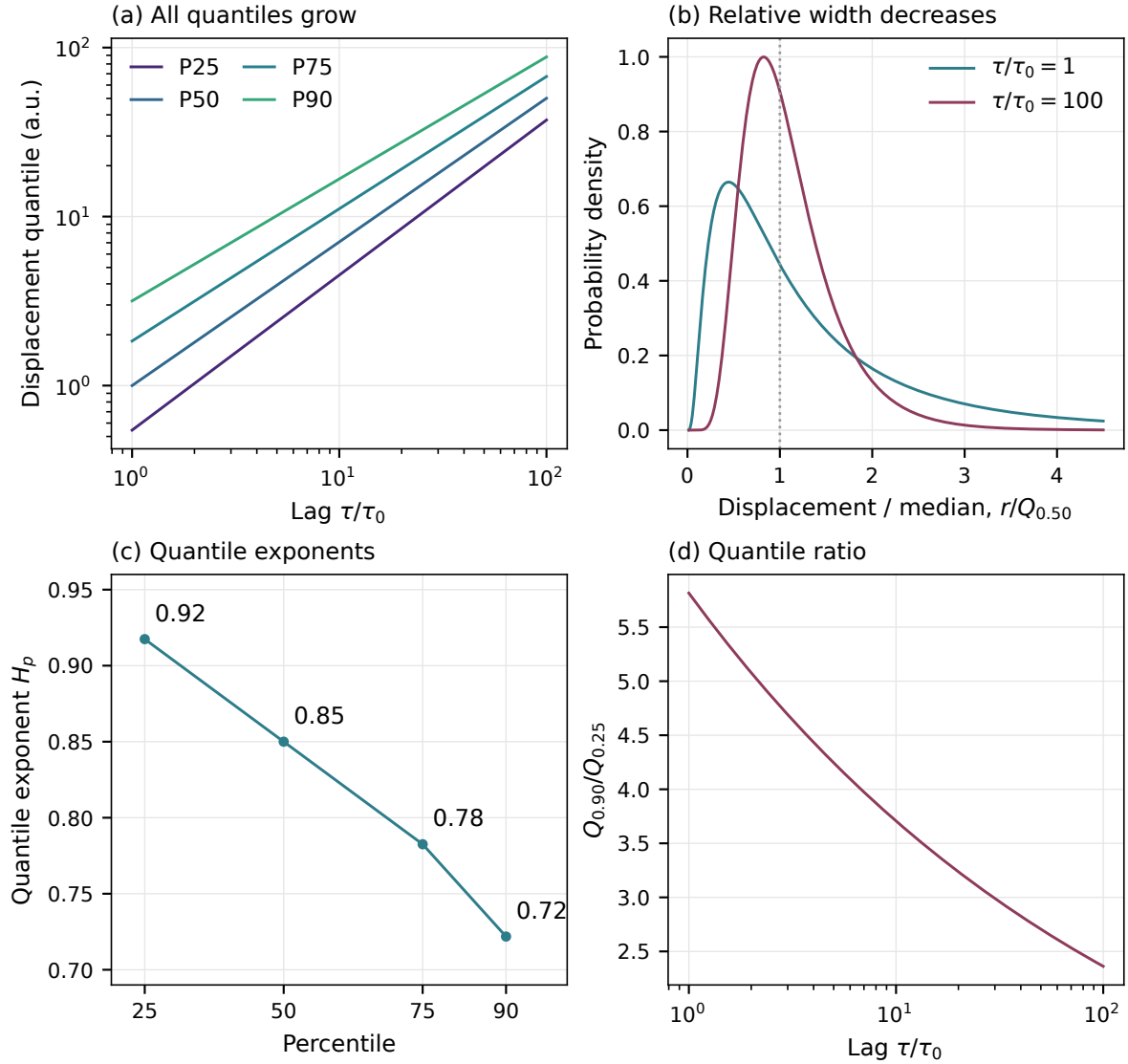


Figure 3.7. Analytical illustration of decreasing quantile exponents, not a fit to flight data. Let $R(\tau) = (\tau/\tau_0)^{0.85} \exp[\sigma(\tau)Z]$, where Z is standard Gaussian and $\sigma(\tau) = 0.9 - 0.1 \log(\tau/\tau_0)$ on the illustrated interval. Then $H_p = 0.85 - 0.1\Phi^{-1}(p)$. All four quantiles grow, but the distribution narrows after division by its median. The declining ratio makes that change explicit.

3.3 Anisotropy, dependence and the environment

3.3.1 East–north balance in the retained archive

The east–north second-moment ratio provides an initial check for directional imbalance in the retained archive. Isotropy would require the *joint* displacement distribution to be unchanged by every rotation; equality of the two marginal distributions alone is insufficient. A first necessary condition is equality of their second moments. For the positions measured from launch, the archive diagnostic is

$$\rho_{EN}(t) = \frac{\langle E_f(t)^2 \rangle_f}{\langle N_f(t)^2 \rangle_f}, \quad \langle E_f(t)^2 \rangle_f = \text{Var}_f[E_f(t)] + \langle E_f(t) \rangle_f^2. \quad (3.19)$$

Thus $\rho_{EN} = 1$ is necessary for isotropy about the launch point, but is not a full test of it. The ratio includes mean displacement as well as spread: it does not isolate anisotropy after drift

removal. Here t is elapsed time since launch, not a lag between two fixes within a flight, and the contributing population changes with t .

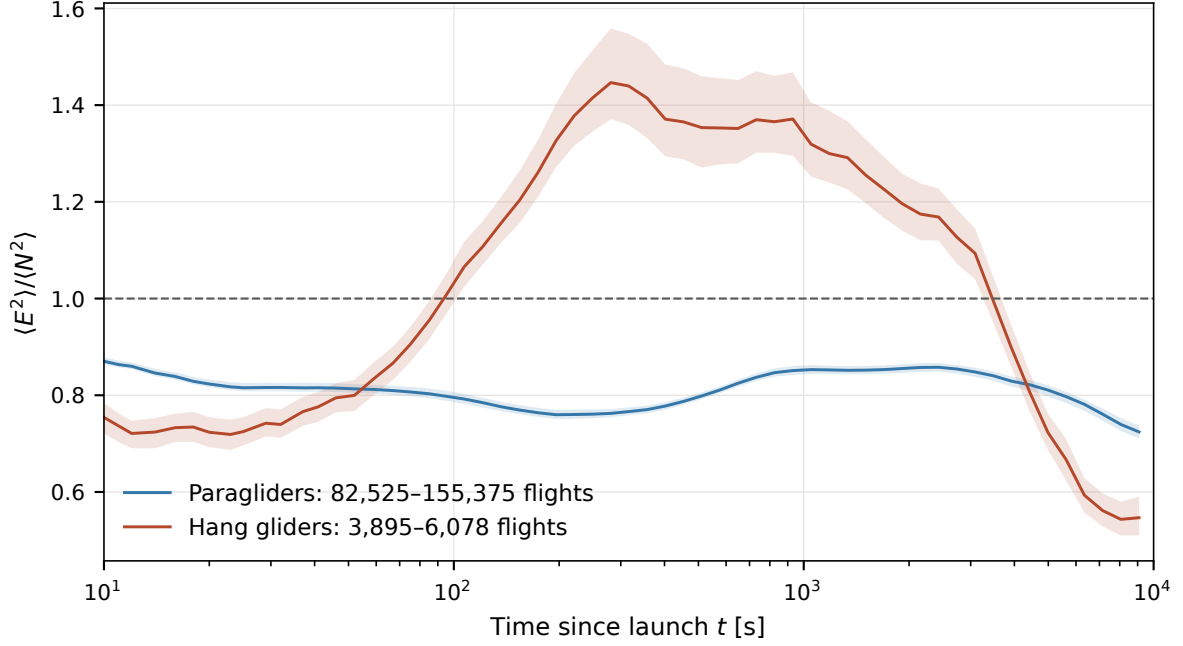


Figure 3.8. Archive east–north second-moment ratio against elapsed time since launch. Curves show the observed ratio; shading gives the pointwise 10–90% range from 200 bootstrap resamples of whole take-off-site-and-day groups. Only times with at least 30 paired flights and 10 contributing clusters are shown; legend counts give the minimum and maximum flight support along each curve. These are display minima, not a precision criterion. Unity is a necessary condition for isotropy, not sufficient evidence of it; these bands are not a simultaneous confidence statement across all times.

Over 10s to 10,000s, the unresampled ratio ranges from 0.72 to 0.87 for paragliders and from 0.54 to 1.45 for hang gliders. These are curve extrema, not bootstrap-band limits. The directional imbalance motivates keeping east, north and the radial displacement separate. It describes the heterogeneous retained archive; it does not identify whether route choice, terrain or wind produces the imbalance. The centred regional analysis below asks how much directional structure remains after subtracting each subgroup’s mean.

3.3.2 Why marginal scaling need not give radial scaling

Good collapses of $|X_E|$ and $|X_N|$ do not establish a common scaling law for the vector. Two failures are possible. First, the component exponents may differ. For centred independent Gaussian components with variances $a\tau^{2H_E}$ and $b\tau^{2H_N}$,

$$\mathbb{E}R^2 = a\tau^{2H_E} + b\tau^{2H_N}, \quad \frac{1}{2} \frac{d \log \mathbb{E}R^2}{d \log \tau} = \frac{H_E a \tau^{2H_E} + H_N b \tau^{2H_N}}{a \tau^{2H_E} + b \tau^{2H_N}}. \quad (3.20)$$

Each marginal has an exact exponent; the radial second moment has a changing effective exponent. This counterexample requires no cross-correlation.

Second, even equal marginal exponents leave the dependence between the components undetermined. The joint law of the rescaled pair may change with lag while both marginal laws remain fixed. A covariance or correlation coefficient detects only part of that change. Conversely, a common scaling law for the *joint* vector implies the same scaling for its modulus, even if the vector is strongly anisotropic. Anisotropy alone therefore does not explain a failed radial collapse.

3.3.3 What regional principal axes measure

For a region g , define the centred displacement covariance

$$\boldsymbol{\mu}_g(\tau) = \mathbb{E}[\mathbf{X} \mid g], \quad \Sigma_g(\tau) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_g)(\mathbf{X} - \boldsymbol{\mu}_g)^\top \mid g] = U_g \text{diag}(\lambda_1, \lambda_2) U_g^\top, \quad (3.21)$$

with $\lambda_1 \geq \lambda_2$. The ratio and whitening below require $\lambda_2 > 0$. The leading eigenvector describes the direction of greatest spread, modulo 180° ; it is an axis, not a signed wind direction. The ratio λ_1/λ_2 measures anisotropy of the covariance. An axis becomes poorly determined when the eigenvalues are close.

At 1,070 s, the Alpine subgroup contains 568 flights and has eigenvalue ratio 2.07, with principal axis 63.1° counterclockwise from east. The Pyrenean subgroup contains 61 flights, ratio 2.20 and axis 155.3° , closer to east–west. At long lag its ellipse becomes much more elongated, with reduced support; route geometry and selection both remain possible contributors. The coastal subgroup contains only 21 flights and has ratio 1.45: low relief is not a guarantee of an isotropic control.

Rotation into the principal frame gives uncorrelated components at the lag at which the covariance was fitted. It does not establish independence or remove temporal correlations. It cannot improve radial collapse because

$$|U_g^\top \mathbf{X}| = |\mathbf{X}|. \quad (3.22)$$

Whitening with the reference covariance, $\mathbf{Z}(\tau) = \Sigma_g(\tau_0)^{-1/2}[\mathbf{X}(\tau) - \boldsymbol{\mu}_g(\tau)]$, changes lengths. It sets the covariance to the identity at the reference lag τ_0 ; it need not do so at other lags, and it does not imply a rotationally invariant distribution. Whitening separately at each lag would enforce a part of the desired collapse by construction. A meaningful control instead fits a transformation at a fixed reference scale and tests it on other scales and flights.

3.3.4 Terrain and wind cannot be separated by differentiating position

The archive kinematic plots give a useful regional contrast. Over their displayed 10 s to 10,000 s interval, the Pyrenean east/north position second-moment ratio has median 1.67; its velocity ratio is 1.19. The Alpine ratios are 0.67 and 0.81, respectively. These are uncentred second moments, as in Eq. (3.19); velocity and acceleration have not had their subgroup means subtracted. The Pyrenean preference for east–west displacement is consistent with motion constrained by the massif. This is an association with geography, not an identification of its sole cause.

The handling demands associated with wing classification motivate an operational proxy for pilot experience. We call the EN A/B group *beginners* and the EN C/D/CCC group *experts*, using the harmonized catalogue classes described in Section 2.4. The split groups together wings with lower and higher experience requirements; it does not assign a measured skill score to each pilot. In particular, an experienced pilot may fly an A/B wing, so the beginners group can contain experienced pilots. The difference between the groups therefore combines pilot selection and equipment effects.

Figure 3.11 shows a clear regional qualification to the experience hypothesis. In the Alps, the experts' position and velocity ratios remain closer to unity across the displayed times. Their median position ratio is 0.70, compared with 0.61 for beginners. In the Pyrenees, experts show a smaller east–west imbalance at early elapsed times, but the position curves cross at a few hundred seconds; at long times both ratios approach five. The velocity ordering also changes. There is therefore no uniform reduction of directional imbalance among experts in that region. Along the Channel coast, expert ratios stay near unity, while the smaller beginners sample develops a larger imbalance and wider bands.

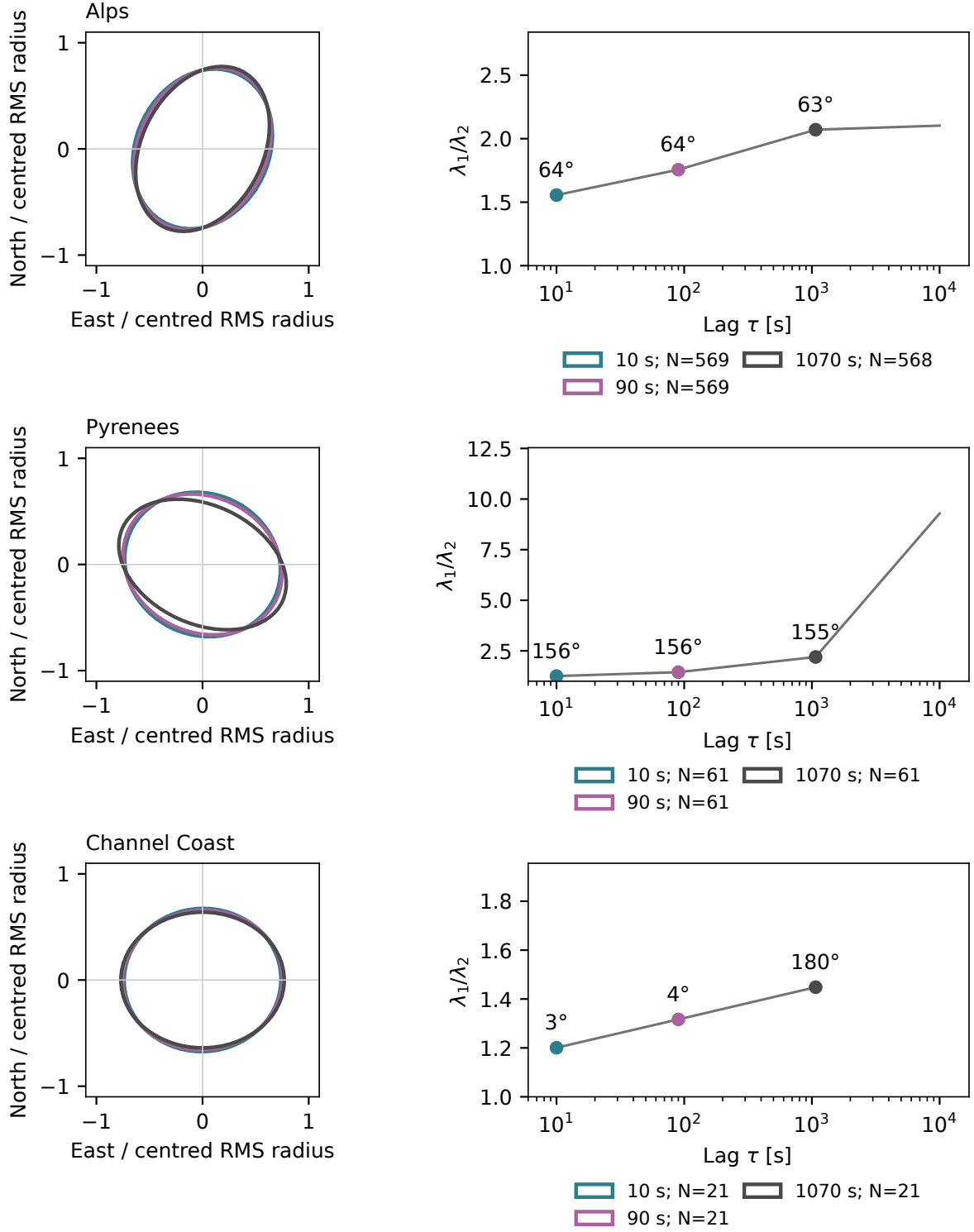


Figure 3.9. Regional principal-axis diagnostics in the paraglider diagnostic subset. Covariances are centred before diagonalisation. The principal direction is an axis modulo 180° and its interpretation depends on the separation of the two eigenvalues. Region counts and lag dependence delimit the evidence. These measurements describe displacement geometry and do not identify wind.

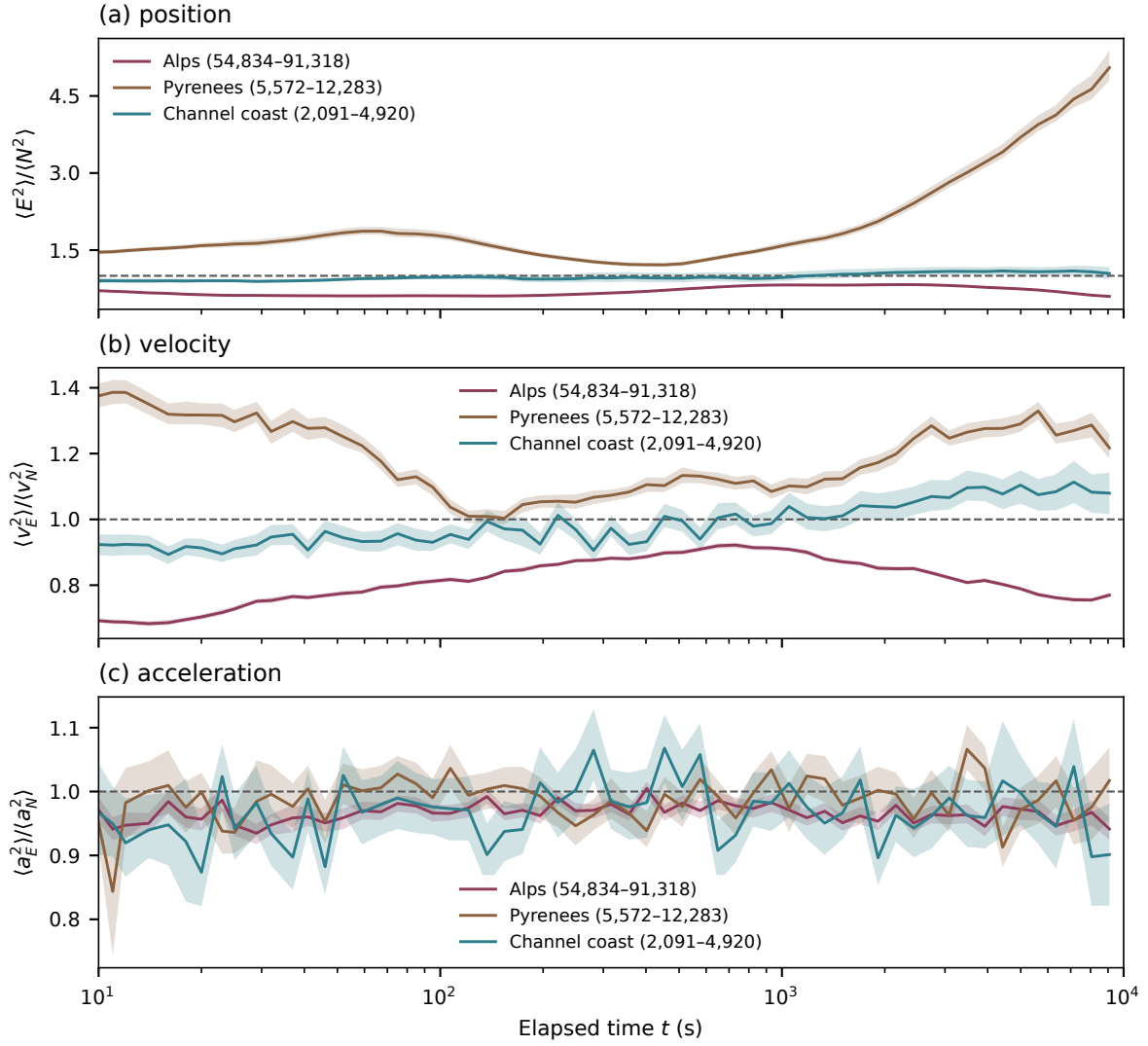


Figure 3.10. Paraglider kinematic anisotropy by take-off region. Position, velocity and acceleration compare different temporal aspects of the same motion, all at elapsed time since launch. Ratios are uncentred second moments, $\langle u_E^2 \rangle / \langle u_N^2 \rangle$, and include mean motion. Lines show the observed ratios and shading the pointwise 10–90% site-day bootstrap range. Both components use the same finite flight rows. Each displayed time has at least 30 flights and 10 clusters; legend counts show the range of contributing flights. Flights with a missing site or date form singleton groups, so unknown shared exposure is not captured by the bands. Panel limits follow the displayed quantity, with unity marked as a common reference; equality of the component moments leaves cross moments and higher-order angular structure untested. No scaling exponent is fitted here.

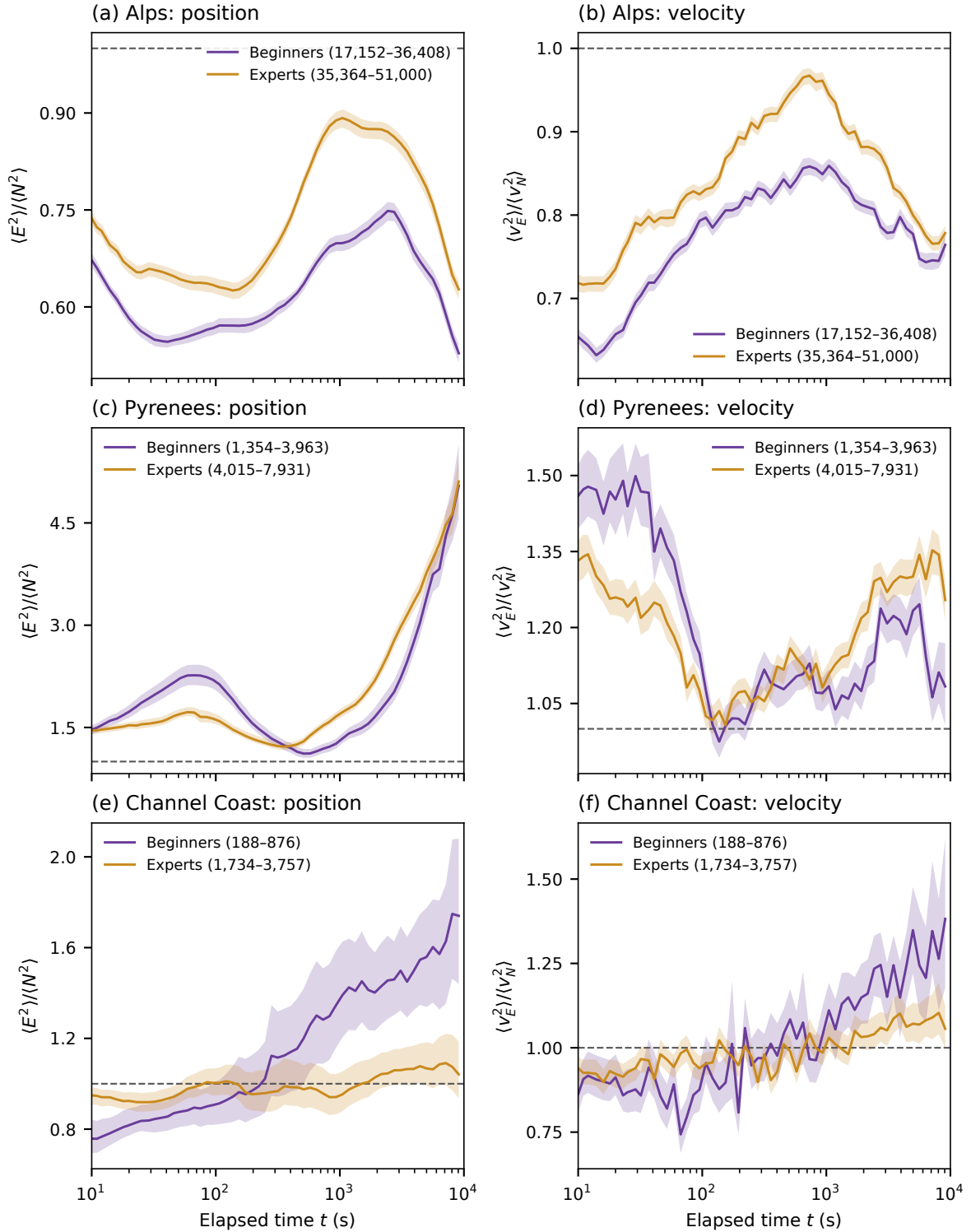


Figure 3.11. Paraglider second-moment ratios within each take-off region for two equipment-based experience groups: beginners (EN A/B, violet) and experts (EN C/D/CCC, ochre). Blue and terracotta are reserved for the two disciplines throughout, so an equipment split never wears them. These names refer to the experience proxy defined in the text. Tandem, non-certified and unclassified records are excluded from this contrast. Lines, pointwise bootstrap bands and coverage minima follow Figure 3.10; panel limits follow the observed range.

A ratio closer to unity describes a more balanced pair of second moments; cross moments and the angular distribution can still be anisotropic. The Alpine contrast is consistent with greater freedom in route choice, but season, launch site, flight duration and wing performance could also account for it. Attributing it specifically to pilot ability would require independent experience information and comparisons matched on flight conditions. In the Pyrenees, even the direction of the group contrast depends on elapsed time.

The kinematic relation

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h \quad (3.23)$$

shows the identifiability problem. Without independent information about one term, ground velocity alone cannot determine the other two. The mean velocity includes route choice, net progress and wind. Terrain affects velocity through constrained headings and pilot decisions, while wind affects position through its time integral. Acceleration removes a strictly constant wind contribution but still contains changing wind and manoeuvres. If the wind field is $\mathbf{w}_h(\mathbf{r}, z, t)$, its derivative along the trajectory is

$$\mathbf{a}_{\text{ground}} = \frac{d\mathbf{v}_{\text{air}}}{dt} + \partial_t \mathbf{w}_h + (\mathbf{v}_{\text{ground}} \cdot \nabla_h) \mathbf{w}_h + \dot{z} \partial_z \mathbf{w}_h. \quad (3.24)$$

Thus even a time-independent wind field contributes to acceleration when the glider crosses spatial gradients.

The defensible next comparison is conditional: principal axes and displacement statistics by region and site-day, then alignment with a terrain axis and an independently estimated wind vector. A thermal-centre drift estimate may help after segmentation, subject to assumptions about thermal motion; it is not an independent wind measurement by definition. Randomly rotating flights would create a rotationally balanced pooled sample while preserving each radial displacement. Such a control can diagnose geographical alignment, but cannot reconstruct the underlying motion in still air or without mountains.

3.4 Which model predictions are informative?

3.4.1 The moment spectrum and the Lévy-walk benchmark

For $q > 0$, define a moment and its growth exponent by

$$M_q(\tau) = \mathbb{E}[R(t, \tau)^q], \quad M_q(\tau) \propto \tau^{\zeta(q)}, \quad \nu(q) = \zeta(q)/q. \quad (3.25)$$

A common scaling law with finite moments gives $\zeta(q) = qH$. Increasing q gives large displacements more weight, so the spectrum tests a different aspect of the distribution from any one quantile. Plain increments are used here: second differences of a straight leg vanish and would change the physical observable being compared.

For an unbiased renewal Lévy walk with independent isotropically chosen headings, constant speed and leg-duration density $\psi(t) \sim t^{-1-\beta}$, the superdiffusive regime $1 < \beta < 2$ predicts

$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & 0 < q < \beta, \\ q + 1 - \beta, & q > \beta, \end{cases} \quad \zeta_{\text{LW}}(2) = 3 - \beta. \quad (3.26)$$

At $q = \beta$ a logarithmic correction accompanies the common power. The bulk and the rare long legs contribute different branches; their meeting produces strong anomalous scaling [4, 23]. This benchmark is an asymptotic model prediction with stated renewal and speed assumptions.

The empirical spectra in Figure 3.13 are compared with these predictions. For q from 0.25 to 4, $\nu(q)$ ranges from 0.849 to 0.878 in paragliders and from 0.839 to 0.861 in hang gliders. The pooled-increment values $\zeta(2) = 1.733$ and 1.693 differ from the equal-flight b_1 in Table 3.1

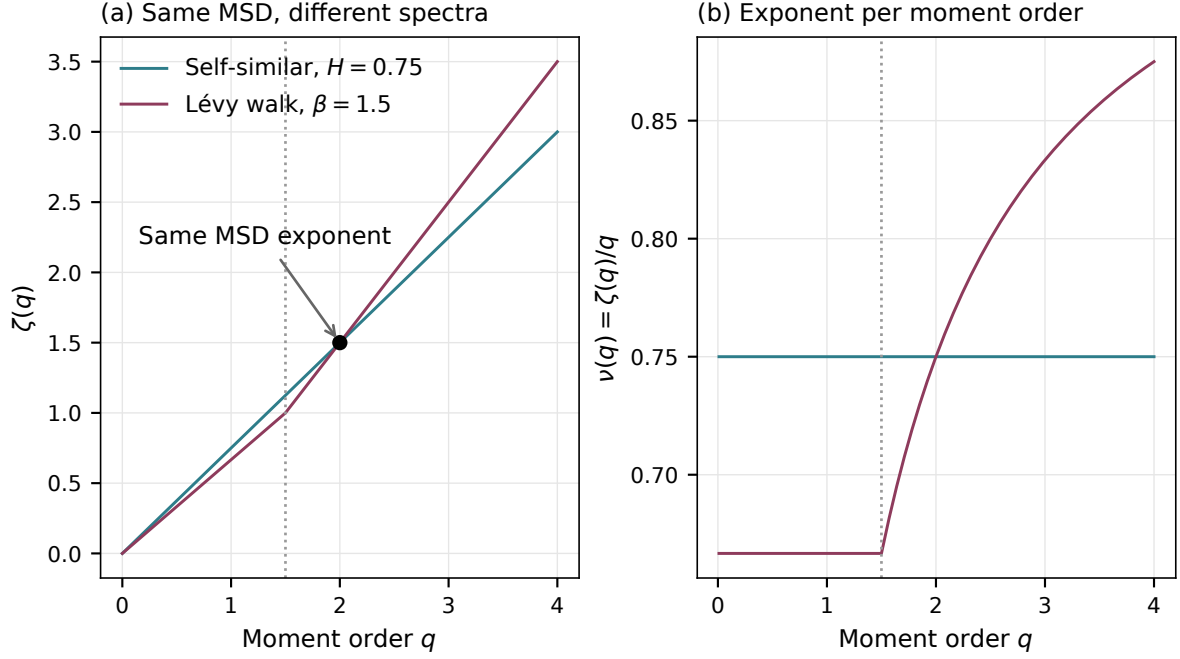


Figure 3.12. Theoretical comparison. A self-similar process with $H = 0.75$ and a standard Lévy walk with $\beta = 1.5$ have the same MSD exponent, $\zeta(2) = 1.5$, but different moment spectra. The dashed vertical line marks $q = \beta$. These are asymptotic predictions, not simulated or empirical estimates.

because the weights differ. The $\beta = 1.5$ curve in the figure is an illustrative reference, not a fitted null. A nearly linear spectrum without the predicted change of branch is evidence against this particular asymptotic description when the sampled scales and moments can resolve the change. It does not exclude all Lévy walks: the ballistic regime $0 < \beta < 1$, truncated leg distributions, correlations and pre-asymptotic behaviour require separate comparisons. Near-linearity must also be compared with the quantiles after controlling their sampling population; it does not independently establish a common scaling law.

A quantitative rejection requires simulating the specified candidate at fitted parameters with the observed durations, sampling and preprocessing, and comparing the *joint* moment and quantile results. The 90th percentile alone need not follow the ballistic front, whose probability changes with time. Its ordering is therefore not a second independent rejection test. High-order moments also require a flight-level sensitivity check before an inferential exclusion.

3.4.2 Non-Gaussianity: what is measured, and what it cannot identify

At each lag with a finite, positive-definite covariance, centre the increment and standardise by that full covariance:

$$Q = (\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}), \quad K_M(\tau) = \mathbb{E}[Q^2] - d(d+2), \quad d = 2. \quad (3.27)$$

This is Mardia's kurtosis excess [24]. A nonsingular Gaussian vector has $Q \sim \chi_d^2$, hence $K_M = 0$, including when it is anisotropic. Positive excess means a larger fourth moment of standardised radius than this Gaussian reference, although it does not by itself imply a power-law tail. Conversely, zero excess satisfies a necessary Gaussian moment condition without establishing Gaussianity. To call a measured difference from zero statistically significant also requires a reference calibrated for the finite sample and its dependence.

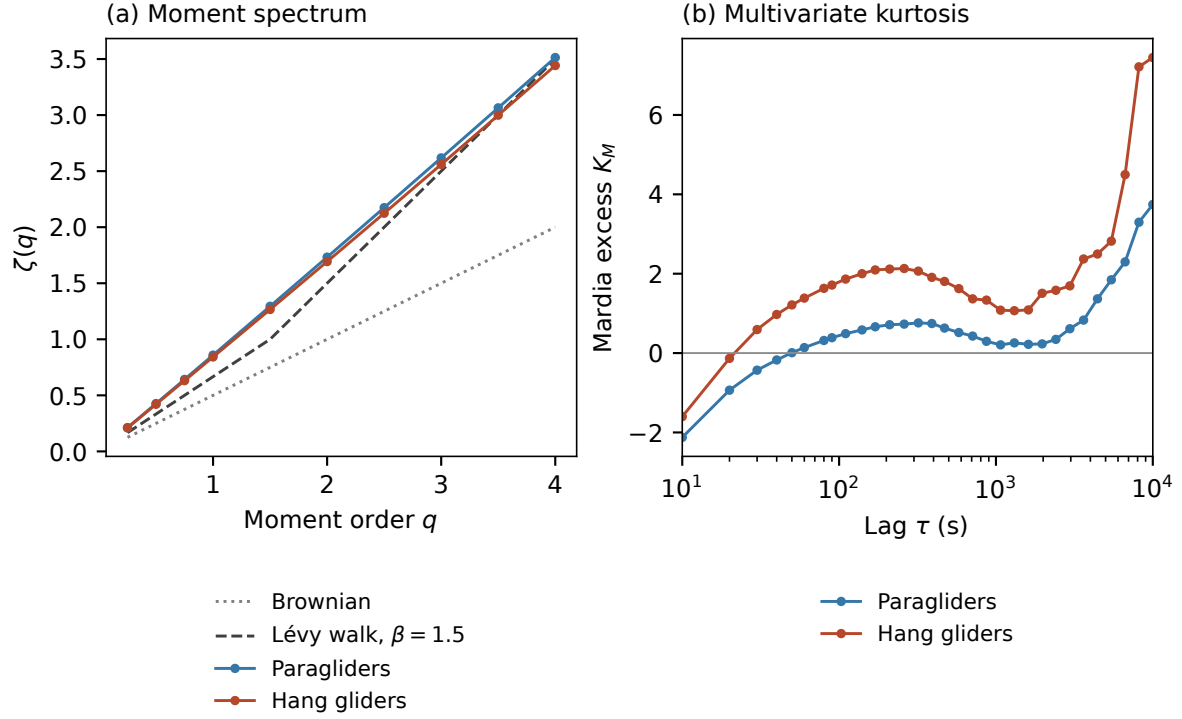


Figure 3.13. Displacement moment spectra and centred multivariate kurtosis in the diagnostic subset, with theoretical moment spectra shown as reference curves. These observables constrain the growth and shape of the increment distribution, respectively, but neither alone identifies the generating process.

For the n pooled increments at a lag, the calculation uses

$$\begin{aligned}\hat{\boldsymbol{\mu}} &= \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i, & \widehat{K}_M &= \frac{1}{n} \sum_{i=1}^n \widehat{Q}_i^2 - 8, \\ \widehat{\boldsymbol{\Sigma}} &= \frac{1}{n} \sum_{i=1}^n (\mathbf{X}_i - \hat{\boldsymbol{\mu}})(\mathbf{X}_i - \hat{\boldsymbol{\mu}})^\top.\end{aligned}\tag{3.28}$$

where $\widehat{Q}_i$ substitutes these estimates into Eq. (3.27). With this covariance denominator, $n^{-1} \sum_i \widehat{Q}_i = 2$ and Jensen's inequality gives $\widehat{K}_M \geq -4$. Negative values are therefore possible: they indicate a smaller fourth moment of standardised radius than the Gaussian reference, not a negative variance. Even a Gaussian sample need not give exactly zero.

The measured K_M is not constant. Across the displayed interval its range is -2.12 to 3.74 for paragliders and -1.60 to 7.45 for hang gliders. The medians over sampled lags are 0.49 and 1.69, respectively; these lag medians are descriptive summaries, not time averages. The change of sign near the shortest lags and the rise where support is smallest discourage replacing the curve with one claim about a tail.

Excess can also arise from heterogeneity even when the conditional motion is Gaussian. For example, let $\mathbf{X} = \sqrt{A} \mathbf{G}$, where $\mathbf{G}$ is a centred Gaussian vector with fixed covariance, $A > 0$ is independent, and both required moments exist. Then

$$K_M = d(d+2) \left[\frac{\mathbb{E}A^2}{(\mathbb{E}A)^2} - 1 \right].\tag{3.29}$$

This is a scale-mixture example, not an identity for arbitrary flights with different means and differently oriented covariance matrices. Subtracting a trace-based width mixture estimate from pooled kurtosis does not recover each flight's kurtosis.

The measured excess therefore tests a *single homogeneous Gaussian law for the pooled increments*. To assess Gaussian fluctuations conditional on environment or phase requires separate centring and covariance estimates in those strata, adequate sample sizes and a matched simulation. The peak of a pooled kurtosis curve may reflect changing mixtures or geometry; it is not automatically a thermal timescale or evidence of central-limit behaviour.

3.4.3 Velocity memory at more than one resolution

A fix-to-fix velocity mixes recorder cadence, smoothing and physical turning. To examine longer scales, define velocity averaged over a duration h ,

$$\mathbf{v}_h(t) = \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}, \quad C_h(\tau) = \frac{\langle \delta \mathbf{v}_h(t) \cdot \delta \mathbf{v}_h(t+\tau) \rangle}{\langle |\delta \mathbf{v}_h|^2 \rangle}. \quad (3.30)$$

The analysis compares $h = 10, 60$ and 300 s. In each selected segment, it forms adjacent block velocities $\mathbf{v}_{f,j}$, subtracts their mean and computes

$$\hat{C}_{f,h}(kh) = \frac{(n_f - k)^{-1} \sum_{j=0}^{n_f-k-1} \delta \mathbf{v}_{f,j} \cdot \delta \mathbf{v}_{f,j+k}}{n_f^{-1} \sum_{j=0}^{n_f-1} |\delta \mathbf{v}_{f,j}|^2}, \quad \hat{C}_h = \frac{1}{N_h(k)} \sum_{f \in \mathcal{F}_h(k)} \hat{C}_{f,h}. \quad (3.31)$$

Here n_f is the number of complete velocity blocks and $\mathcal{F}_h(k)$ contains segments with $n_f \geq 8$, nonzero velocity variance and $k \leq \lfloor n_f/4 \rfloor$. A mean of normalised correlations gives each contributing flight equal weight; it differs from normalising a covariance pooled across all flights. Lags smaller than h share averaging intervals and can generate correlation from overlap alone; interpretation uses non-overlapping intervals. Correlations at different h answer different questions and need not coincide. Subtracting a mean estimated on a finite segment can also depress the long-lag correlation.

Figure 3.14 places the positive correlations on logarithmic axes, where a power-law interval would satisfy

$$C_h(\tau) \simeq A_h \tau^{-\gamma_h} \implies \log C_h(\tau) \simeq \log A_h - \gamma_h \log \tau. \quad (3.32)$$

Curvature therefore matters as much as the apparent slope. The companion panels retain the sign, since a negative correlation cannot be represented on a logarithmic ordinate and taking its absolute value would change the physical question. The plotted quantity is the mean of normalised flight correlations in Eq. (3.31), rather than a pooled velocity autocovariance. Within one flight, normalisation by the zero-lag covariance leaves a positive power-law exponent unchanged; after averaging heterogeneous flights, the weighting still matters.

At a separation of 600 s, the three resolutions give correlations 0.144, 0.198 and 0.272 for paragliders; the hang-glider values are 0.130, 0.186 and 0.265. The sample means remain positive beyond individual fixes and minute-scale averaging, without a calibrated uncertainty statement at this separation. Their log-log curves bend, and the hang-glider estimate becomes negative at long separation; neither figure supports assigning one power-law decay across the full interval. A larger normalised correlation after averaging also reflects the suppression of faster fluctuations, not necessarily an increase in the underlying memory time.

For a stationary centred velocity with variance $\sigma_v^2 = \langle |\delta \mathbf{v}|^2 \rangle$, integrating velocity over the displacement interval and using stationarity gives Taylor's relation [25],

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s) C(s) ds. \quad (3.33)$$

The velocity-variance factor is required because C is dimensionless. This relation concerns the centred displacement and the same stationary population; it cannot directly identify the growth of an uncentred, heterogeneous archive MSD.

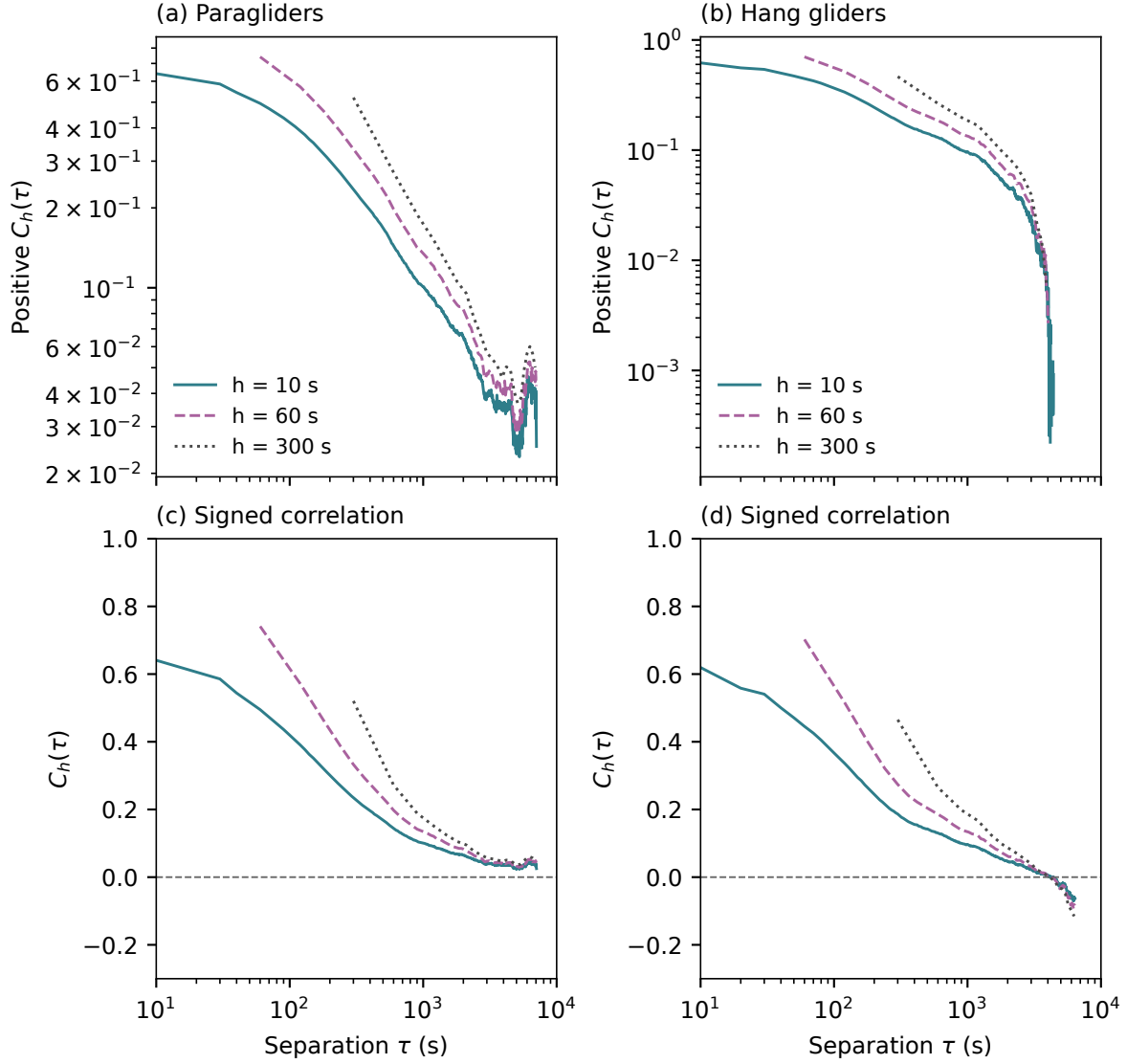


Figure 3.14. Velocity memory at averaging widths $h = 10, 60$ and 300 s. The upper panels use log–log axes for positive correlations; nonpositive values interrupt the curves. The lower panels show the same estimates with a linear vertical axis, preserving zero crossings and anticorrelation. At least 20 contributing flights are required, with separation no shorter than h and no longer than one quarter of each contributing velocity record. A visually straight part of the upper curves is a possible scaling interval, not evidence of an asymptotic law.

For example, an exponentially correlated velocity, $C(s) = e^{-s/t_p}$, gives

$$M_{2,c}(\tau) = 2\sigma_v^2 t_p \left[\tau - t_p(1 - e^{-\tau/t_p}) \right]. \quad (3.34)$$

It is ballistic for $\tau \ll t_p$ and diffusive for $\tau \gg t_p$. An intermediate superdiffusive slope can therefore arise in a finite-persistence model. A distribution of t_p across flights broadens the crossover. Such a model is not excluded merely because a fitted MSD slope exceeds one.

If instead $C(s) \sim s^{-\gamma}$ with $0 < \gamma < 1$ *asymptotically*, Eq. (3.33) gives $M_{2,c} \propto \tau^{2-\gamma}$. Measurements on a finite interval cannot establish convergence or divergence of the integral to infinity. Consequently the measured memory is reported at the observed scales, without declaring it non-integrable. Its persistence also does not prove correlation between successive glides: that comparison requires the episode boundaries of Chapter 4.

3.4.4 A model comparison with explicit limits

Table 3.3 collects the useful exclusions and the hypotheses that remain open. The scope is the pooled data over the observed interval. A candidate can fail there and still describe a restricted phase, environment or limiting time regime.

Table 3.3. Model predictions and the scope of the present comparison. “Challenged” means that a displayed diagnostic disagrees descriptively with the simple prediction; it does not stand for a calibrated statistical rejection.

Candidate	Prediction to confront	Present scope
Homogeneous Brownian motion	$\zeta(q) = q/2$, Gaussian increments, independent disjoint increments	Challenged as a description across the measured interval by displacement growth, shape and memory. A long-time diffusive limit remains untested.
Constant drift plus Brownian motion	$M_2 = \mathbf{v}_d ^2 \tau^2 + 4D\tau$ in 2D; drift-free $V_2 \propto \tau$	A control for apparent superdiffusion. Log-log curvature in V_2 or non-Gaussian residuals require more than a fitted mean velocity; environmental mixtures remain possible.
Homogeneous fractional Brownian motion	Gaussian joint law, common $H_p = H$, $\zeta(q) = qH$	Challenged as a single pooled law when quantile slopes differ and centred kurtosis is nonzero. Conditional Gaussian models are not thereby excluded.
Exponential velocity persistence	Eq. (3.34): ballistic–diffusive crossover	Remains a candidate to test jointly against MSD and coarse velocity correlations; a broad distribution of persistence times is a separate hypothesis.
Standard renewal Lévy walk, $1 < \beta < 2$	Two-branch spectrum in Eq. (3.26)	The missing expected branch change challenges this benchmark. Finite-record simulations are needed before claiming exclusion; generalisations are not covered.
Unbounded Lévy flight	Instantaneous jumps; divergent second moment for stable index below two	Unsuitable as a literal finite-speed flight mechanism. Truncated or effective coarse models require their own finite-scale predictions.

An uncoupled renewal model with long resting times is also not established simply by naming climb a waiting phase: a glider moves while climbing. Tests of waiting times, renewal independence and coupling to the next displacement belong to the segmented analysis, where a renewal hypothesis can be confronted with the observed episode durations and their dependence.

3.5 Flight duration and equipment composition

Longer flights may select different equipment, pilots and conditions. A change in their MSD can therefore reflect a change of population rather than an effect of observation length. We examine both duration thresholds and their equipment composition, using the identified segment curves from the full archive: 154,984 paraglider and 6,059 hang-glider flights have segments of at

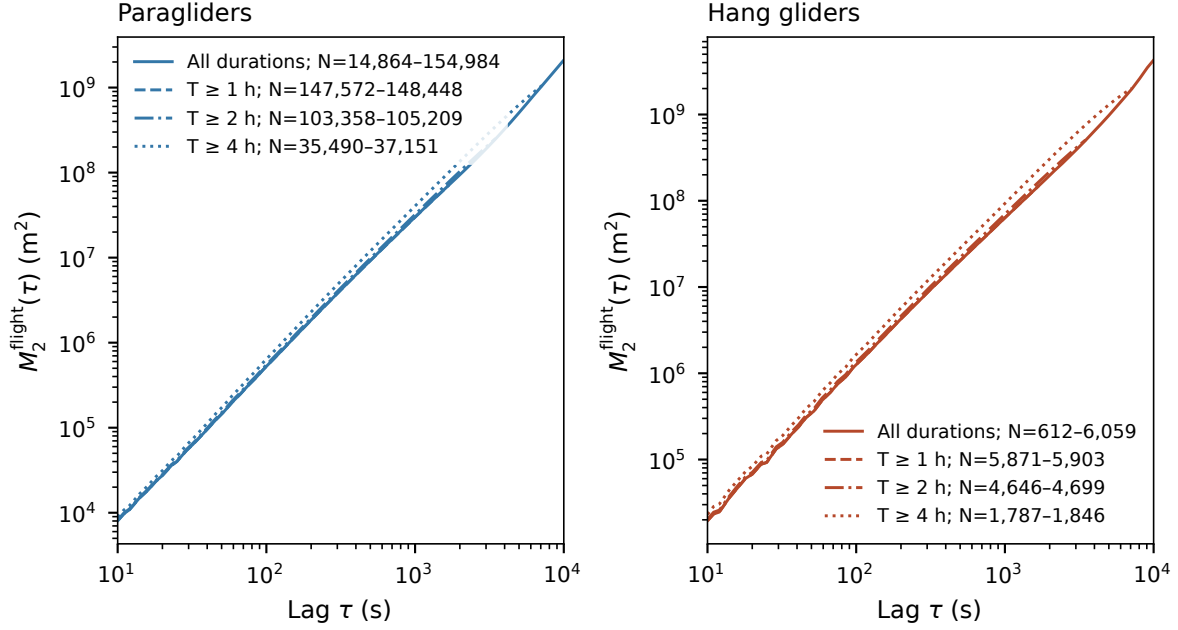


Figure 3.15. Time-averaged squared displacement grouped by retained flight duration. The groups are nested; legend counts give the range of distinct contributing flights over each displayed curve. At least 30 flights are required per lag. Each threshold curve stops at $\tau = T_0/2$, while the all-duration curve extends to 10,000 s. As lag approaches the threshold’s half-duration, shorter flights progressively disappear from the reference too, bringing the populations closer. For uninterrupted records, no flight shorter than T_0 can contribute beyond $T_0/2$; segmentation further restricts support. Convergence there is a selection effect, not evidence of duration independence.

least eight fixes with native cadence at most 10 s. Here T_f is the sum of the retained segment spans of flight f ; gaps and discarded time are not part of this observed duration.

First combine the available segment curves within each flight, weighting them by the number of admissible time origins, then give every contributing flight equal weight:

$$m_f(\tau) = \frac{\sum_s n_{fs}(\tau) m_{fs}(\tau)}{\sum_s n_{fs}(\tau)}, \quad M_{T_0}(\tau) = \frac{1}{N_{T_0}(\tau)} \sum_{f: T_f \geq T_0, n_f(\tau) > 0} m_f(\tau). \quad (3.35)$$

For an eligible segment of N_{fs} fixes at interval δ_{fs} , $n_{fs} = N_{fs} - \text{round}(\tau/\delta_{fs})$; it is zero outside that segment’s stored lag support, which ends at $\lfloor N_{fs}/2 \rfloor \delta_{fs}$. No displacement crosses a segment boundary. This equal-flight average differs from the equal-segment archive average introduced earlier. Requested lags are rounded to each segment’s native grid.

For paragliders the comparison is repeated within the beginners (EN A/B) and experts (EN C/D/CCC) groups defined above (Fig. 3.16). This asks whether the duration contrast remains within each experience proxy, and whether the two groups have different growth rates. Longer flights need not contain the same proportions of beginners and experts, so their pooled MSD can change even when neither group’s conditional dynamics depends on duration.

Without duration conditioning, the descriptive slopes b of $M_2 \propto \tau^b$ over 10 s to 10,000 s are 1.728 for beginners and 1.788 for experts. For a direct comparison of all eight group–duration cells, Table 3.4 uses the shorter common interval 10 s to 1,695 s. Its upper limit cannot exceed half the one-hour threshold. This secondary comparison does not replace the main 10 s to 10,000 s analysis. A straight-line fit summarizes growth over its stated interval; its residual quantifies curvature or departures from that fit, and is not a sampling confidence interval or a test of a Hurst exponent.

On the common interval, the expert slope exceeds the beginner slope in every duration cell.

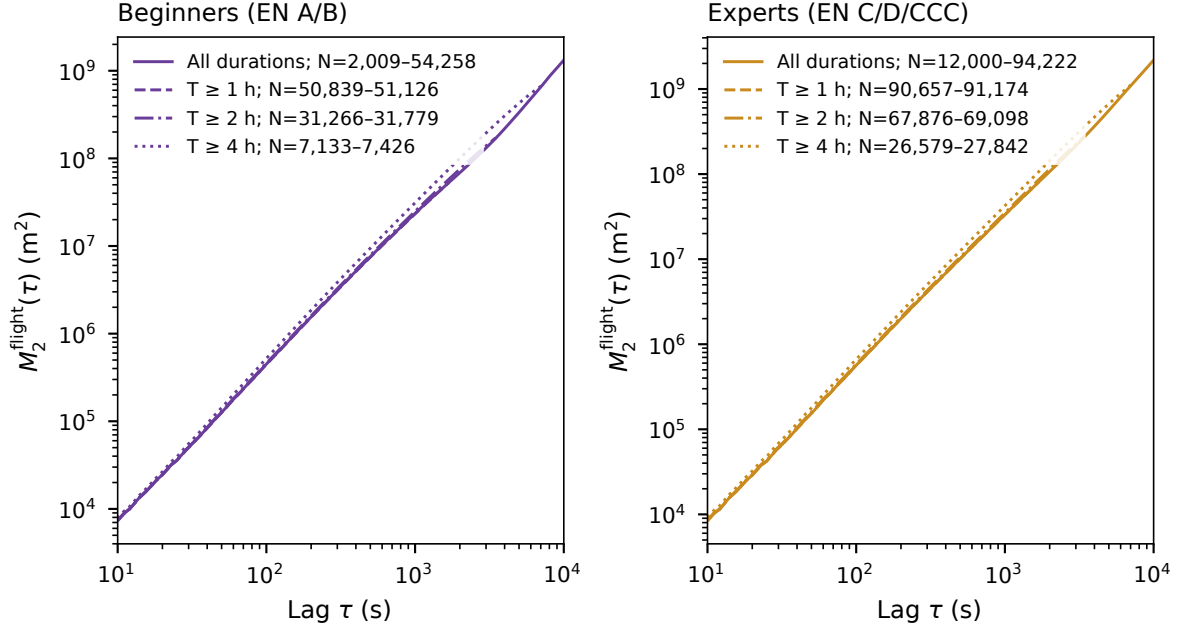


Figure 3.16. The duration comparison repeated for beginners (EN A/B) and experts (EN C/D/CCC), using the same estimator, lag limits and minimum support as Fig. 3.15. Counts refer to distinct flights, and overlapping groups do not constitute independent samples. Tandem, non-certified and unknown equipment is omitted from this contrast.

Table 3.4. Duration and equipment contrasts on a common supported lag interval, 10 s to 1,695 s. Flight counts include every eligible flight in the cell; the contributing count decreases with lag. The residual is the root-mean-square deviation in $\log_{10} M_2$ from the fitted line.

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,258	1.755	0.022
Beginners	$T \geq 1$ h	51,126	1.754	0.022
Beginners	$T \geq 2$ h	31,779	1.767	0.020
Beginners	$T \geq 4$ h	7,426	1.806	0.017
Experts	All	94,222	1.803	0.017
Experts	$T \geq 1$ h	91,174	1.802	0.017
Experts	$T \geq 2$ h	69,098	1.811	0.016
Experts	$T \geq 4$ h	27,842	1.836	0.015

Within beginners it increases from about 1.755 for all durations to 1.806 for $T \geq 4$ h; within experts the corresponding values are 1.803 and 1.836. Thus both an equipment contrast and a within-class duration contrast are present descriptively. They do not measure a causal effect of pilot experience, and the nested samples cannot be treated as independent comparisons.

To separate within-class behaviour from class proportions, write the exact mixture identity over the two experience proxies,

$$M_{T_0}^{\text{known}}(\tau) = \sum_c p_c(\tau | T_0) M_{T_0,c}(\tau), \quad p_c(\tau | T_0) = \frac{N_{T_0,c}(\tau)}{\sum_j N_{T_0,j}(\tau)}. \quad (3.36)$$

The weights depend on lag as well as duration. A standardized comparison instead holds them fixed at $\pi_c = N_{0,c} / \sum_j N_{0,j}$, the class proportions among all eligible known-equipment flights:

$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau). \quad (3.37)$$

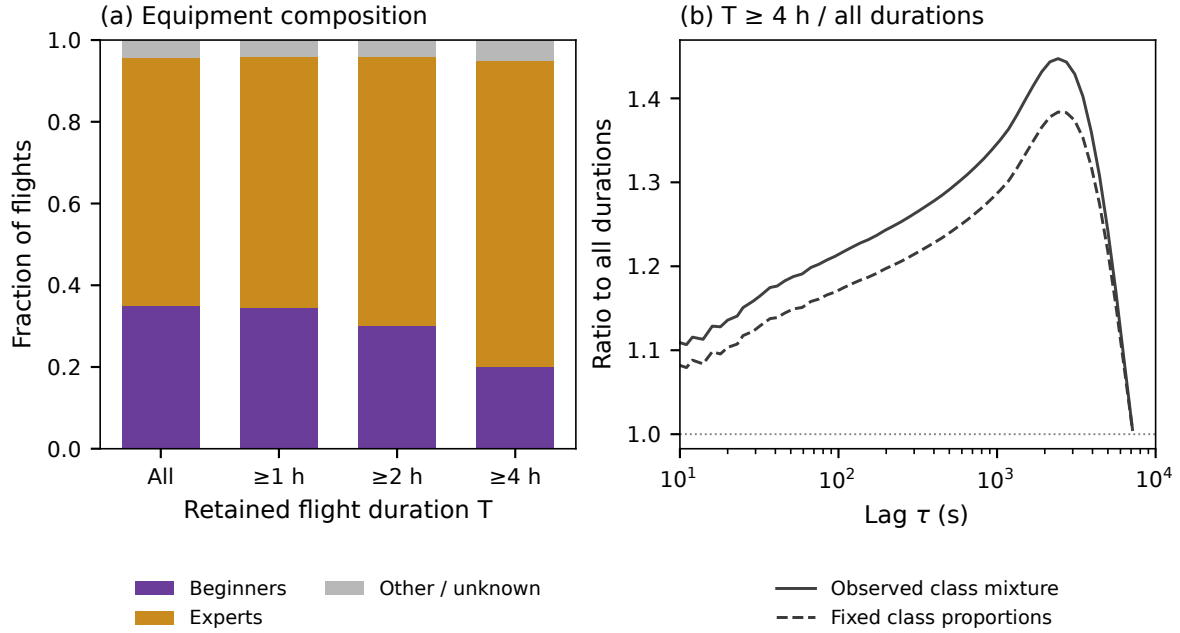


Figure 3.17. Equipment composition and its contribution to the duration comparison. Left: proportions of flights in the nested duration cohorts, including other or unknown equipment (6,504 flights in the full eligible sample). Right: the four-hour/all-duration MSD ratio among known classes, using the observed lag-dependent mixture or fixed class weights. Unity denotes equal MSD amplitudes at that lag, not equal trajectories or established statistical independence.

Figure 3.17 compares the four-hour/all-duration ratio before and after this adjustment. Both ratios exclude the same unknown or other equipment; otherwise changing the population would be confounded with changing the weights. The adjustment is displayed only where every class has at least 30 contributing flights.

Among flights in the two known classes, the expert share increases from about 63.5% for all durations to 78.9% for $T \geq 4$ h; the latter cohort still contains 7,426 beginner-proxy flights, so it is not made up of experts alone. Standardisation reduces the four-hour/all-duration MSD ratio at 576 s from about 1.30 to 1.25, and at 1,695 s from 1.42 to 1.35. The two-class composition therefore accounts for part of the amplitude contrast under this descriptive adjustment, while a substantial within-class contrast remains. The return towards unity near 7,200 s coincides with the loss of shorter flights from the reference; it cannot demonstrate duration independence.

The MSD therefore varies with both duration and equipment in this archive. These associations do not isolate pilot skill or the effect of duration itself, because conditions, routes and pilot selection can still differ within each class.

3.6 Chapter summary and constraints for the phase analysis

Over the measured interval, displacement grows faster than the linear MSD of homogeneous Brownian motion, but this alone does not distinguish persistent motion from longer-range correlations. The constant-velocity control retains curvature and effective slopes near two, so it has not delivered a well-defined intrinsic Hurst exponent. Fixing the quantile population exposes changes in relative spread that a single fitted exponent obscures, while the regional analysis reveals a pronounced Pyrenean east–west preference and no universal reduction of anisotropy among experts. Duration contrasts persist within both equipment groups and after fixing their proportions. These results constrain a model jointly; none identifies the physical process in isolation.

The later model comparison must reproduce the displacement moments and quantiles over stated lags, changes in relative distribution width, anisotropy by region, coarse velocity memory, and differences associated with open and closed tasks. The second difference is retained as a control with a precise invariance: it cancels a constant velocity. It does not identify that velocity as wind or reconstruct an environmentally neutral flight.

The next chapter introduces phase labels. A subsequent analysis must measure durations, displacements, altitude changes and dependencies within and between episodes. There is an exact bookkeeping constraint on that work. Let c partition *all* admissible windows into phase histories, including windows that cross a boundary. Then

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau | c), \quad w_c = \frac{n_c}{\sum_{c'} n_{c'}}, \quad \sum_c w_c = 1, \quad (3.38)$$

for a window-weighted observable. A flight-weighted version first applies the identity within each flight. Restricting to windows wholly inside one of three phases omits transition windows and will not, in general, reconstruct the global moment. The exponents are not weighted sums of phase exponents because both amplitudes and weights depend on lag.

Any exhaustive partition satisfies this identity, including one with incorrect labels. Reconstructing the global moments therefore checks the calculation; independent labels are still needed to assess the phase assignment.

Chapter 4

Flight phases

The global observables of Chapter 3 constrain the motion of a complete flight, but because they mix directed travel, searching and altitude recovery, they do not identify which behaviour contributes to a particular scaling law. Segmentation provides the missing conditional description: given a phase, how does the glider move, how long does that phase last, and what happens next?

The aim here is to assign these phases reproducibly and establish how the labels can be checked, bearing in mind that kinematics provide indirect evidence of behaviour: a coloured climb on a trajectory remains an inference. We first build the probabilistic model and define its measured features, then examine its numerical behaviour and the independent annotations needed to assess its accuracy.

4.1 What the three phases mean

We retain the three-state description used by Vilpellet et al. [2], with labels $\mathcal{S} = \{\text{T}, \text{S}, \text{C}\}$:

Transition (T).

Sustained directed travel between lift regions, usually with limited turning and altitude loss.

Search (S).

Exploratory turns, reversals or recentering without a sustained established climb. Search can be brief or absent between a transition and a climb.

Climb (C).

Sustained exploitation of lift, usually combining positive mean vertical speed and circling.

These definitions concern observable flight geometry rather than a pilot's unrecorded intention, and allow departures from a fixed cycle: a glider may encounter lift directly, leave an unsuccessful search, or interrupt a climb. Ridge soaring and straight flight in rising air may also be ambiguous within this three-state vocabulary. Such cases should remain visible during validation instead of being forced to confirm the proposed interpretation.

4.2 From a Markov chain to hidden-state inference

The phase of a flight is not observed directly: a recorder measures motion, and different behaviours can produce overlapping ranges of speed and turning. A hidden Markov model represents this distinction through two coupled sequences. The latent sequence Q_k describes the state at decision time k , while the observed vector $\mathbf{Y}_k$ carries the measurements associated with



Figure 4.1. Schematic plan views of the three phase interpretations. The drawings are illustrations, not measured trajectories or decision rules. Climbing circles can drift, and searching can cover substantial ground; neither phase is assumed stationary.

that state. The construction and inference below follow the standard HMM formulation [26, Secs. II–III]; its use for behavioural interpretation requires additional care, as discussed by McClintock et al. [27].

4.2.1 Constructing the probability law

Begin with a chain on a finite state space $\mathcal{S}$. Its initial distribution and transition probabilities satisfy

$$\begin{aligned} \Pr(Q_1 = i) &= \pi_i, \quad \sum_i \pi_i = 1, \\ \Pr(Q_{k+1} = j \mid Q_{1:k-1}, Q_k = i) &= a_{ij}, \quad \sum_j a_{ij} = 1. \end{aligned} \quad (4.1)$$

Here $a_{ij} = \Pr(Q_{k+1} = j \mid Q_k = i)$ and all probabilities are nonnegative. The Markov assumption concerns the hidden state: its present value contains the information about its history used to predict the next state. A constant matrix A adds time homogeneity; it is a modelling choice, not a consequence of the observations being sampled regularly.

Now attach an emission density $b_i(\mathbf{y})$ to each state, with $\int b_i(\mathbf{y}) d\mathbf{y} = 1$. Conditional on the complete state sequence, the emissions are assumed independent and the density at time k depends only on Q_k . The chain rule then gives

$$p(q_{1:K}, \mathbf{y}_{1:K} \mid \theta) = \pi_{q_1} b_{q_1}(\mathbf{y}_1) \prod_{k=1}^{K-1} a_{q_k q_{k+1}} b_{q_{k+1}}(\mathbf{y}_{k+1}), \quad (4.2)$$

where θ collects π , A and the emission parameters. This is also a generative recipe: draw the first state, generate its observation, then alternate a state transition with a new observation. Although emissions are independent conditional on the states, the observed sequence is generally correlated and need not itself be first-order Markov.

For continuous features, a common emission model is a multivariate Gaussian,

$$b_i(\mathbf{y}) = \frac{\exp[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu}_i)^\top \Sigma_i^{-1}(\mathbf{y} - \boldsymbol{\mu}_i)]}{(2\pi)^{d/2} |\Sigma_i|^{1/2}}, \quad \Sigma_i \succ 0. \quad (4.3)$$

Its full covariance describes dependence among simultaneous features within a state. Gaussianity is an additional assumption about the emission shape; it is not part of the definition of an HMM. Nor does a state number determine a behavioural name: permuting all state labels consistently leaves the observation law unchanged.

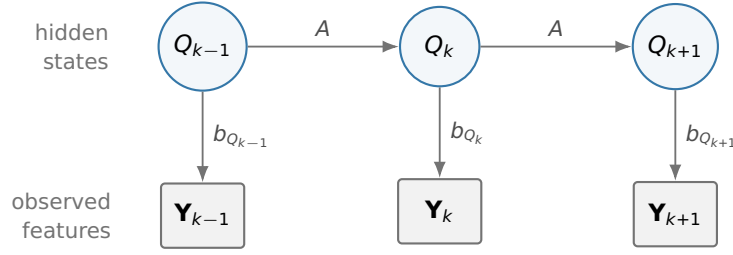


Figure 4.2. Conditional structure of an HMM. Horizontal arrows carry state persistence and transitions; vertical arrows describe how a state generates a measurement. Absence of an arrow between observations represents conditional independence given the hidden sequence, not independence of the observed features over time.

4.2.2 Likelihood and uncertainty about the states

Three tasks must be distinguished: evaluating how well given parameters explain an observed sequence, inferring its hidden states, and estimating the parameters themselves. For the first, the states must be summed out,

$$\mathcal{L}(\theta) = p(\mathbf{y}_{1:K} \mid \theta) = \sum_{q_{1:K} \in \mathcal{S}^K} p(q_{1:K}, \mathbf{y}_{1:K} \mid \theta). \quad (4.4)$$

With continuous observations this is a density evaluated at the recorded vectors, rather than the probability of those exact vectors. Direct evaluation would enumerate $|\mathcal{S}|^K$ paths, but the Markov factorization allows their shared prefixes and suffixes to be accumulated recursively [26, Sec. III-A].

For one valid block of K observations, define forward and backward quantities

$$\alpha_1(i) = \pi_i b_i(\mathbf{y}_1), \quad \alpha_{k+1}(j) = b_j(\mathbf{y}_{k+1}) \sum_i \alpha_k(i) a_{ij}, \quad \mathcal{L} = \sum_i \alpha_K(i), \quad (4.5)$$

$$\beta_K(i) = 1, \quad \beta_k(i) = \sum_j a_{ij} b_j(\mathbf{y}_{k+1}) \beta_{k+1}(j). \quad (4.6)$$

Their cost is $O(K|\mathcal{S}|^2)$, in contrast to enumerating $|\mathcal{S}|^K$ paths. The posteriors used in Baum–Welch are

$$\gamma_k(i) = \frac{\alpha_k(i) \beta_k(i)}{\mathcal{L}}, \quad \xi_k(i, j) = \frac{\alpha_k(i) a_{ij} b_j(\mathbf{y}_{k+1}) \beta_{k+1}(j)}{\mathcal{L}}. \quad (4.7)$$

Here $\alpha_k(i)$ is the joint density of the observations up to k and state i , whereas $\beta_k(i)$ is the conditional density of the remaining observations given that state. Their product combines evidence on both sides of the decision. Thus $\gamma_k(i)$ is a smoothed state probability, while $\xi_k(i, j)$ gives the expected weight of a transition $i \rightarrow j$. Summing γ_k over states, or ξ_k over both indices, gives one; no most-probable path needs to be selected to calculate them.

4.2.3 Decoding a whole sequence

Choosing $\arg \max_i \gamma_k(i)$ separately at every time maximizes the expected number of correct individual states, but those choices need not form the most probable joint sequence and can even violate a forbidden transition. Viterbi instead retains the best path ending in each state [26, 28]. In logarithmic form,

$$d_1(j) = \log \pi_j + \log b_j(\mathbf{y}_1), \quad (4.8)$$

$$d_{k+1}(j) = \log b_j(\mathbf{y}_{k+1}) + \max_i [d_k(i) + \log a_{ij}], \quad (4.9)$$

$$\psi_{k+1}(j) = \arg \max_i [d_k(i) + \log a_{ij}]. \quad (4.10)$$

After choosing $\hat{q}_K = \arg \max_j d_K(j)$, backtracking through $\hat{q}_k = \psi_{k+1}(\hat{q}_{k+1})$ recovers the optimal sequence. Zero probabilities have log score $-\infty$; ties require a consistent convention. The forward recursion adds competing paths, while Viterbi takes their maximum, which explains why a decoded path and posterior uncertainty are different outputs. Both are conditional on the adopted model, so a high posterior does not independently establish that a behavioural interpretation is correct.

4.2.4 Learning from unlabelled observations

If states were observed, transitions could be counted and each Gaussian fitted to its assigned measurements. With hidden states, Baum–Welch replaces those unknown assignments by their posterior weights and iterates the calculation [29]. This is an expectation–maximization procedure: at iteration r , compute

$$\mathcal{Q}(\theta \mid \theta^{(r)}) = \mathbb{E}_{Q|\mathbf{y}, \theta^{(r)}}[\log p(Q, \mathbf{y} \mid \theta)], \quad \theta^{(r+1)} \in \arg \max_{\theta} \mathcal{Q}(\theta \mid \theta^{(r)}). \quad (4.11)$$

The E step uses forward–backward probabilities; the M step maximizes the resulting expected complete-data log likelihood. Its monotonicity follows from the EM lower-bound argument under an exact M step [30], but does not imply a global optimum. For B observation blocks treated as independent in the likelihood and indexed by b , the unregularized Gaussian M step gives the following updates, provided each denominator is positive:

$$\pi_i^{\text{new}} = \frac{1}{B} \sum_{b=1}^B \gamma_{b,1}(i), \quad a_{ij}^{\text{new}} = \frac{\sum_b \sum_{k=1}^{K_b-1} \xi_{b,k}(i, j)}{\sum_b \sum_{k=1}^{K_b-1} \gamma_{b,k}(i)}, \quad (4.12)$$

$$\boldsymbol{\mu}_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) \mathbf{y}_{b,k}}{\sum_{b,k} \gamma_{b,k}(i)}, \quad \boldsymbol{\Sigma}_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) (\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})(\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})^{\top}}{\sum_{b,k} \gamma_{b,k}(i)}. \quad (4.13)$$

Initial-state counts use only the first observation of each block, and transition counts stop at its penultimate observation, so no transition is invented across a gap. These formulas explain the statistical fit; numerical scaling, regularization, initialization and stopping rules must also be specified for the actual estimator. The next sections make those choices for flight observations, rather than treating the general HMM theory as evidence that this particular three-state interpretation is correct.

4.3 From the cleaned flight to observations

The input is the final trajectory of Chapter 2, with local position (E, N, z) and the smoothed velocity and acceleration components. Each preprocessing segment defines a separate observation sequence, within which feature windows and hidden-state transitions cannot cross an acquisition gap or an excluded quality interval.

A common decision interval $\Delta = 10$ s gives the transition matrix the same physical meaning for every flight, restricting the analysis to native cadences no slower than this interval. At each decision time t_k , we first compute each scalar feature at native fixes and linearly interpolate those values. Denoting this interpolant by $\mathcal{I}[f]$, we average it over a centred window of width $W = 30$ s:

$$\bar{f}(t_k) = \frac{1}{W} \int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[f](u) du, \quad \mathbf{X}_k = (\bar{v}_z, \bar{v}_h, \overline{|\omega|}, C_{\omega})(t_k). \quad (4.14)$$

The window suppresses fix-scale fluctuations while preserving manoeuvres over tens of seconds; its width is a chosen measurement scale rather than a measured universal thermalling period. Although boundaries lie on a 10-s grid, the overlapping 30-s windows prevent a claim of independent 10-s temporal resolution.

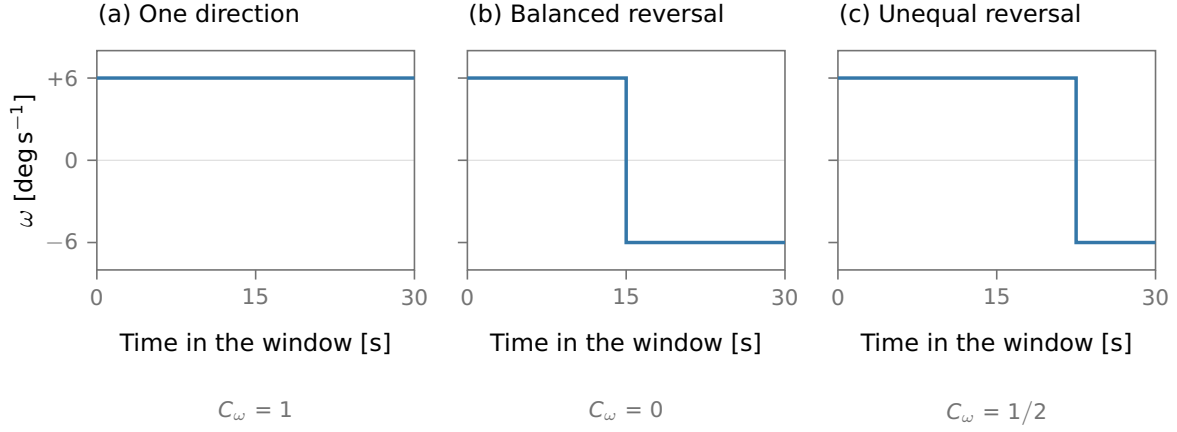


Figure 4.3. Turning amount and directional coherence measure different properties. All three idealized windows have $|\omega| = 6$ degrees per second, but their coherence differs because opposite turns cancel in the signed integral. The jumps illustrate limiting waveforms; the data calculation integrates piecewise-linear interpolants of native-fix values as specified in Eqs. (4.14) and (4.16). No phase label follows from coherence alone.

The horizontal speed and signed turn rate are

$$v_h = \sqrt{v_E^2 + v_N^2}, \quad \omega = \frac{v_E a_N - v_N a_E}{v_E^2 + v_N^2} \quad (v_h \geq 1 \text{ m s}^{-1}). \quad (4.15)$$

Below the stated speed floor the implementation sets $\omega = 0$, because numerical heading is unreliable when the horizontal velocity nearly vanishes. The fourth feature measures whether turning retains its sign:

$$C_\omega(t_k) = \frac{\left| \int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[\omega](u) du \right|}{\int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[|\omega|](u) du}, \quad 0 \leq C_\omega \leq 1, \quad (4.16)$$

The denominator is an accumulated angle in radians. The code sets $C_\omega = 0$ if it is at most machine epsilon (2^{-52} in float64), and clips floating-point round-off to $[0, 1]$. Turning in one direction throughout the window gives $C_\omega = 1$ even when its rate varies: coherence is not a measure of circularity or of lift by itself. At a reversal between two fixes, $\mathcal{I}[|\omega|]$ differs from $|\mathcal{I}[\omega]|$: the former is the quantity the code integrates. The pointwise inequality $|\mathcal{I}[\omega]| \leq \mathcal{I}[|\omega|]$ proves the stated bound on C_ω .

For example, equal-duration turns at $+6$ and -6 degrees per second cancel in the numerator and give $C_\omega = 0$, despite a large mean absolute turn rate. Keeping the positive sign throughout gives $C_\omega = 1$; spending three quarters of a window at $+6$ and one quarter at -6 gives $C_\omega = 1/2$ in the ideal piecewise-constant limit. Thus two different questions are measured: how much the heading turns, and how much opposite turns cancel. Figure 4.3 makes this distinction explicit.

A complete safe window is required. Windows touching reconstructed altitude, a derivative influenced by that reconstruction, or an edge derivative from preprocessing are unclassified, as are unsupported feature edges. Influence includes the entire Savitzky–Golay vertical fitting window, including its off-centre boundary fits. The support includes the native fixes bracketing interpolated window endpoints. Consequently even an unsafe fix just outside the nominal integration interval can invalidate it. Grey in the figures means unavailable support; there is no posterior confidence threshold that turns uncertain classifications grey.

Table 4.1. Quantities shown in the phase time-series figures. Feature windows are centred and 30 s wide; the plotted quantities retain physical units, before standardization.

Axis	Mathematical quantity	Unit / range
Altitude	$z(t_k)$ from the cleaned trajectory	m
Vertical speed	$\bar{v}_z(t_k)$, Eq. (4.14)	m s^{-1} , signed
Horizontal speed	$\sqrt{v_E^2 + v_N^2}(t_k)$	m s^{-1} , ≥ 0
Absolute turn rate	$(180/\pi) \omega (t_k)$	deg s^{-1} , ≥ 0
Coherence	$C_\omega(t_k)$, Eq. (4.16)	dimensionless, $[0, 1]$
Posterior	$\gamma_k(i)$ for $i \in \{\text{T, S, C}\}$	$[0, 1]$, sum = 1
Phase strip	$\hat{q}_k$ from the joint Viterbi path	T, S, C or unavailable

4.4 A probabilistic segmentation, and its assumptions

4.4.1 The fitted hidden Markov model

A separate continuous Gaussian HMM is fitted for each discipline, since speed and turning scale differ between paragliders and hang gliders. Write $Q_k \in \mathcal{S}$ for the hidden phase and $\mathbf{Y}_k$ for the features standardized by training-set means and standard deviations. More explicitly, for the n observations actually selected for fitting and feature j ,

$$\bar{x}_j = \frac{1}{n} \sum_{k=1}^n x_{kj}, \quad s_j^2 = \frac{1}{n} \sum_{k=1}^n (x_{kj} - \bar{x}_j)^2, \quad Y_{kj} = \frac{x_{kj} - \bar{x}_j}{s_j}. \quad (4.17)$$

A numerically constant feature is rejected; the population divisor n , rather than $n - 1$, matches the stored standardizer. The same $\bar{x}_j, s_j$ are used on held-out observations. The model parameters are an initial distribution π , a transition matrix A , and a full Gaussian covariance and mean for each component:

$$a_{ij} = \Pr(Q_{k+1} = j \mid Q_k = i), \quad \sum_j a_{ij} = 1, \quad b_i(\mathbf{y}) = \mathcal{N}_d(\mathbf{y}; \boldsymbol{\mu}_i, \Sigma_i). \quad (4.18)$$

Full covariance allows simultaneous vertical speed, horizontal speed and turning to be correlated within a phase. It does not model residual correlation *between* successive observations. The factorization in Eq. (4.2) assumes first-order phase memory, time-homogeneous parameters and conditionally independent emissions. The last is only approximate for overlapping windows; posterior confidence may therefore be sharper than independent observations would justify. Gaussian emissions also place some probability on physically impossible feature values, such as negative speed. They are a classification approximation, not a generative model of flight physics.

Baum–Welch applies expectation–maximization to the unlabelled training likelihood [29,30]. Ten fixed-seed initializations are compared by training likelihood. Whole valid blocks enter the fit; the final observation of one block never contributes a transition to the next. The fitted scaler is frozen for validation and test flights.

A numerical component has no behavioural name. At present the component with greatest mean vertical speed is provisionally called climb. Among the other two components, transition maximizes $\mu_{v_h} - \mu_{|\omega|} - \frac{1}{2}\mu_{C_\omega}$ in standardized feature coordinates; the remaining component is called search. Human train labels will resolve this permutation. In particular, “search” currently also collects observations unexplained by the other two components: its behavioural interpretation requires a direct check.

4.4.2 Numerical estimation in the fitted model

Equations (4.12)–(4.13) describe unregularized maximum likelihood. The stored models were fitted with `hmmlearn` 0.3.3 [31]. Its default initial/transition Dirichlet parameters are one, and its mean weight is zero, so those updates reduce to the expressions above. For full covariance the default scalar `covars_prior` is 0.01, broadcast to every matrix entry. With $N_i = \sum_{b,k} \gamma_{b,k}(i)$, the implemented covariance update is therefore

$$\Sigma_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) (\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})(\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})^\top + 0.01 \mathbf{1}\mathbf{1}^\top}{N_i}. \quad (4.19)$$

The added matrix is rank one: it does not impose a positive lower bound on every covariance eigenvalue. In particular, `min_covar` = 10^{-3} is added to the initial covariance diagonal; it is not enforced as an eigenvalue floor after every M step. This distinction matters for the very narrow coherence component.

Exact unregularized EM cannot decrease its likelihood, but convergence of the parameters to a stationary point requires further regularity conditions; a mixture can approach a singular covariance. The software’s stopping flag is also true when the iteration cap is reached or the last likelihood change is negative. That flag in an old artifact is not evidence of convergence. New fits record convergence only for a finite, non-negative last gain below the tolerance, reject non-finite final scores, and still retain all restart scores for inspection. For restart r , write $\ell_r = \sum_b \log \mathcal{L}_b(\hat{\theta}_r)$ on the same training observations. Its reported relative score is $(\ell_r - \max_u \ell_u)/n \leq 0$, in natural-log units per fitted observation, whereas the final EM gain is $\ell^{(m)} - \ell^{(m-1)}$ before normalization. These are numerical fit diagnostics, not accuracy scores. Ten fixed-seed restarts are ranked by finite training likelihood; a restart can be selected without satisfying the stated convergence criterion. Training uses scaled forward–backward arithmetic; current Viterbi and posterior decoding use logarithms to avoid underflow.

4.4.3 Why the current decoder differs from the original fit

The original fit used all four features. Its climb component concentrates coherence extremely close to one: the physical standard deviations are approximately 0.0000889 for paragliders and 0.0000604 for hang gliders (Table 4.3). A circling climb with a small turn reversal can therefore receive negligible climb likelihood. More persistent transitions alone cannot repair this emission problem.

The current decoder integrates coherence out of the original four-dimensional Gaussian. If $\mathbf{U} = (Y_1, Y_2, Y_3)$ and $V = Y_4$, then

$$b_i^{(3)}(\mathbf{u}) = \int_{\mathbb{R}} b_i^{(4)}(\mathbf{u}, v) dv = \mathcal{N}_3(\mathbf{u}; \boldsymbol{\mu}_{i,U}, \Sigma_{i,UU}). \quad (4.20)$$

Thus the retained mean entries and principal covariance submatrix define the new likelihood. This is marginalization, not conditioning on perfect coherence, and not a new three-dimensional fit. Coherence remains recorded and plotted for inspection. A separately refitted three-dimensional model remains a validation comparison: the original parameters were selected using the problematic fourth feature.

The coherence mean and width in Table 4.3 undo the training standardization: $\mu_{C,i}^{\text{phys}} = \bar{x}_4 + s_4 \mu_{i,4}$ and $\sigma_{C,i}^{\text{phys}} = s_4 \sqrt{\Sigma_{i,44}}$. They describe a fitted Gaussian component, not the sample mean and standard deviation of decisions labelled climb.

A second, independent choice softens the phase sequence. With $s_i = a_{ii} < 1$ and $q_{ij} = a_{ij}/(1 - s_i)$ for $j \neq i$, the decoding matrix is

$$\tilde{a}_{ii} = \tilde{s}_i, \quad \tilde{a}_{ij} = (1 - \tilde{s}_i)[(1 - w)q_{ij} + wq_{ij}^*], \quad j \neq i. \quad (4.21)$$

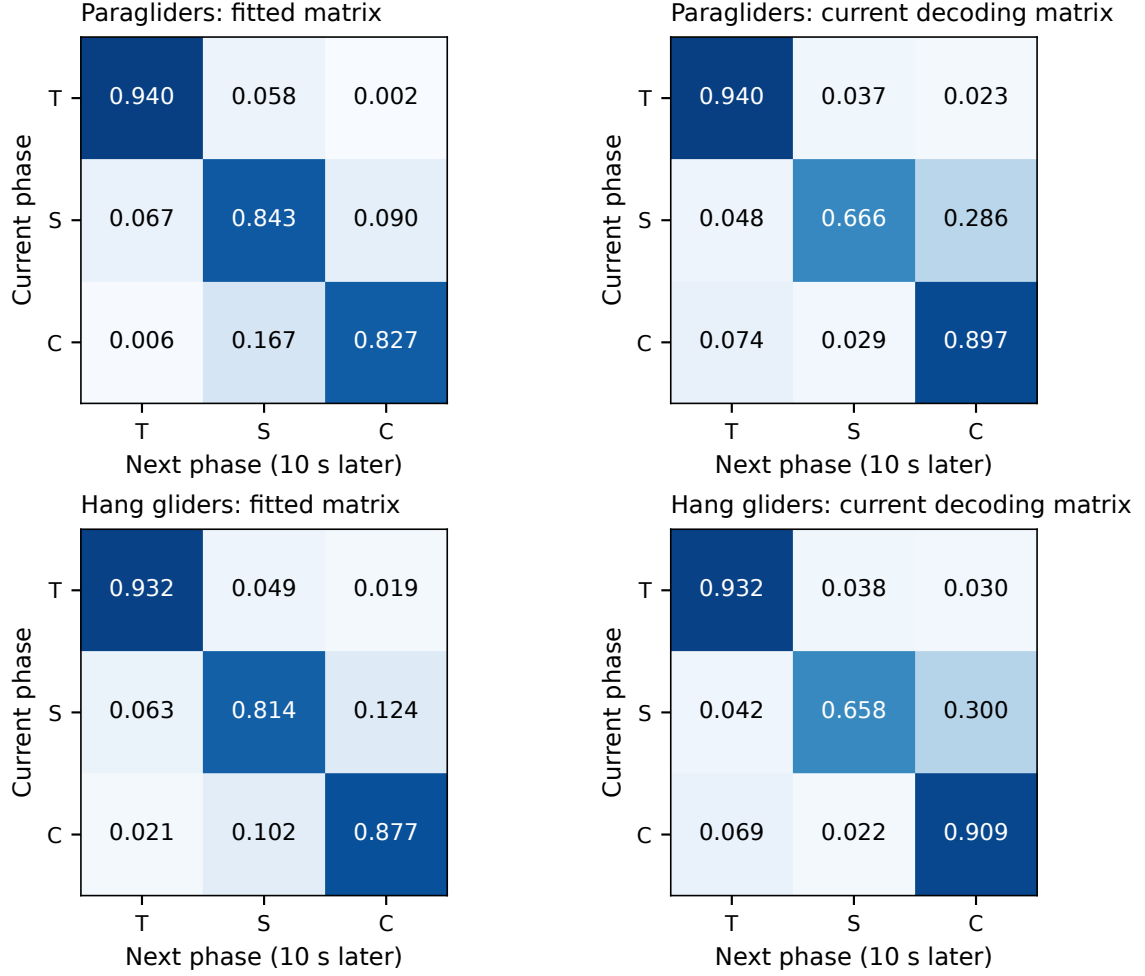


Figure 4.4. Original fitted and current decoding transition matrices, ordered as transition (T), search (S), climb (C). Every entry is a probability per 10-s decision. The right column includes the declared sequence preference and cannot be read as an empirical frequency of pilot behaviour.

Here $w = 0.75$. The target exits from transition split equally between search and climb, allowing direct entry into lift; from search and climb, the next phase in $T \rightarrow S \rightarrow C \rightarrow T$ receives target probability 0.95. The persistence rule is

$$\tilde{s}_i = \begin{cases} s_i + w(e^{-\Delta/\theta_S} - s_i), & i = S, \\ s_i + w \max(0, e^{-\Delta/\theta} - s_i), & i = T, C, \end{cases} \quad \theta_S = 20 \text{ s}, \quad \theta = 120 \text{ s}. \quad (4.22)$$

For an absorbing fitted row ($s_i = 1$), the code starts from equal exit weights and caps the new persistence at $1 - \epsilon_{\text{mach}}$ to retain possible exits. These scales and weights are provisional decoder settings. They are neither fitted behavioural durations nor minimum run lengths. Figure 4.4 shows both matrices so that learned structure and imposed preference can be distinguished. Once a semantic prior is active, changing component names requires decoding again, rather than merely recolouring an existing path.

For both the original and modified model, Viterbi chooses the entire most probable path rather than the most likely phase at each isolated time [28]:

$$\hat{q}_{1:K} = \arg \max_{q_{1:K}} p(q_{1:K} \mid \mathbf{y}_{1:K}), \quad \gamma_k(i) = \Pr(Q_k = i \mid \mathbf{y}_{1:K}). \quad (4.23)$$

The hard path and posterior γ use the same effective transition matrix and marginal emission.

Table 4.2. Archive-scale HMM fit and decoding census. A sequence is a contiguous classifiable feature block; restart numbers are one-based.

Discipline	Fit flights	Fit sequences	Fit points	Decoded points	Restart
Paragliders	972	11,604	953,078	171,721,035	4/10
Hang gliders	874	8,784	1,000,000	7,305,190	10/10

Table 4.3. Support and numerical diagnostics of the segmentation. Excluded time uses all retained cleaned-segment duration as its denominator. Coherence widths belong to the original four-dimensional climb component.

Quantity	Paragliders	Hang gliders
Emitted archive decisions	171,721,035	7,305,190
Excluded cleaned duration	8.28%	5.93%
Climb C_ω mean (original fit)	0.9999986	0.9999996
Climb C_ω standard deviation	0.0000889	0.0000604
Final EM log-likelihood change	-0.223	-0.253
Annotated intervals supplied	0	0

“Smoothed posterior” refers to inference using observations before and after k , not to an extra filter applied to probabilities.

4.5 What has been checked

4.5.1 Archive support and reproducible examples

The original models were fitted and applied to both archives. Table 4.2 records the actual fitting and decoding counts. Fitting uses a deterministic subset of whole training blocks up to the memory cap; longer selected blocks contribute more observations, so this is not equal weighting of flights. The saved archive uses the current decoder. The four-dimensional path is reconstructed from the same fitted parameters as a comparison. The figures in this chapter use the current configuration and saved features, with complete valid blocks decoded before an eight-minute display excerpt is taken. A shortened figure therefore does not shorten the sequence used for inference.

The selected fits do not both satisfy a finite, non-negative last gain below tolerance (Table 4.3). The software stopping flags therefore cannot establish convergence of these fits; covariance and refitting sensitivity remain to be checked.

Coverage has a separate denominator from classification accuracy:

$$f_{\text{unclassified}} = 1 - \frac{\text{duration covered by classified decision cells}}{\text{total retained cleaned-segment duration}}. \quad (4.24)$$

Decision cells $[t_k - \Delta/2, t_k + \Delta/2)$ are clipped at segment boundaries; acquisition gaps contribute to neither duration. For exclusion reason r , the plotted percentage is $100 \sum_k |I_k| \mathbf{1}\{r_k = r\} / D_{\text{clean}}$, with I_k the clipped decision cell and D_{clean} the denominator in Eq. (4.24). Feature-edge status takes precedence over a quality mask if both apply. Segments with native cadence above 10 s contribute their full duration to the cadence category; remaining tails outside any cell form the final category. The categories are disjoint. The denominator does include retained segments too sparsely sampled for classification. The excluded fractions are 8.28% for paragliders and 5.93% for hang gliders, chiefly from unsafe feature support (Fig. 4.5). These masks are shared by the two decoders compared on that snapshot. The feature masks include the propagated reconstruction support of the vertical Savitzky–Golay derivatives. Their

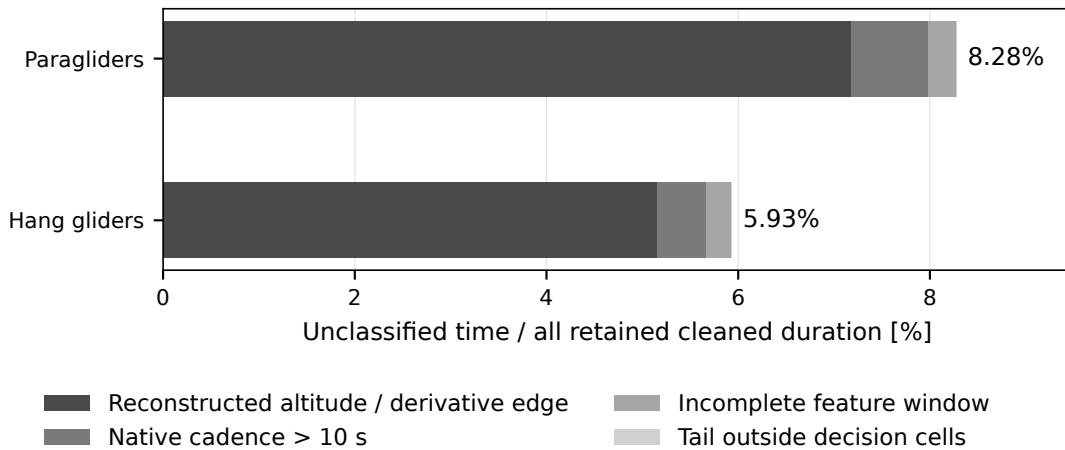


Figure 4.5. Why retained cleaned time lacks a phase label. Fractions use the same duration denominator in Eq. (4.24); they are not fractions of flights or fixes. Windows touching reconstructed altitude or edge derivatives dominate the excluded time.

coverage says how much time is available for subsequent phase statistics, not whether any phase label is correct or whether exclusion is behaviourally unbiased.

The two examples are selected by a fixed priority from validation flights, independently of their apparent phase composition. Recomputing labels with the stored model reproduces the archived paths. Their current labels and posterior curves then use Eqs. (4.20)–(4.22). The audit records the selected identities, model hashes, configuration, and changed-label counts. Thus each plotted label can be traced to its observations and decoder settings.

The paraglider excerpt shows sustained turning and altitude recovery after an initial transition, with a climb label that persists through brief changes in vertical speed and coherence. The hang-glider excerpt is less clear-cut: part of a sustained ascent is assigned to search, and the posterior divides between search and climb. Positive vertical speed is therefore not equivalent to the fitted climb state. This could reflect a difference in turning behaviour, but the figure alone cannot resolve the behavioural name of that component. It gives a concrete interval for human validation rather than an example selected to display apparently perfect classification.

The development example in Fig. 4.8 separates the two decoder changes. On its full preprocessing segment, the four-dimensional reference assigns 6 decisions to climb, the sequence-only variant assigns 6, and the current three-dimensional marginal assigns 32. The comparison shows how the coherence coordinate and sequence preference affect the same observations. The example helped motivate the change, so the comparison is a regression diagnostic, not independent evidence of improved accuracy. Neither a longer green interval nor a smaller number of switches is, by itself, a success criterion.

4.5.2 The manual check that is still needed

The prepared blinded pack contains 40 candidate flights: per discipline, four training candidates, eight validation candidates and eight test candidates. Human reference intervals are needed before reporting accuracy, confusion matrices or a definitive physical meaning for the search state. No human reference intervals were supplied to this audit, so it reports no independent classification scores. The interactive tool is available: select an altitude interval and mark transition, search or climb; edits are saved automatically on the 10-s decision grid. The plan view and kinematics are visible, while model predictions are hidden. The pack must match

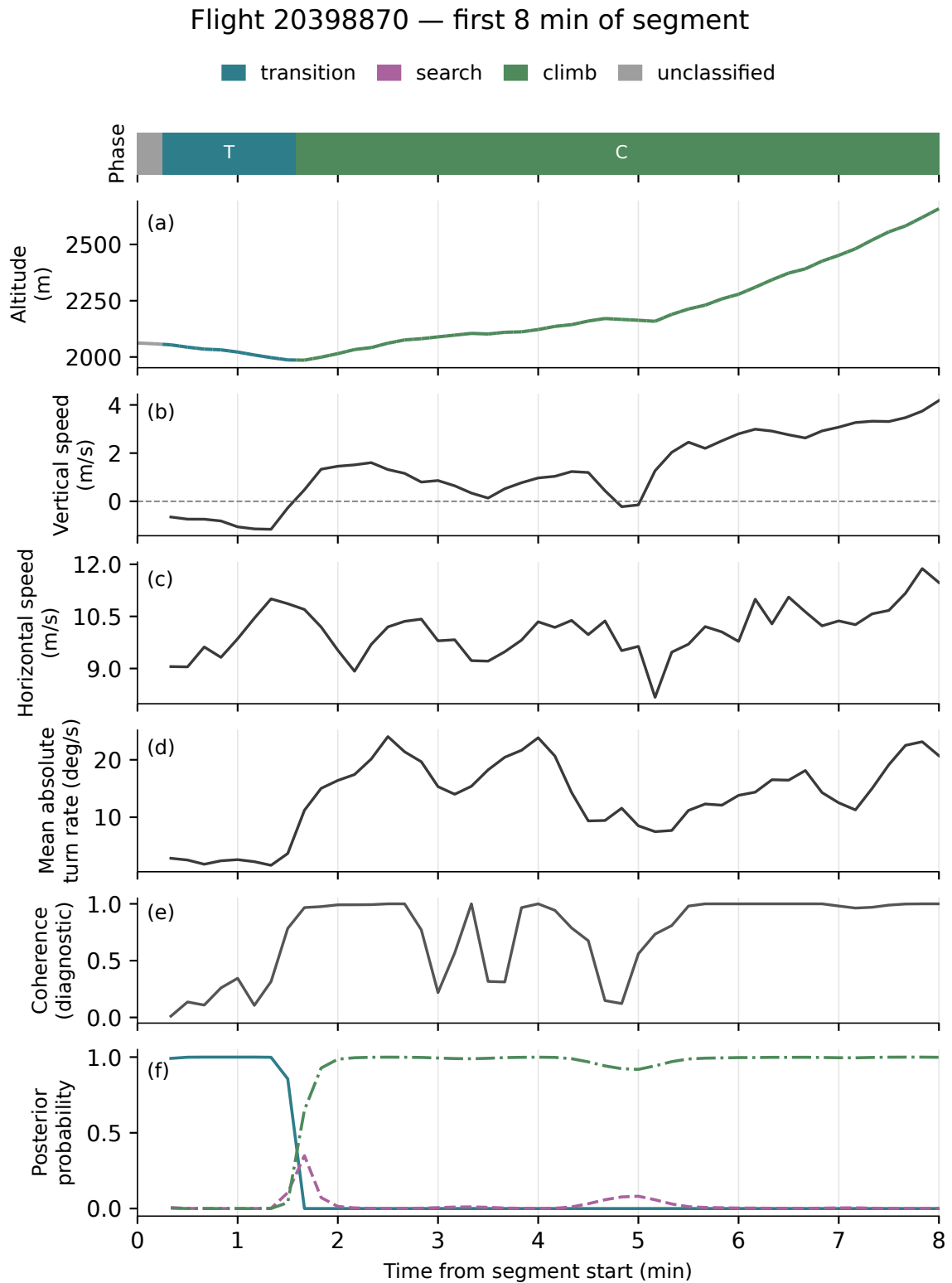


Figure 4.6. Eight minutes of a validation paraglider segment, decoded using the current three-feature marginal and sequence policy. The phase strip, altitude, separate feature axes and posterior probabilities share a time axis marked at one-minute intervals; decisions remain 10 s apart. Table 4.1 defines every displayed quantity. Coherence is shown as a diagnostic and does not enter the current emission likelihood. Grey indicates unavailable support. Phase names remain provisional.

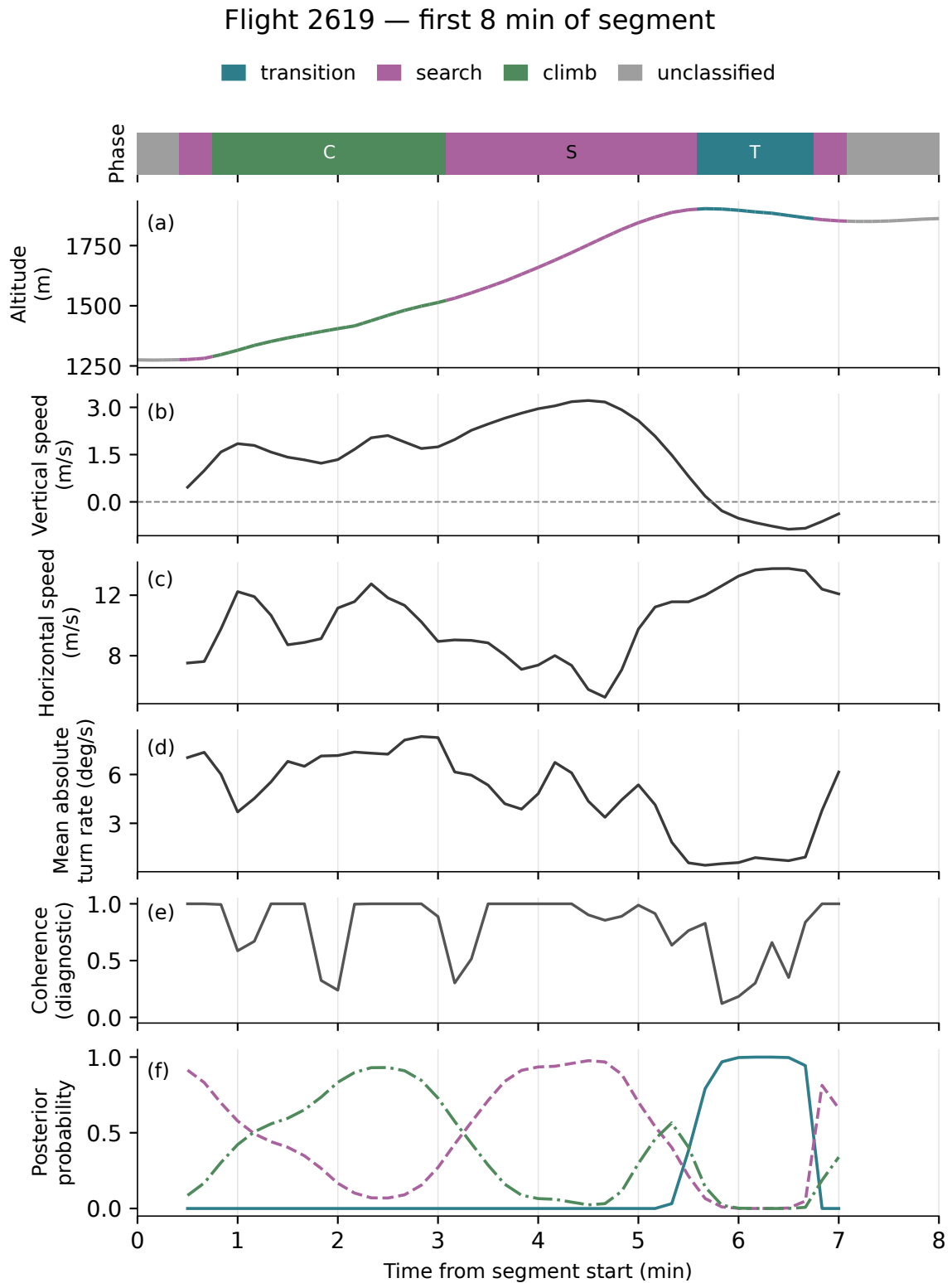


Figure 4.7. The corresponding eight-minute hang-glider validation excerpt. It uses its own fitted model and standardizer, with the same feature definitions and display convention as Fig. 4.6. A high posterior expresses certainty within the adopted model; it is not an independently measured probability of correct behaviour.

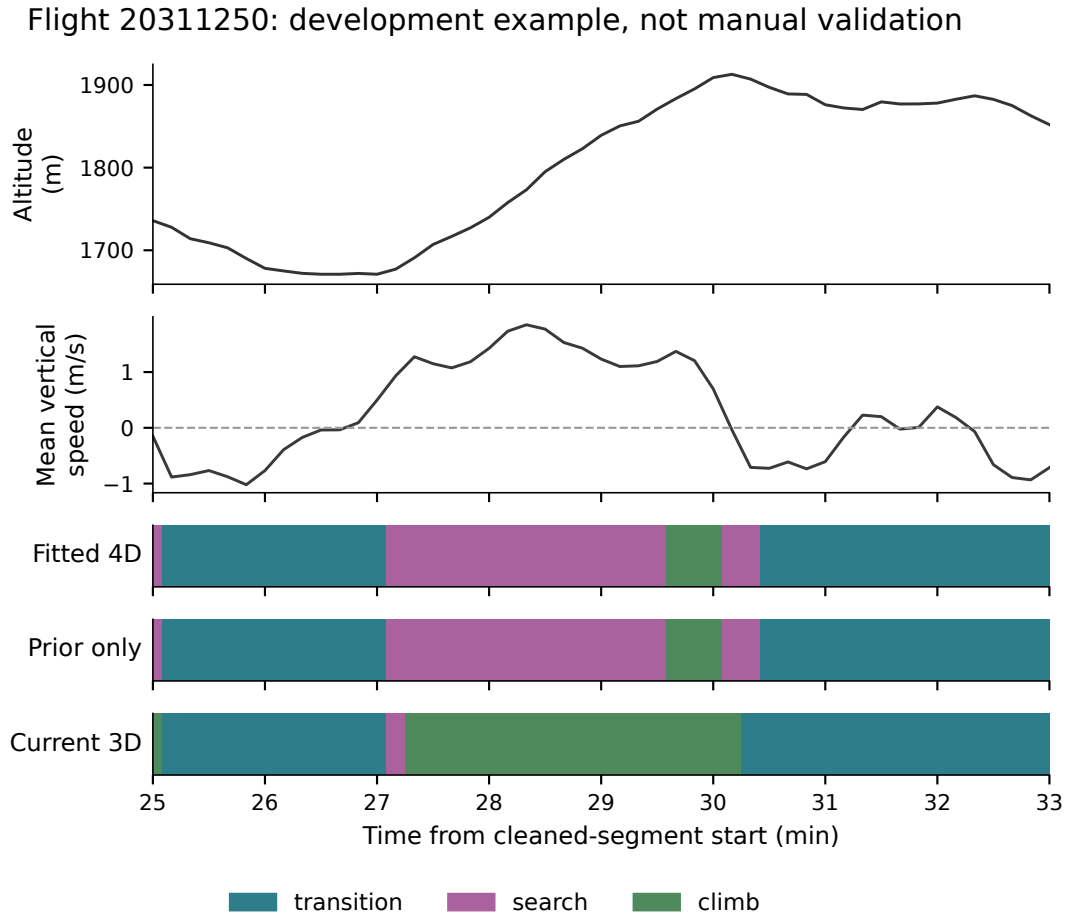


Figure 4.8. An eight-minute development excerpt from flight 20311250. The aligned phase strips compare the fitted four-dimensional reference, a change of sequence policy alone, and the current three-dimensional marginal with that policy. The altitude and vertical speed provide physical context. No human reference labels are present.

the cleaned archive used for evaluation; the regeneration procedure creates a new dated pack while preserving the previous pack and any labels already written.

Clearly interpretable intervals should be labelled across the whole candidate window; ambiguous boundaries can remain unlabelled. The fraction left unlabelled must accompany the scores, since evaluating only easy interiors would otherwise conceal the difficult part of segmentation. A second independent annotation of a subset will measure human agreement and boundary ambiguity. That agreement study has not yet been performed.

The deterministic 70/15/15% split operates at flight level, so no flight is shared between fitting and nominal holdout sets. This alone does not guarantee an untouched test: development flights 20275040 and 20279877 were inspected despite belonging to the nominal test partition. They, and hang-glider example flight 975, are excluded from the prepared annotation test pack and must remain outside final accuracy claims. Repeated pilots, days or sites can also make flights statistically related; generalization to a new region or season requires an additional grouped holdout, rather than being inferred from a random flight split.

Train annotations resolve state naming; validation annotations compare the original four-dimensional model, the marginal decoder with and without the sequence preference, and a refitted three-dimensional model. Model choices are then frozen before test scores are opened. For human label i and inferred label j , let n_{ij} count classified, annotated decisions. The core

metrics are

$$\begin{aligned} \text{Precision}_i &= \frac{n_{ii}}{\sum_j n_{ji}}, & \text{Recall}_i &= \frac{n_{ii}}{\sum_j n_{ij}}, \\ F_{1,i} &= \frac{2n_{ii}}{\sum_j n_{ji} + \sum_j n_{ij}}, & F_{1,\text{macro}} &= \frac{1}{3} \sum_i F_{1,i}. \end{aligned} \quad (4.25)$$

If a precision, recall or F1 denominator is zero, the evaluator assigns that metric zero and retains the phase in the three-state macro average. The corresponding accuracy is $\text{Acc} = \sum_i n_{ii} / \sum_{ij} n_{ij}$ on the same annotated, classified decisions; the evaluator requires at least one annotated, classified decision. On that nonempty set all these classification scores lie in $[0, 1]$. The confusion matrix reveals whether search is systematically confused with climbing or transition; macro-F1 gives a rare phase the same weight as a frequent one. Accuracy alone can be dominated by the most occupied phase. Bootstrap intervals [21] obtained by resampling whole flights preserve within-flight dependence, although they do not remove dependence between related flights. The existing evaluator provides these classification metrics. Boundary errors, annotation coverage and between-observer agreement remain additional validation measurements to complete; no values are supplied before labels exist.

4.6 What the segmentation will let us measure

The next empirical chapter will characterize all three phases before selecting a stochastic transport model. For a phase occurrence $I = [a, b)$, useful primitive quantities are its duration $D = b - a$, horizontal path length L , net displacement R , and altitude change:

$$L = \int_a^b v_h(t) dt, \quad R = \|\mathbf{r}(b) - \mathbf{r}(a)\|, \quad \Delta z = z(b) - z(a), \quad \mathcal{P} = R/L \in [0, 1] \quad (L > 0). \quad (4.26)$$

Here $\mathbf{r} = (E, N)$ and $v_h = \|\dot{\mathbf{r}}\|$ refer to the same horizontal curve; this consistency is needed for the triangle inequality $R \leq L$. The discrete implementation must use a matching position-based path length rather than assume that separately smoothed derivatives satisfy the inequality exactly. The ratio $\mathcal{P}$ distinguishes local circling from directed progress without assuming that either occurs in a particular phase. The ground displacement of a climb can be substantial in wind; search is not automatically an immobile waiting state. Distributions of $(D, L, R, \Delta z)$ and phase-conditioned velocity correlation will quantify those differences.

A phase-conditioned MSD requires an explicit conditioning convention. One useful choice keeps the *entire* displacement interval inside a single occurrence of phase i :

$$M_i(\tau) = \frac{\sum_I \int_{a_I}^{b_I - \tau} \|\mathbf{r}(t + \tau) - \mathbf{r}(t)\|^2 dt}{\sum_I (D_I - \tau)_+}, \quad I \text{ an occurrence of phase } i. \quad (4.27)$$

Integrals for $D_I \leq \tau$ contribute zero, and the observable is defined only when its denominator is positive. This weights eligible time origins; long occurrences contribute more, and short occurrences disappear at large lags. A plot must therefore show the contributing occurrences and durations, and be checked against equal-occurrence or equal-flight weighting. Conditioning only the two endpoints would define a different observable because it could cross other phases.

Prior phase-resolved results [2] supply comparisons, not values that the segmenter must be tuned to recover. Agreement in phase MSDs or time budgets cannot independently validate a classifier designed around the same kinematic expectations. Differences in sampling, geography, wind and the phase definitions must be considered before attributing a disagreement to the dynamics.

4.6.1 Keeping model assumptions separate from measured physics

An HMM imposes a geometric latent dwell-time law. With decoding persistence $\tilde{a}_{ii}$, its model prediction is

$$\Pr(D_i = m\Delta) = \tilde{a}_{ii}^{m-1}(1 - \tilde{a}_{ii}), \quad \mathbb{E}[D_i] = \frac{\Delta}{1 - \tilde{a}_{ii}}. \quad (4.28)$$

for $m = 1, 2, \dots$ and $0 \leq \tilde{a}_{ii} < 1$. A Viterbi run-length histogram is not a sample of observed physical durations independent of this assumption. It depends on the emissions and the decoder as well as on behaviour. Runs cut by a gap, quality mask or feature boundary are censored; merely discarding them can bias the surviving duration distribution. Testing non-geometric phase durations requires human reference intervals and a sensitivity analysis against a model with explicit duration distributions, such as a hidden semi-Markov model [32]. The imposed exit preferences likewise prevent interpreting the current decoded phase order as unqualified evidence of physical memory.

The later transport model must reproduce both these phase observables and the global constraints of Chapter 3. In particular, the duration of a moving leg and its displacement should be studied jointly. A constant-speed Lévy walk couples a leg's length to that *same leg's travel time*; dependence between its length and a *subsequent* climb or search is a separate question. A stereotyped phase cycle by itself also does not disprove renewal at the level of complete cycles. Renewal requires independence of successive cycle variables, which must be tested at the chosen event scale.

4.7 Chapter summary

The chapter has built the hidden-state model from its probability law, connected each feature to a measured kinematic quantity, and separated parameter fitting from sequence decoding. The current labels use a three-feature marginal of a fitted four-feature Gaussian HMM and an explicit sequence preference. Coverage, numerical diagnostics, short trajectory excerpts and a decoder comparison make those choices inspectable, without supplying an independent estimate of behavioural accuracy. The current archive leaves 8.28% of retained paraglider time and 5.93% of hang-glider time unclassified. The narrow coherence components and negative final likelihood changes remain numerical limitations, while the ascent labelled search in the hang-glider example exposes a behavioural ambiguity that posterior probabilities cannot settle.

The next step is the independent phase validation. Once its errors and boundary uncertainties are known, phase durations, displacements and correlations can be used alongside the global observables to test a physical transport model.

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Appendices

The appendices supply the spectral and geodetic definitions used in Chapter 2. They are numbered 2.A and 2.B in their order of first citation.

Appendix 2.A

Power spectral density and Welch's method

The spectra in Chapter 2 compare fluctuations in recorded altitude. They contain both flight dynamics and measurement error. A spectral estimate does not separate those contributions without further assumptions.

2.A.1 From covariance to a spectrum

Let $x_n = z_n - \mathbb{E}[z_n]$ be a real, uniformly sampled, second-order stationary process, with sampling interval $\Delta t = 1/f_s$ and covariance $C_k = \mathbb{E}[x_n x_{n+k}]$. When the covariance is summable, its two-sided spectral density is

$$S_2(f) = \Delta t \sum_{k=-\infty}^{\infty} C_k e^{-2\pi i f k \Delta t}, \quad |f| \leq f_s/2. \quad (2.A.1)$$

For real signals S_2 is even. The one-sided density doubles the positive-frequency contribution, giving

$$\int_0^{f_s/2} S_1(f) df = C_0 = \text{Var}(x_n). \quad (2.A.2)$$

Its units are $\text{m}^2 \text{Hz}^{-1}$ for an altitude signal. The Nyquist frequency $f_s/2$ limits distinguishable sampled frequencies: physical motion above that frequency may be aliased into the observed band.

A full flight altitude series need not be stationary. Equation (2.A.2) therefore gives the stationary reference interpretation; it is not a claim that the variance of an entire climbing and descending flight equals the integral of its detrended Welch estimate. Here the spectrum is a finite-record diagnostic, computed after detrending shorter windows.

2.A.2 The estimator used here

For window j of length L , subtract its least-squares linear trend and denote the residuals by $y_{j,n}$. With a periodic Hann taper w_n , define

$$I_j(f_k) = \frac{\left| \sum_{n=0}^{L-1} w_n y_{j,n} e^{-2\pi i k n / L} \right|^2}{f_s \sum_{n=0}^{L-1} w_n^2}, \quad f_k = k f_s / L. \quad (2.A.3)$$

This is the two-sided normalisation. For a one-sided estimate, positive bins are doubled except for the Nyquist bin when L is even; the zero-frequency bin is also left unchanged. Welch's

estimate averages the resulting periodograms over K overlapping windows [33]:

$$\hat{S}_1(f_k) = \frac{1}{K} \sum_{j=1}^K I_{j,1}(f_k). \quad (2.A.4)$$

The implementation uses $L = 256$, half-window overlap, linear detrending and the arithmetic mean within each flight. The frequency spacing is f_s/L ; tapering broadens the effective resolution. Details of the eligible intervals and the separate aggregation across flights are given in [Choice of altitude channel](#).

An individual periodogram has substantial sampling variability. Under suitable stationarity and weak-dependence conditions, averaging independent windows reduces its variance approximately by K . Overlapping windows are correlated, so their effective number is smaller. Increasing L improves frequency resolution but leaves fewer windows to average; keeping L fixed also retains smoothing bias as the record grows. The mean Welch estimator is not, by itself, resistant to arbitrarily large outliers.

Linear detrending suppresses slow variation within each window; it does not remove all physical motion or isolate sensor noise. Likewise, a high-frequency plateau can be compatible with measurement noise without uniquely identifying its cause. Comparisons between GNSS and barometric spectra must account for missing-channel selection, sampling, interpolation and the response of any subsequent filter.

Appendix 2.B

Geodetic transforms: from (φ, λ, h) to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.12), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis a , flattening f , $b = a(1 - f)$ the semi-minor axis, $e^2 = f(2 - f) = 1 - b^2/a^2$ the first eccentricity squared [15].

In the exact transform, h denotes ellipsoidal height. The GNSS channel in an arbitrary HPG record need not follow this convention (Sec. 2.7.1). For orthometric height H , the relation is $h = H + \mathcal{N}(\varphi, \lambda)$, where $\mathcal{N}$ is geoid undulation. The pipeline currently inserts the recorded GNSS value as a height proxy; it does not infer a datum or apply a geoid correction. A height error δh perturbs a horizontal displacement from the origin by approximately $|\delta h|d/R_\oplus$ on a short, level route. This is a sensitivity estimate, not an established error bound for every recorder. Moreover, a spatially varying geoid correction does not cancel exactly from altitude increments.

2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre O : the Z axis along the polar axis, the X axis through the intersection of the equator and the prime meridian, Y completing the right-handed triad. By symmetry, a point at longitude λ lies in the meridian plane obtained by rotating the X - Z plane by λ ; writing p for the distance from the polar axis, its ECEF coordinates are

$$X = p \cos \lambda, \quad Y = p \sin \lambda. \quad (2.B.1)$$

The problem therefore reduces to the meridian ellipse: find (p, Z) of a point at geodetic latitude φ and height h .

The surface point. The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \quad (2.B.2)$$

The geodetic latitude is defined through the surface *normal* (Figure 2.9). The gradient of the left-hand side of (2.B.2) is proportional to $(p/a^2, Z/b^2)$, so the outward normal at a surface point (p_s, Z_s) makes an angle φ with the equatorial plane given by

$$\tan \varphi = \frac{Z_s/b^2}{p_s/a^2} \quad \implies \quad Z_s = \frac{b^2}{a^2} p_s \tan \varphi = (1 - e^2) p_s \tan \varphi. \quad (2.B.3)$$

Substituting (2.B.3) into (2.B.2), with $b^2 = a^2(1 - e^2)$,

$$\frac{p_s^2}{a^2} \left(1 + (1 - e^2) \tan^2 \varphi \right) = 1 \implies p_s^2 = \frac{a^2 \cos^2 \varphi}{\cos^2 \varphi + (1 - e^2) \sin^2 \varphi} = \frac{a^2 \cos^2 \varphi}{1 - e^2 \sin^2 \varphi}, \quad (2.B.4)$$

that is,

$$p_s = N(\varphi) \cos \varphi, \quad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad (2.B.5)$$

and, back through (2.B.3),

$$Z_s = (1 - e^2) N(\varphi) \sin \varphi. \quad (2.B.6)$$

The quantity $N(\varphi)$ so defined is the *prime-vertical radius of curvature*; a computation shows that, away from the poles, it is the distance along the normal from the surface point to the polar axis (the segment $Q-P$ of Figure 2.9): the normal direction in the meridian plane is $(\cos \varphi, \sin \varphi)$, and following it inward from (p_s, Z_s) a length N lands on the axis, $p_s - N \cos \varphi = 0$ by (2.B.5). At the poles the normal lies on the polar axis, so this intersection construction is degenerate; the formula for N remains defined by continuity.

Adding the height. The height h of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi). \quad (2.B.7)$$

Adding $h \hat{\mathbf{u}}$ to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h) \cos \varphi \cos \lambda, \quad Y = (N + h) \cos \varphi \sin \lambda, \quad Z = (N(1 - e^2) + h) \sin \varphi, \quad (2.B.8)$$

which is Eq. (2.12). The asymmetry between the horizontal factor $(N + h)$ and the vertical one $(N(1 - e^2) + h)$ is the ellipsoid's flattening at work: on a sphere ($e = 0$) both reduce to $R + h$ and the geodetic and geocentric latitudes coincide.

How far the two latitudes differ. The geocentric latitude ψ of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at $h = 0$,

$$\tan \psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2) \sin \varphi}{N \cos \varphi} = (1 - e^2) \tan \varphi, \quad (2.B.9)$$

the classic relation: the two agree only where $e = 0$, or at the equator and the poles, where $\tan$ is 0 or infinite. To see the size of the gap, write $\psi = \varphi - \Delta$ and expand for small e^2 . Using $\tan(\varphi - \Delta) \simeq \tan \varphi - \Delta(1 + \tan^2 \varphi)$ in (2.B.9),

$$\tan \varphi - \Delta(1 + \tan^2 \varphi) \simeq (1 - e^2) \tan \varphi \implies \Delta \simeq \frac{e^2 \tan \varphi}{1 + \tan^2 \varphi} = e^2 \sin \varphi \cos \varphi, \quad (2.B.10)$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \quad (2.B.11)$$

which is the expression quoted in Sec. 2.7.5. At this order it is largest at $\varphi = 45^\circ$, where it equals $e^2/2 = 3.347 \times 10^{-3}$ radian, or $0.192^\circ \approx 11.5'$. Mistaking one latitude for the other therefore displaces a fix along the meridian by $R_\oplus (e^2/2) \approx 21$ km at mid-latitudes—the ~ 20 km quoted in the body. The error vanishes at the equator and at the poles; the leading-order approximation is symmetric about 45° .

2.B.2 ECEF to local ENU

The local frame at the origin fix $(\varphi_0, \lambda_0, h_0)$ is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing toward increasing latitude; Up, along the ellipsoid normal (2.B.7). Away from the poles, East is the direction in which the position (2.B.1) changes under λ alone,

$$\hat{\mathbf{e}}_E = \frac{\partial \mathbf{X} / \partial \lambda}{\|\partial \mathbf{X} / \partial \lambda\|} = (-\sin \lambda_0, \cos \lambda_0, 0), \quad (2.B.12)$$

At a pole, longitude fixes a conventional orientation of the horizontal axes; the displayed vector still defines that convention although the derivative vanishes. Up is $\hat{\mathbf{u}}$ of (2.B.7) evaluated at (φ_0, λ_0) , and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin \varphi_0 \cos \lambda_0, -\sin \varphi_0 \sin \lambda_0, \cos \varphi_0). \quad (2.B.13)$$

The ENU components of a displacement $\Delta \mathbf{X}$ are its projections on the triad, $E = \hat{\mathbf{e}}_E \cdot \Delta \mathbf{X}$ and likewise for N and U ; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix}}_R \Delta \mathbf{X}, \quad (2.B.14)$$

whose rows are orthonormal by construction, so R is a rotation ($R^\top R = 1$) [16]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same φ that enters Eq. (2.12), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

2.B.3 Size of the tangent-plane error

Two different distances must be distinguished. On a sphere, an arc of length d subtends a three-dimensional chord

$$c = 2R_\oplus \sin \frac{d}{2R_\oplus} = d - \frac{d^3}{24 R_\oplus^2} + O(d^5 / R_\oplus^4). \quad (2.B.15)$$

The horizontal EN projection from a frame anchored at one endpoint is instead

$$d_{EN} = R_\oplus \sin(d / R_\oplus), \quad d - d_{EN} = \frac{d^3}{6 R_\oplus^2} + O(d^5 / R_\oplus^4). \quad (2.B.16)$$

This deficit is 0.033 m at 20 km, 0.51 m at 50 km, and about 33 m at 200 km. These estimates assume surface points and an origin at one endpoint. For increments between two remote points in a fixed frame, projection distortion also depends on their distance and bearing from the frame origin. It cannot be bounded from increment length alone. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta $\approx d^2 / (2R_\oplus)$, which reaches tens of metres already at $d = 20$ km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the adopted altitude channel itself (Sec. 2.7.5, Sec. 2.7.1), and the U component serves only the geometric construction.

2.B.4 Size of the spherical-approximation error

Sec. 2.7.2 computes inter-fix distances by the haversine formula, Eq. (2.2), which treats the Earth as a sphere of fixed radius $R_\oplus = 6,371$ km regardless of latitude or heading. This section bounds what that costs against the WGS84 ellipsoid.

Local metric. For a short step at geodetic latitude φ , the ellipsoid and the sphere give

$$ds_{\text{ell}}^2 = M(\varphi)^2 d\varphi^2 + N(\varphi)^2 \cos^2 \varphi d\lambda^2, \quad ds_{\text{sph}}^2 = R_\oplus^2 (d\varphi^2 + \cos^2 \varphi d\lambda^2). \quad (2.B.17)$$

Define $d\sigma = ds_{\text{sph}}/R_\oplus$ and the bearing α in this spherical metric by $d\varphi = d\sigma \cos \alpha$ and $\cos \varphi d\lambda = d\sigma \sin \alpha$. Then

$$R_{\text{eff}}(\varphi, \alpha) \equiv \frac{ds_{\text{ell}}}{d\sigma} = \sqrt{M(\varphi)^2 \cos^2 \alpha + N(\varphi)^2 \sin^2 \alpha}, \quad (2.B.18)$$

interpolating between the meridian value $M(\varphi)$ ($\alpha = 0$, north–south) and the prime-vertical value $N(\varphi)$ of Eq. (2.B.5) ($\alpha = 90^\circ$, east–west) [15]. $N(\varphi)$ is already in hand; $M(\varphi)$ follows the same route, differentiating the surface point $(p_s(\varphi), Z_s(\varphi))$ of Eqs. (2.B.5)–(2.B.6) with respect to φ and using $ds = M d\varphi$ along the meridian:

$$\frac{dp_s}{d\varphi} = -\frac{N(\varphi)(1-e^2) \sin \varphi}{1-e^2 \sin^2 \varphi}, \quad \frac{dZ_s}{d\varphi} = \frac{N(\varphi)(1-e^2) \cos \varphi}{1-e^2 \sin^2 \varphi}, \quad (2.B.19)$$

so that $M(\varphi)^2 = (dp_s/d\varphi)^2 + (dZ_s/d\varphi)^2$ collapses, via $\sin^2 \varphi + \cos^2 \varphi = 1$, to

$$M(\varphi) = \frac{N(\varphi)(1-e^2)}{1-e^2 \sin^2 \varphi} = \frac{a(1-e^2)}{(1-e^2 \sin^2 \varphi)^{3/2}}. \quad (2.B.20)$$

Comparison to the sphere. The reference radius used in Eq. (2.2), $R_\oplus = 6,371$ km, agrees with the ellipsoid’s mean radius $R_m = (2a + b)/3 = a(1 - f/3) = 6,371.009$ km to within 10 m. The fractional difference, expressed as ellipsoidal length relative to spherical length, is approximately $[R_{\text{eff}}(\varphi, \alpha) - R_m]/R_m$, bounded between its values at $\alpha = 0$ and $\alpha = 90^\circ$. Expanding Eqs. (2.B.5) and (2.B.20) to first order in the flattening f (using $e^2 \simeq 2f$),

$$\frac{N(\varphi)}{R_m} - 1 \simeq f \left(\sin^2 \varphi + \frac{1}{3} \right), \quad \frac{M(\varphi)}{R_m} - 1 \simeq f \left(3 \sin^2 \varphi - \frac{5}{3} \right), \quad (2.B.21)$$

both of order the flattening $f = 1/298.257223563 \approx 0.335\%$ at every latitude: from $-5f/3 \approx -0.56\%$ at the equator ($\varphi = 0$, meridian direction) to $+4f/3 \approx 0.45\%$ at the poles ($\varphi = 90^\circ$, where $M = N$ by symmetry). Thus 0.3% is a regional estimate, not a global bound.

At the latitudes flown. Most of the archive launches from the Alps box $\varphi \in [43.8^\circ, 46.6^\circ]$ used to classify take-off sites (Sec. 2.9, Fig. 2.20). Evaluating Eq. (2.B.21) across that range moves the bound by under 0.02 percentage points, so its midpoint $\varphi = 45^\circ$ is representative:

$$\frac{N(45^\circ)}{R_m} - 1 \simeq \frac{5f}{6} \approx 0.28\%, \quad \frac{M(45^\circ)}{R_m} - 1 \simeq -\frac{f}{6} \approx -0.06\% \quad (2.B.22)$$

(numerically, $N(45^\circ) = 6,388.84$ km and $M(45^\circ) = 6,367.38$ km against $R_m = 6,371.01$ km). At 45° , the sphere underestimates an east–west ellipsoidal step by about 0.28% and overestimates a north–south step by about 0.06%. Across the latitude interval above the absolute discrepancy remains below 0.3%, or roughly 3 cm on a 10 m step. The bound only opens up toward the 0.45–0.56% extremes for the small remainder of flights nearer the equator or at higher latitude (Sec. 2.9, “elsewhere”), corresponding to at most roughly 6 cm on a 10 m step. The uncertainty of a GNSS increment must be assessed separately, since errors at successive fixes can be correlated.